The 2023 Mathematics Compendium

114 developed mathematical treatments and 14 connecting arguments and extensions

Developed from the mathematics-only retrospective of the 2023 conversations. Prepared 20 October 2026.

The purpose is to turn earlier questions into actual mathematics: definitions, hypotheses, explicit results, derivations, worked examples and reproducible checks. Every original M identifier is retained. The accompanying HTML edition offers search and filtering; the source archive contains the verification programs and their outputs.

The historical evidence is fragmentary. These are newly authored developments of the recovered questions, not newly discovered 2023 transcripts. Classical theorems and illustrative computations are not claimed as original discoveries. Completing a worked treatment does not settle every open extension attached to it.

Historical labels: U means a user question was recovered in the search evidence; A means an assistant passage survives without the original user wording; T means only a dated topic-level fragment is available. None indicates a complete conversation transcript.

A written proof, a finite exhaustive computation and a floating-point check have different scopes. The proofs below are mathematical exposition, not proof-assistant formalizations. General literature dependencies remain visible. The source ledger records each program’s actual check.

Contents and scope

Part S develops connecting arguments and extended results. The first six concern observable recurrences, reversible information-preserving charts, finite observation algebras, heat-kernel recovery bounds, conservative graph dynamics and Gaussian projection geometry. The additional developments supply explicit classifications and finite certificates.

A Arithmetic nonassociative algebra and exact wave dynamics · M001 to M018

B Harmonic analysis graph invariants and arithmetic geometry · M019 to M037

C Representations geometric phases and dynamical models · M038 to M056

D Probability control relativity and foundations · M057 to M077

E Analysis geometry and computational certificates · M078 to M099

F Singularities integrable systems and quantum field mathematics · M100 to M114

The complete index follows the main treatments. The source archive also contains the original revised-question table, verification records and exact certificates. The HTML edition provides direct entry links.

Connections and extended developments

S001 Minimal recurrences connect observations to finite realizations

Classical framework with a proved synthesis and worked example

A finite-state linear description and a recurrence for an observed sequence are two presentations of the same algebraic structure. The useful dimension is the reachable and observable dimension, which can be smaller than the dimension of the original matrix. This links the Cayley-Hamilton, observability, Fourier and exact-certificate questions without claiming that every nonlinear model has finite closure.

Definitions and assumptions

Work over a field F. A scalar sequence has a linear representation $y_n=cA^nb$ with finite-dimensional A, row c and column b. Its infinite Hankel matrix is $H=(y_{i+j})$ for $i,j\geq 0$.

The rank of the infinite Hankel matrix is the supremum of ranks of its finite submatrices. A recurrence of order r means $y_{n+r}=\Sigma_{j=0}^{r-1}a_j\, y_{n+j}$ for every $n\geq 0$.

Hankel rank equals the minimum linear realization dimension

For a scalar sequence with a finite linear representation, the minimum possible state dimension equals rank H. It also equals the minimum order of an everywhere-valid forward recurrence, with the zero sequence assigned order zero.

Derivation or proof

For any representation H factors as O R, where O has rows cA^i and R has columns A^jb. Therefore rank H cannot exceed the state dimension. Restrict first to the reachable subspace $\operatorname{span}\{A^jb\}$, then quotient by its invisible subspace $\{v : cA^iv=0 \text{ for all } i\}$. The resulting representation is reachable and observable. In a minimal r-dimensional scalar realization, the first r Krylov columns are independent: the first linear dependence would make their earlier span A-invariant and stop further growth. The analogous statement holds for the first r observation rows. Their product is the leading $r\times r$ Hankel block, which is consequently invertible, proving $\operatorname{rank} H\geq r$. Cayley-Hamilton gives a recurrence of order at most r. Conversely, an order-r recurrence has an r-dimensional companion realization, so no smaller recurrence is possible. These arguments also account for finite transients, which can contribute nilpotent modes.

Worked example

Take $y_n=2^n+3^n$, generated by $A=\operatorname{diag}(2,3)$, $b=(1,1)^T$ and $c=(1,1)$.

The first values are 2,5,13,35,97. The leading 2×2 Hankel matrix is $[[2,5],[5,13]]$, whose determinant is 1. The polynomial $(z-2)(z-3)=z^2-5z+6$ gives $y_{n+2}=5y_{n+1}-6y_n$. Thus a two-state representation exists and the nonzero Hankel determinant rules out a one-state representation. If instead $A=\operatorname{diag}(2,-2)$, then $y_n=2^n+(-2)^n$ has dimension two, while the even subsequence $y_{2k}=2\cdot 4^k$ has dimension one.

A short recurrence can be a certificate of exact observable compression. Subsampling can also erase modes, so reduced apparent dimension need not mean that the original dynamics became simpler.

Scope of the result

Finite prefix agreement alone never proves an infinite recurrence without a model-class or order bound. Given any inspected prefix, a sequence that agrees there and changes later is indistinguishable on that prefix. Numerical Hankel rank additionally requires tolerances and perturbation analysis.

Further development

Given an exact matrix family $A(\theta)$, determine where the reachable-observable dimension changes by computing rank minors. This is a finite algebraic question for a specified polynomial parameterization. For nonlinear polynomial dynamics, the degree-growth obstruction in M018 identifies an important class where a polynomial closure containing the state coordinate cannot be finite.

Related entries: M016, M017, M018, M038

References: R001

S002 A reversible observation transform preserves Fisher information

Classical framework with a proved synthesis and worked example

The Gudermannian can turn an unbounded coordinate into a bounded angle and simplify a formula. An exact reversible change of observation cannot itself create statistical information. This distinction turns the old activation and geometry questions into an analysis of representation, quantization, noise and numerical precision.

Definitions and assumptions

Let X have a regular density $p_\theta(x)$, with θ-independent support and square-integrable score $s_\theta=\partial_\theta \log p_\theta$. Fisher information is $I_X(\theta)=E_\theta[s_\theta s_\theta^T]$.

Let g be a θ-independent differentiable bijection with nonzero Jacobian on its domain. $Y=g(X)$ is the same observation expressed in another coordinate.

Fisher information is invariant under an invertible parameter-independent observation chart

Under the stated change-of-variable and regularity assumptions, $I_Y(\theta)=I_X(\theta)$. Additional quantization or a parameter-independent noisy channel cannot increase Fisher information when the score-conditioning identity and the usual differentiation conditions hold.

Derivation or proof

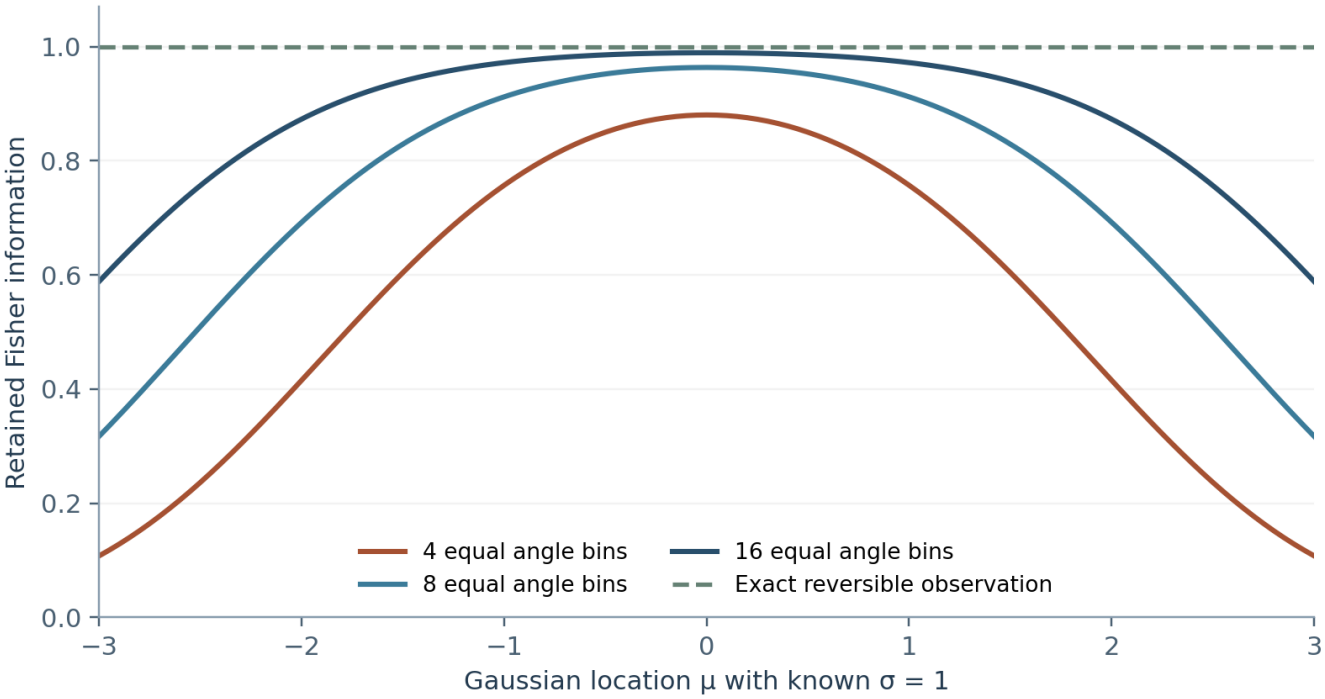
The transformed density is $q_\theta(y)=p_\theta(g^{-1}(y))|\det Dg^{-1}(y)|$. The Jacobian factor is independent of θ, so differentiating its logarithm with respect to θ gives zero. Consequently the transformed score is $s_\theta(g^{-1}(y))$, and change of variables in its second moment proves equality. For a further observation Z produced through a θ-independent channel, differentiation under the integral gives $s_Z(Z)=E[s_X(X)|Z]$. Conditional Jensen yields $E[s_Z s_Z^T]\preceq E[s_X s_X^T]$, since their difference is $E[\mathrm{Cov}(s_X|Z)]$. This is a statement about observations; reparameterizing θ has its own Jacobian transformation law and is a different operation.

Worked example

Let $X\sim N(\mu,\sigma^2)$ with $\sigma>0$ known and take $Y=\mathrm{gd}(X)=\arctan(\sinh X)$, so $-\pi/2<Y<\pi/2$.

The inverse is $x=\operatorname{artanh}(\sin y)$, with $dx/dy=\sec y$. Therefore $q_\mu(y)=\exp(-(\operatorname{artanh}(\sin y)-\mu)^2/(2\sigma^2))\sec y/(\sigma\sqrt{(2\pi)})$. Its μ-score is $(\operatorname{artanh}(\sin y)-\mu)/\sigma^2$, and its squared expectation is $1/\sigma^2$, exactly the original information. If Y is then rounded into a finite set of bins, the retained information becomes $\Sigma_j (\partial_\mu P_j)^2/P_j$ and is at most $1/\sigma^2$. The bundled calculation evaluates this finite-bin quantity directly from Gaussian CDFs.

The coordinate transformation is mathematically useful while its statistical information is unchanged. A benefit under a fixed finite-bit or hardware budget is possible, but it must be demonstrated for that budget and distribution.



Exact reversible observations preserve Fisher information. Finite uniform angle bins lose information in a distribution-dependent way. Curves evaluate Gaussian bin probabilities directly; they are illustrative floating-point calculations.

Scope of the result

The theorem does not forbid a transform from improving optimization behavior, finite-precision conditioning or a restricted approximation class. It also does not apply unchanged to a transform depending on an unknown θ. A bounded activation used inside a noninvertible network is not merely an observation chart.

Further development

Compare candidate transforms at a fixed bit budget by optimizing retained information under a declared prior or worst-case parameter range. Include saturation, endpoint roundoff and added sensor noise. This yields a concrete design objective rather than a claim that bounded coordinates are intrinsically more informative.

Related entries: M010, M014, M015, M047, M048, M078

References: R002, R003

S003 Finite observations generate an algebra of distinguishable classes

Classical framework with a proved synthesis and worked example

The common core of exact classification, graph signatures and probe selection is the partition induced by the available observations. On a finite state set, that partition has a complete algebraic description. This is stronger and more precise than assigning a complexity score to each state.

Definitions and assumptions

Let Ω be a finite set and $f_1,\dots,f_m:\Omega\to\mathbb{R}$ be exact observations, with $m\geq 1$. States x and y are equivalent when $f_i(x)=f_i(y)$ for every i.

Let A be the unital algebra generated by the f_i under real linear combinations and pointwise products. Its elements are polynomial functions of the observation vector.

The observation algebra is exactly the set of functions constant on observation classes

If the observations induce K distinct classes, then A has dimension K and contains the indicator of each class. If every observation is invariant under a group action, no function in A can distinguish two states in the same orbit.

Derivation or proof

Every polynomial in the observations is constant on each equivalence class, giving one inclusion. For the reverse inclusion choose a linear combination $h=\sum a_i f_i$ whose values $h_1,\dots,h_K$ on the classes are all distinct. Such coefficients exist because equality on any two distinct observation vectors excludes only one proper hyperplane, and finitely many proper hyperplanes cannot cover $\mathbb{R}^m$. The Lagrange polynomial $\ell_j(h)=\prod_{k\neq j}(h-h_k)/(h_j-h_k)$ is the indicator of class j. These K indicators span every function constant on the classes and are independent. This also gives a constructive integer choice: take $a_i=T^{(i-1)}$. Each pairwise equality is a nonzero polynomial in T of degree at most $m-1$, so among more than $(m-1)K(K-1)/2$ distinct integers at least one choice avoids all equalities. Group invariance is preserved by sums and products, proving the orbit obstruction.

Worked example

Take $\Omega=\{(0,0),(1,0),(0,1),(1,1)\}$ and first observe $f(x,y)=x+y$.

The observed values $0,1,1,2$ give $K=3$ classes. Swapping coordinates preserves f, so no polynomial in f can separate $(1,0)$ from $(0,1)$. Add $g(x,y)=x$. The combined observation separates all four states; $h=f+2g$ takes values $0,3,1,4$. Its four Lagrange polynomials recover the four individual state indicators exactly. The verification script checks the indicator table using rational arithmetic.

An ambiguity caused by the observation partition cannot be repaired by increasingly elaborate postprocessing of the same data. A new probe must split the ambiguous class.

Scope of the result

The finite-set hypothesis is essential to this elementary algebraic completeness statement. Exact equality, noisy equality and approximate separability are different questions. A coordinate identity or formal feature space may include states that no physical object realizes.

Further development

For a specified family of admissible probes, define each probe’s covered pairs as those it separates. Minimum-cost separation of a finite target set becomes a weighted set-cover problem on state pairs. The certificate is exact only for that declared target population; it makes no automatic claim about unseen states or faults.

Related entries: M017, M020, M021, M040, M047, M099

S004 Heat observations separate exact recovery from stable recovery

Classical framework with a proved synthesis and worked example

The heat-kernel questions contain two distinct issues. A compressed signature can lose information exactly, whereas the full kernel can be complete but poorly conditioned for reconstruction. The exact collision in M020 and the bounds below make both statements concrete.

Definitions and assumptions

Let L and M be real symmetric positive semidefinite matrices. At t>0 define K_L(t)=exp(−tL). Matrix norms below are Euclidean operator norms.

For a labeled undirected weighted graph, L is its combinatorial Laplacian. A heat trace, the diagonal of the heat kernel and the full heat kernel are different observations.

Forward heat evolution is Lipschitz while its logarithmic inverse can amplify noise

> For t≥0, ||exp(−tL)−exp(−tM)||≤t||L−M||. For t>0, the full positive-definite kernel determines L uniquely by L=−log(K_L(t))/t. If t>0 and ||L||,||M||≤Λ, then ||L−M||≤e^(tΛ)||K_L(t)−K_M(t)||/t.

Derivation or proof

Differentiate F(s)=exp(−(t−s)L)exp(−sM) with respect to s. Since F(t)−F(0)=exp(−tM)−exp(−tL), integration gives exp(−tL)−exp(−tM)=$\int_0^t$exp(−(t−s)L)(M−L)exp(−sM)ds. Each exponential has norm at most one because the generators are positive semidefinite; integrating norms proves the first bound. For positive-definite A and B with eigenvalues at least a>0, the resolvent integral for the logarithm and the identity $U^{-1}-V^{-1}=U^{-1}(V-U)V^{-1}$ give ||log A−log B||≤||A−B||$\int_0^\infty$$(a+s)^{-2}$ds=||A−B||/a. The spectral lower bound for either heat matrix is e^(−tΛ); substitute a and divide by t. The principal logarithm is unique for a positive-definite matrix, which proves exact completeness of the full labeled kernel.

Worked example

The 4×4 rook graph and the Shrikhande graph have the same Laplacian spectrum 0,4 with multiplicity 6, and 8 with multiplicity 9, but different clique numbers.

Both are vertex-transitive, so every diagonal heat value equals (1+6e^(−4t)+9e^(−8t))/16 for every t. Thus even all-time vertex heat signatures cannot distinguish these two graphs. Their full heat matrices do differ up to graph isomorphism, because equality after a vertex relabeling would force equality of their recovered Laplacians. For the full-kernel inverse bound, e^(tΛ)/t is minimized at t=1/Λ when Λ>0 and the absolute kernel-error budget is held fixed.

There can be exact blindness in a compressed signature and severe numerical sensitivity in a complete observation. Completeness and robustness must be evaluated separately.

Scope of the result

The inverse bound applies to exact heat kernels of symmetric positive semidefinite generators with a common norm bound. An arbitrary noisy matrix need not remain positive definite; reconstructing a nearest valid generator is a separate constrained problem. The time 1/Λ minimizes this particular upper bound, not every practical estimator’s risk.

Further development

Combine the exact observation partition with perturbation bounds: first determine which states are distinguishable in principle, then select times and probes that provide a specified margin under the declared noise model. This avoids optimizing a highly precise score on a fundamentally incomplete signature.

Related entries: M020, M022, M040, M068, M084

References: R004

S005 Positivity conservation entropy and spectra share one graph update

Classical framework with a proved synthesis and worked example

A reaction-diffusion discretization becomes much more informative when its algebraic and probabilistic properties are checked together. For a finite undirected graph, one elementary step provides mass conservation, positivity, an entropy inequality and an explicit stability boundary.

Definitions and assumptions

Let a_ij=a_ji≥0, a_ii=0 define a connected finite weighted graph, with degrees d_i=Σ_j a_ij and d_max>0. Let L=diag(d)−A and P_h=I−hL.

For row probability vectors p, the update is p↦pP_h. The uniform distribution u is stationary whenever P_h is doubly stochastic.

An explicit step is a Markov update precisely within its positivity range

For 0≤h≤1/d_max, P_h is nonnegative and doubly stochastic, so it conserves probability and cannot decrease Shannon entropy. At h≤1/(2d_max), all eigenvalues of P_h are nonnegative. For h>0 in that latter range, every zero-sum vector contracts in Euclidean norm by at most 1−hλ₂(L), where λ₂ is the spectral gap.

Derivation or proof

Off-diagonal entries of P_h are $h a_{ij} \geq 0$, diagonal entries are $1-h d_i$ and each row sums to one. Symmetry gives column sums one. With $\varphi(s)=s \log s$ and $\varphi(0)=0$, Jensen applied to each column gives $\varphi((pP)_j) \leq \Sigma_i P_{ij} \varphi(p_i)$; summing and using row sums proves $\Sigma \varphi(pP) \leq \Sigma \varphi(p)$, which is the entropy statement. Symmetric diagonalization gives eigenvalues $1-h\lambda_i(L)$. The bound $\lambda_{max}(L) \leq 2d_{max}$ follows from $x^T L x = \Sigma_{(i<j)} a_{ij}(x_i-x_j)^2 \leq 2\Sigma_i d_i x_i^2$. Therefore $h \leq 1/(2d_{max})$ makes all update eigenvalues nonnegative. On the orthogonal complement of constants, the largest is $1-h\lambda_2$. Squaring gives the corresponding residual-energy contraction.

Worked example

For a periodic one-dimensional grid with diffusion coefficient D and mesh size Δx, the nearest-neighbor generator has degree $2D/\Delta x^2$.

The condition $h \leq 1/d_{max}$ becomes $r = D\Delta t/\Delta x^2 \leq 1/2$, the usual positivity condition for the explicit stencil. Taking $r \leq 1/4$ also makes every Fourier multiplier nonnegative. At the endpoint $r=1/2$ on an even cycle, the alternating mode has multiplier -1 and does not decay: positivity and conservation hold, but convergence to equilibrium can fail through periodicity. A two-state swap gives the simplest example: $p=(1,0)$ alternates with $(0,1)$, preserving zero Shannon entropy.

A valid nonnegative conservative update need not be a strictly mixing update. The spectral calculation identifies the missing hypothesis and quantifies convergence once it is supplied.

Scope of the result

Reaction terms, directed edges, nonuniform stationary measures and adaptive time steps need additional analysis. Entropy monotonicity for the uniform measure does not automatically transfer to arbitrary dynamics; a relative entropy against the correct stationary measure is often the relevant quantity.

Further development

Design a discretization or graph update to satisfy a joint contract: mass, positivity, symmetry, entropy and a specified decay rate. Then verify each property algebraically before measuring prediction performance.

Related entries: M023, M026, M044, M045, M053, M080, M091

References: R005

S006 Whitening turns sensor questions into projection geometry

Classical framework with a proved synthesis and worked example

The squared-residual, covariance and observation questions admit an exact common model for Gaussian mean changes. In this model one can calculate precisely what a linear sensor panel preserves and what it loses. The same calculation gives a robust margin and an impossibility statement for universal dimension reduction.

Definitions and assumptions

Let $X \sim N(\mu+\delta,\Sigma)$, where the baseline μ and positive-definite covariance Σ are known and δ is an unknown mean shift. Let B be a $k \times d$ matrix of full row rank and observe $Y=BX$.

Whiten the shift as $d_0=\Sigma^{(-1/2)}\delta$. Define $C=B\Sigma^{(1/2)}$ and the orthogonal projection $P=C^{T}(CC^{T})^{-1}C$ onto the row space of C.

A linear panel retains exactly the projected noncentrality

> The statistic $T_B=(Y-B\mu)^{T}(B\Sigma B^{T})^{-1}(Y-B\mu)$ has a noncentral χ^2_k law with noncentrality $\lambda_B=||P d_0||^2$. Full observation has $\lambda_{full}=||d_0||^2$. Hence $\lambda_B \le \lambda_{full}$, with equality exactly when d_0 lies in the retained row space. If $k<d$, some nonzero shifts have $\lambda_B=0$.

Derivation or proof

Write $X-\mu=\Sigma^{(1/2)}(Z+d_0)$ with Z standard normal. Multiplying by $(CC^{T})^{(-1/2)}C$ gives a $k \times d$ matrix with orthonormal rows. It maps Z to a standard k-dimensional normal, while the squared norm of the transformed shift is $d_0^{T}C^{T}(CC^{T})^{-1}Cd_0=||P d_0||^2$. This proves the distribution and noncentrality formula. The Pythagorean identity $||d_0||^2=||P d_0||^2+||(I-P)d_0||^2$ proves the inequality and equality condition. If $k<d$, the projection has a nontrivial kernel, giving a nonzero invisible shift. For perturbations $d=d_0+e$ with $||e|| \le \varepsilon$, triangle inequality gives $||Pd|| \ge \max(||Pd_0||-\varepsilon,0)$. Equality is attained by choosing e opposite to Pd_0 within the retained subspace, or by cancelling it completely when ε is large enough. Thus the exact worst-case noncentrality is $\max(||Pd_0||-\varepsilon,0)^2$.

Worked example

Take $\Sigma=I_3$ and let B retain the first two coordinates. Then $P=\mathrm{diag}(1,1,0)$.

A shift $\delta=(3,4,0)$ has $\lambda_{full}=\lambda_B=25$. A shift $\delta=(0,0,5)$ also has $\lambda_{full}=25$ but $\lambda_B=0$: the selected panel sees the null distribution exactly. For the first shift with whitened uncertainty radius $\varepsilon=2$, the worst-case retained noncentrality is $(5-2)^2=9$, attained by perturbation $e=(-6/5,-8/5,0)$. Under the null, $T_B \sim \chi^2_2$, with mean 2 and variance 4; the full statistic has three degrees of freedom. Comparing only raw residual scores without this calibration can be misleading.

Exact optimization for a declared finite list of shifts is useful and checkable. Preserving every possible shift while reducing dimension is impossible in this model. Robustness can be expressed as a proved geometric margin rather than a hope that selected probes generalize.

Scope of the result

Known Gaussian covariance, mean-shift alternatives and linear measurements are strong assumptions. Estimated baselines change calibration, non-Gaussian tails can invalidate χ^2 laws and nonlinear features need their own analysis. The noncentrality comparison alone is not a universal theorem about every detector at every significance level.

Further development

For a finite target family $\{d_j\}$, choose a minimum-cost panel subject to $\max(||P_B d_j||-\varepsilon_j,0)^2 \ge \tau_j$. Solve a small instance exactly and report every target and uncertainty set. This produces a concrete experimental-design certificate whose scope is visible.

Related entries: M017, M047, M060, M061, M062, M078

S007 Exact finite polynomial closure for triangular maps and its resonance strata

A complete theorem for the specified triangular family with exact examples; no novelty claim

The univariate degree-growth obstruction in M018 has a substantial positive counterpart. Every triangular polynomial map T(x,y)=(ax,by+p(x)) admits an exact finite linear feature representation containing the original coordinate functions. Its smallest such space can be classified completely by collisions among the eigenvalues a^k. Repeated nonlinear eigenvalues reduce the required feature count, while resonances with b determine the only possible nontrivial Jordan chain. The same calculation yields an exact iterate formula and a criterion for removing the forcing by a polynomial fiber translation. An invertible coupled counterexample demonstrates why this is a theorem about a specified family, not a universal finite-closure principle.

Definitions and assumptions

Work over F=R or C with the polynomial ring F[x,y]. Let a,b∈F and p(x)=Σ_(k=0)^d c_kx^k, with zero coefficients omitted from its support. The Koopman operator is U_Tf=f∘T.

A feature space is an F-linear subspace W of polynomials with U_TW⊂W. It is not required to be closed under pointwise products. A finite-dimensional algebra containing x would already be impossible because it would contain every x^n.

Require W to contain 1,x,y. Let q(x)=Σ_(k≥2)c_kx^k, S={k≥2:c_k≠0}, Λ={a^k:k∈S}, and v_λ(x)=Σ_(k∈S,a^k=λ)c_kx^k for λ∈Λ.

On the diagonal subspace E=span{1,x,v_λ:λ∈Λ}, group all terms of p by their eigenvalue: p_μ(x)=Σ_(k≥0,a^k=μ)c_kx^k. For k=0 use a^0=1, including a=0. Missing groups mean the zero polynomial.

Minimal feature dimension, resonance normal form and exact iterates

The unique smallest invariant linear feature space containing 1,x,y is W_min=span{1,x,y,v_λ:λ∈Λ}, and dim W_min=3+|Λ|. Its action is U1=1, Ux=ax, Uv_λ=λv_λ and Uy=by+p. Put z=y+Σ_(μ≠b)p_μ/(b−μ). Then Uz=bz+p_b and Up_b=bp_b. Consequently U|W_min is diagonalizable exactly when p_b=0; if p_b≠0 it has precisely one size-two Jordan block at b, with every other block of size one. Writing D={1,a,b}∪Λ as a set, its minimal polynomial is ∏_(μ∈D)(t−μ), multiplied by one further factor (t−b) when p_b≠0. Its characteristic polynomial, with multiplicities retained, is (t−1)(t−a)(t−b)∏_(λ∈Λ)(t−λ).

For n≥0 the exact state iterate is x_n=a^n x and y_n=b^n y+Σ_μ C_n(b,μ)p_μ(x), where C_0=0 and C_n(b,μ)=Σ_(j=0)^(n−1)b^(n−1−j)μ^j. For b≠μ this is (b^n−μ^n)/(b−μ); for b=μ≠0 it is n b^(n−1). The convolution sum is the definition at b=μ=0 and removes any 0^0 ambiguity: C_1=1 and C_n=0 for n≥2.

Derivation or proof

The proposed space is invariant because each v_λ is a polynomial eigenfunction, while Uy=by+c_0+c_1x+Σv_λ belongs to it. Every invariant W containing 1,x,y must contain q=Uy−by−c_0−c_1x and hence U^jq for every j. The distinct-eigenvalue decomposition is U^jq=Σ_(λ∈Λ)λ^jv_λ. Its first |Λ| rows form a Vandermonde matrix with nonzero determinant, including when one eigenvalue is zero. Therefore W contains each v_λ. Equivalently, applying the Lagrange spectral projector ∏_(ν≠λ)(U−νI)/(λ−ν) to q extracts v_λ. This proves minimality. The nonlinear group polynomials have disjoint nonempty monomial supports of degree at least two, so they are mutually independent and independent of 1,x,y. Thus the dimension formula follows even if a nonlinear eigenvalue coincides with 1 or a; that coincidence does not permit discarding the nonlinear vector.

The restriction to E is diagonal in the displayed basis, though its eigenvalues may repeat. Decompose p into its E-eigencomponents p_μ. Direct calculation gives U(y+h)−b(y+h)=p+(U−b)h. Taking h=Σ_(μ≠b)p_μ/(b−μ) cancels every nonresonant component and leaves p_b. If p_b=0, z is an additional eigenvector independent of E and completes an eigenbasis. If p_b≠0, z maps to p_b under U−b and p_b maps to zero, giving a Jordan chain of length two. No chain can be longer because E is diagonal. Replacing a basis of E_b by one containing p_b leaves exactly one such chain and all remaining vectors as eigenvectors. The characteristic and minimal polynomial formulas now follow, including the case in which b equals several listed diagonal eigenvalues.

The iterate formula follows by solving y_(n+1)=b y_n+p(a^n x) and expanding the forcing into p_μ(a^j x)=μ^j p_μ(x). Induction produces the convolution sum; multiplying it by b−μ telescopes to b^n−μ^n. Resonance makes every term of the sum equal b^(n−1), which explains the factor n. Finally a polynomial fiber change H(x,y)=(x,y+h(x)) removes p entirely exactly when c_k+(a^k−b)h_k=0 for every coefficient k. This is possible precisely when all resonant c_k vanish, equivalently p_b=0, and the displayed h supplies it. All such h differ by a polynomial satisfying h(ax)=bh(x), so resonant eigenfunctions describe the freedom in the coordinate change. This is a criterion for that specified class of coordinate changes, not a classification of arbitrary polynomial conjugacies.

Worked example

The twelve exact replay cases include generic real eigenvalues, a=0, a=1, a=−1, affine forcing, zero forcing and the complex fourth root of unity a=i. All coefficients and all calculations are rational or Gaussian-rational. The following examples expose different structural regimes.

For a=2, b=4 and p=x²+x³, Λ={4,8}, so W_min=span{1,x,y,x²,x³} has dimension five. The b=4 component is x² and creates a size-two Jordan chain. Taking z=y−x³/4 gives Uz=4z+x². The exact iterate is y_n=4^n y+n4^(n−1)x²+(8^n−4^n)x³/4 for n≥1. This is a nonlinear map with a finite exact linear feature evolution, and the resonant term has the predicted n factor.

For a=−1, b=1 and p=x²+2x³+3x⁴, the even powers form v_+=x²+3x⁴ and the odd power forms v_−=2x³. Dimension is five rather than the six obtained by retaining every monomial separately. The shifted coordinate z=y+x³ satisfies Uz=z+v_+, so the minimal polynomial is (t−1)²(t+1). For a=0, b=0 and p=3+2x+x²+2x⁵, the single nonlinear group is q=x²+2x⁵, dimension is four, and z=y−3 satisfies Uz=2x+q and U²z=0. This is the zero-eigenvalue Jordan case, not a reason to discard q.

For a=i and p=x²+x⁶+2x³+3x⁷+x⁴+x⁸, six nonlinear monomials collapse into the three residue groups modulo four, so dimension is six. Their eigenvalues are −1,−i,1. Choosing b=−1 creates a resonant chain; choosing b=2 does not. Conversely, a=2,b=3,p=3+2x+x²+x³ has no resonance: z=y+3/2+2x−x²−x³/5 gives the actual two-coordinate linear dynamics x′=2x,z′=3z. Nevertheless

any invariant linear space required to contain the original y still needs the nonlinear readout features and has dimension five.

For contrast, the coupled invertible map G(x,y)=(y,y²−x) has inverse G^(−1)(x,y)=(x²−y,x). If Y_n=U_G^n y, then Y_0=y, Y_1=y²−x and Y_(n+1)=Y_n²−Y_(n−1). Induction gives total degree 2^n with leading monomial y^(2^n); the lower-degree subtracted term cannot cancel it. The Y_n are therefore linearly independent, so no finite invariant polynomial feature space containing y exists. Invertibility by itself neither guarantees nor obstructs finite closure.

Finite closure, feature dimension, eigenvalue collision and Jordan resonance are separate questions and all are decidable for the triangular family. The verifier independently forms polynomial compositions and exact ranks, checks closure and generation, verifies every asserted Jordan relation and minimal-polynomial factor, and compares eight iterates with the closed formula in twelve cases. The coupled-map calculation illustrates the proved degree-growth obstruction rather than relying on a fitted approximation.

Scope of the result

The classification holds for the specified triangular polynomial family over R or C and for invariant linear spaces containing 1,x,y. It does not classify all polynomial systems, all finite smooth feature embeddings or all nonlinear conjugacies. A different observation requirement, nonlinear output recovery, restriction to a finite orbit or quotient relations among states can change the minimal representation problem. At an exact root of unity the dimension can drop, while parameters arbitrarily close to that stratum retain the generic dimension and may produce ill-conditioned Vandermonde coordinates. Exact algebraic dimension therefore does not imply robust numerical recoverability.

Further development

Parameter strata are explicit. For fixed nonzero nonlinear support S, real a outside {0,1,−1} gives |Λ|=|S|. At a=0 or a=1, every nonlinear term belongs to one group, so a genuinely nonlinear p gives dimension four. At a=−1 the count is the number of occupied parities, giving dimension four or five. Over C, a≠0 that is not a root of unity gives distinct nonlinear eigenvalues. If a has finite multiplicative order h, group exponents by their residue modulo h and count the occupied residues. Coefficient hyperplanes c_k=0 change the support. For a fixed finite degree, collision loci satisfy a^k=a^l for some supported distinct exponents, so the possible collision strata form a finite union of zero and root-of-unity loci. When the nonlinear support has at most one exponent, there are no pairwise collision strata. Jordan resonance occurs exactly when (b=1 and c_0≠0), or (b=a and c_1≠0), or (b=a^k for at least one supported k≥2). Because these components use independent monomials, they cannot cancel each other when eigenvalues coincide. These statements classify the exact dimension and Jordan strata of this family. A useful further investigation is a perturbation analysis near a collision or resonance, rather than an unsupported claim that nearby numerical features share the lower-dimensional exact closure.

Related entries: M003, M015, M016, M017, M018, S001, S003

References: R006, R007

S008 An exact robust Gaussian sensor panel with a strict greedy gap and a blind direction

Constructed finite model proved and exhaustively certified

This worked design turns the projection formula in S006 into a complete exact optimization problem. Six coordinate channels, six declared mean-shift centers, nonuniform costs and six uncertainty balls give a unique cost13 optimum. A fully specified gain-per-cost greedy rule instead costs16. Every channel participates in an inclusion-minimal feasible panel, no single channel is cost-dominated, and none of the six fault constraints is implied by the others. A deliberately withheld direction is nevertheless invisible to the optimum panel. All of these claims concern this constructed mathematical instance, with no empirical dataset or application-performance claim.

Definitions and assumptions

Let $X\sim N(\mu+d+e,I_6)$, with a known baseline μ and deterministic mean shift $d+e$. A coordinate panel $S\subseteq\{A,B,C,D,E,F\}$ observes the selected coordinates. Its centered squared statistic $T_S=\Sigma_{i\in S}(X_i-\mu_i)^2$ has a noncentral $\chi^2_{|S|}$ law with noncentrality $\lambda_S(d+e)=\Sigma_{i\in S}(d_i+e_i)^2$. The empty panel is allowed in the finite enumeration but cannot meet the positive target.

The six costs and coverage sets are A:(6,{1,3,5}), B:(7,{1,2,3}), C:(6,{4}), D:(6,{2,3,5,6}), E:(7,{2,4,5,6}), F:(4,{1,6}). Define the six rational centers by $d^{(j)}_i=2$ when channel i covers target j and zero otherwise. Thus $d^{(1)}=(2,2,0,0,0,2)$, $d^{(2)}=(0,2,0,2,2,0)$, $d^{(3)}=(2,2,0,2,0,0)$, $d^{(4)}=(0,0,2,0,2,0)$, $d^{(5)}=(2,0,0,2,2,0)$, and $d^{(6)}=(0,0,0,2,2,2)$.

For each center, the declared uncertainty set is the full Euclidean ball $||e||_2\le1$ in six-dimensional whitened coordinates. Feasibility means $\inf_{||e||\le1}\lambda_S(d^{(j)}+e)\ge1$ for every $j=1,\dots,6$. Costs add across selected channels. Noncentrality is the declared design criterion; panels of different sizes have different null degrees of freedom and require their own significance calibration.

The greedy heuristic starts with no channels. At each step it chooses the unused channel maximizing the number of newly satisfied target constraints divided by its cost; ties would be settled alphabetically. It stops when all six constraints are satisfied. There are no maximizing ties along the path in this instance.

Exact robust-margin reduction and a unique finite optimum

For any orthogonal coordinate projection P_S and $\varepsilon\ge0$, $\inf_{||e||\le\varepsilon}||P_S(d+e)||^2=\max(||P_Sd||-\varepsilon,0)^2$. Consequently the declared $\varepsilon=1$,target1 problem is feasible exactly when every center has selected energy at least4. The unique minimum-cost panel is {A,E}, of cost13. Greedy selects D,F,C and has cost16, a strict ratio16/13. The optimum is blind to the withheld center $d^*=2e_D$, whose full-observation noncentrality is4.

Derivation or proof

Triangle inequality and $||P_Se||\le||e||$ give $||P_S(d+e)||\ge\max(||P_Sd||-\varepsilon,0)$. If $||P_Sd||>\varepsilon$, choose $e=-\varepsilon P_Sd/||P_Sd||$. If $||P_Sd||\le\varepsilon$, choose $e=-P_Sd$. Both perturbations are allowed and attain the bound, including the zero case. For positive target1 and $\varepsilon=1$, the inequality $\max(\sqrt{\lambda}-1,0)^2\ge1$ is equivalent to $\lambda\ge4$. Every nonzero center coordinate is2, so the energy condition is equivalent to selecting at least one channel from that target’s coverage support. The statistical robustness problem has therefore reduced exactly, without approximation, to this declared weighted set-cover instance.

The panel A,E covers all targets and costs13. Any feasible panel containing C needs at least two additional channels: no pair C plus one other channel covers all six targets. The cheapest possible two other channels cost4+6, so such a panel costs at least16. A feasible panel not containing C must contain E, because only C and E cover target4. At total cost at most13, E can be accompanied by at most one other channel, since the two cheapest remaining costs are4 and6. E alone fails; E,F misses target3; E,D misses target1; E,A succeeds and costs13; E,B exceeds the budget. Thus A,E is the unique feasible panel at cost at most13. This gives an analytic optimality and uniqueness certificate in addition to the enumeration.

Initially D has gain/cost4/6=2/3, uniquely larger than E’s4/7 or any other ratio. After D, the missing targets are1 and4. F supplies target1 at ratio1/4, uniquely beating A’s1/6 and the other choices. After D,F only target4 remains; C’s1/6 beats E’s1/7. Greedy therefore returns D,F,C of total cost16. No claim is made that this heuristic is worst possible, or that the example estimates its usual behavior.

For the withheld vector2e_D, P_{A,E}d*=0. Its selected Gaussian law is exactly the same as the baseline law. Hence every test using only those observed coordinates, including repeated independent samples, has the same rejection probability under this alternative as under the null. Full observation retains norm2 and worst-case noncentrality$(2-1)^2=1$ under the same unit uncertainty. More generally, any rank-k linear panel with k<6 has a nonzero kernel. Selecting an appropriately scaled shift in that kernel proves that no such panel can preserve all mean-shift directions. This is an information obstruction, not a failure of the finite optimization certificate.

Worked example

The portable supplement_verify.py enumerates every one of the$2^6=64$ coordinate panels. It records the selected channels, integer cost, six integer center energies, covered-target list and feasibility flag. It also records the exact greedy candidates at every step, inclusion-minimal panels and witnesses that each target constraint is necessary.

There are34 feasible panels. The optimum A,E has center energies(4,4,4,4,8,4). Its six exact worst-case noncentralities are$(1,1,1,1,9-4\sqrt{2},1)$. For the five energy4 centers, subtracting one unit in their sole active retained coordinate attains1. For target5, subtract$(1/\sqrt{2},1/\sqrt{2})$ in coordinates A,E; this perturbation has norm1 and attains$9-4\sqrt{2}$. Greedy D,F,C has center energies(4,4,4,4,4,8), and its minimum robust margin is also1 for the declared targets, at the higher cost16. A withheld center2e_D has selected noncentrality0 in A,E but4 in D,F,C and the full panel. For the simple Gaussian center alternative, the selected Kullback-Leibler divergence from baseline is half the selected noncentrality, so it is zero for the withheld direction in the optimum panel.

To see that the six channels and six constraints are substantive, inspect the additional exact witnesses. No channel has its entire target set contained in that of another channel of equal or smaller cost. Each channel belongs to at least one inclusion-minimal feasible panel. For each target j, selecting all channels that do not cover j satisfies the other five targets and fails exactly j. Thus no target constraint is implied by the other five. The verifier checks all these finite properties rather than relying on a picture of the incidence matrix.

The finite optimum is exact, has a strict cost advantage over the stated greedy rule, and remains unable to guarantee unlisted directions. The constructive counterexample is compatible with all the declared-target certificates. Exact optimization, robustness to a specified ball, detector calibration and universal observability are distinct questions that can now be stated and checked separately.

Scope of the result

This is a deliberately constructed six-channel Gaussian location model with known identity covariance and additive deterministic mean uncertainty. Its costs, centers and radius are mathematical inputs, not measured quantities. The objective constrains worst-case noncentrality; it is not an empirical accuracy guarantee, and equal noncentralities in different degrees of freedom do not by themselves imply equal test power. Estimated covariance, covariance-changing faults, temporal dependence, nonlinear sensors and unmodeled shifts need separate analysis. The withheld vector demonstrates a scope boundary and does not invalidate optimality for the declared family. The general robust-margin formula is classical projection geometry, and the small exact instance is an illustrative certificate rather than a claim of new general optimization theory.

Further development

Replace identity covariance by a fixed rational positive-definite matrix Σ and let each nonempty panel measure specified linear combinations with a full-row-rank matrix B. Then $B\Sigma B^T$ is invertible and $\lambda_S = d^T K_B d$, where $K_B = B^T(B\Sigma B^T)^{-1}B$; state the uncertainty ball in whitened coordinates. When B and Σ are rational, K_B is rational. The Euclidean orthogonal projector in whitened coordinates is instead $P = \Sigma^{(1/2)} K_B \Sigma^{(1/2)}$, which need not be rational: for $\Sigma = \mathrm{diag}(1,2)$ and $B = (1,1)$, K_B is the all-ones matrix divided by3, whereas $P = [[1,\sqrt{2}],[\sqrt{2},2]]/3$. Redundant measurement rows require a separately stated pseudoinverse and observation-rank convention. For finitely many candidate full-row-rank panels, exhaustive enumeration still certifies the optimum, although robust thresholds can involve algebraic numbers. A further exact task is to classify the parameter cells in costs or radii where the optimal panel changes, while retaining the nullspace impossibility for any universal dimension reduction.

Related entries: S006, M017, M060, M061, M062, M078, M106

S009 Exact tree synchronization certificates and observable phase information

Full nondegenerate tree branch classification and quantitative bounds proved here; rational examples verified; no general cyclic or global-convergence claim

A finite tree makes the nonlinear locking problem unusually exact: every edge flow is fixed by a cut sum, the cohesive phase differences follow by arcsine, and stability is certified by a positive weighted Laplacian. The same calculation separates dynamic stability from observability. Relative-phase measurements lose common phase and common frequency, while even a stable synchronized star has exact nonlinear motions invisible to a center-only sensor. The locking criterion is classical; the derivation, rational certificate, sensitivity bounds and observation examples here are an explicit synthesis, not a claim of a newly discovered synchronization theorem.

Definitions and assumptions

Let $T=(V,E)$ be an undirected connected tree with $n \geq 2$ vertices, arbitrarily oriented edges and incidence matrix $B \in R^{(n \times (n-1))}$. A column has -1 at its tail and $+1$ at its head. Every edge has coupling $K_e > 0$.

Use the symmetric Kuramoto equation $\dot{\theta} = \omega - B\,\mathrm{diag}(K)\sin(B^T\theta)$, componentwise sine. Since $1^T B = 0$, mean frequency is $\Omega = (1/n)1^T\omega$. In the rotating frame set $b = \omega - \Omega 1$, so $1^T b = 0$ and $\dot{\theta} = b - B\,\mathrm{diag}(K)\sin(B^T\theta)$.

General phase states lie on $(R/2\pi Z)^n$ modulo the diagonal common-phase circle. A cohesive lift has $\delta=B^{T}\theta$ with $|\delta_e|<\pi/2$. Fixing $1^{T}\theta=0$ removes the common-phase direction locally. For $0<\gamma<\pi/2$ one can instead impose the closed bound $|\delta_e|\le\gamma$.

For each oriented edge e, remove it and let H_e be the component containing its head. Define the cut flow $f_e=\Sigma_{(i\in H_e)}b_i$. At a locked state the flow equation is $Bf=b$.

The linearization at a cohesive locked state is $\dot\eta=-L_w\eta$, where $L_w=B\,\mathrm{diag}(w)B^{T}$ and $w_e=K_e\cos\delta_e=\sqrt{(K_e^2-f_e^2)}>0$. Let λ_2 denote the least positive Laplacian eigenvalue.

Necessary and sufficient tree locking, local stability, boundary sensitivity and observability

If every $|f_e|<K_e$, there are exactly $2^{(n-1)}$ locked equilibria on the phase torus modulo common phase. Each edge independently selects $\arcsin(f_e/K_e)$ or $\pi-\arcsin(f_e/K_e)$, modulo 2π. An equilibrium with q negative edge cosines has exactly q unstable linear directions on the phase quotient; there are $C(n-1,q)$ such equilibria. Thus precisely one branch, with all edge cosines positive, is locally exponentially stable and cohesive. A state with $|\delta_e|\le\gamma<\pi/2$ exists exactly when $|f_e|\le K_e\sin\gamma$ for every edge. If any $|f_e|>K_e$ there is no locked equilibrium; if exactly r edges saturate and the others are strictly feasible, the count becomes $2^{(n-1-r)}$ and the quotient Hessian has r zero directions. For the stable branch, edge-flow sensitivity is $\partial\delta_e/\partial f_e=1/w_e$. The rate bounds are $\min_e(w_e)\lambda_2(BB^{T})\le\lambda_2(L_w)\le n\,w_e/[s(n-s)]$ for every edge separating s and $n-s$ vertices. A full relative-phase panel leaves only common phase unobservable and a uniform natural-frequency shift unidentifiable; a restricted center-only star sensor can miss additional exact nonlinear trajectories.

Derivation or proof

First prove the cut formula. A connected tree has $n-1$ edges and B has full column rank. One elementary proof removes a leaf: a zero-divergence flow has zero flow on its only incident edge, and induction removes all edges. The column image is contained in $1^{\perp}$ and has dimension $n-1$, so it equals $1^{\perp}$. Thus $Bf=b$ has exactly one solution. Summing the divergence equation over H_e cancels every internal edge and leaves $+f_e$ at the unique crossing edge, giving the displayed cut sum. Conversely those sums produce the unique solution; this can also be constructed by repeatedly eliminating leaves. No nonlinear approximation has been made.

At an equilibrium, $f_e=K_e\sin\delta_e$. Strict cohesion forces $|f_e|<K_e$ and gives the unique principal value $\delta_e=\arcsin(f_e/K_e)$. On a tree, any collection of real edge differences lifts to node phases: choose a root phase and integrate differences along the unique root-to-node path. There are no cycle-sum compatibility conditions. Subtracting the node mean fixes the chosen gauge. This proves sufficiency and uniqueness modulo common phase. Replacing the interval by $[-\gamma,\gamma]$ gives precisely $|f_e|\le K_e\sin\gamma$ because sine is strictly increasing there. For a graph with cycles, the integration step can fail even when individual edge bounds hold, so this proof must not be extended merely by reusing its inequalities.

For the full branch classification, consider phase differences modulo 2π rather than restricting to the principal interval. Each strict scalar equation $\sin\delta_e=f_e/K_e$ has exactly two distinct circle solutions, with cosines of opposite sign. A tree integrates every edge assignment uniquely modulo common phase, so there are exactly $2^{(n-1)}$ equilibria. On $1^{\perp}$ the linear map $T=B^{T}$ is an isomorphism. The quotient Hessian quadratic form is $x\mapsto(Tx)^{T}\mathrm{diag}(K_e\cos\delta_e)(Tx)$, and Sylvester's law of inertia therefore gives exactly one negative direction per negative edge cosine, one positive direction per positive cosine and one zero direction per saturated edge. The dynamics has the negative of this Hessian as its Jacobian, so q negative cosines give q unstable eigenvalues. Choosing the q negative

branches among n−1 edges gives C(n−1,q). All-positive cosines give the unique local minimum; every other nondegenerate branch is a saddle or the all-negative repeller. If r edges saturate, each of those sine equations has only one circle solution, giving 2^(n−1−r) equilibria with r extra linear zero directions. These boundary statements classify the algebraic count and Hessian inertia only; higher-order behavior at a zero eigenvalue is not inferred.

For local stability, differentiate the vector field. Its negative Jacobian is L_w. For every x, $x^{T}L_wx=\Sigma_e w_e(B^{T}x)_e^2$. With positive w and connected T, this vanishes exactly when x is constant. Hence the common-phase eigenvalue is zero and every mean-zero eigenvalue is positive. This proves transverse linear stability, and the nonlinear conclusion follows directly from a local gradient estimate. On a mean-zero cohesive lift define $U(\theta)=-b^{T}\theta-\Sigma_e K_e \cos((B^{T}\theta)_e)$; the rotating-frame dynamics is $-\nabla U$. Around a strict cohesive equilibrium choose a sufficiently small convex ball inside a domain $|\delta_e|\le\gamma<\pi/2$. Its Hessian is bounded below on $1^{\perp}$ by mI with $m=\cos\gamma\cdot\lambda_2(B\,\mathrm{diag}(K)B^{T})>0$. If $x=\theta-\theta^*$, then $d||x||^2/dt\le-2m||x||^2$ in that ball. The vector field points inward on its boundary, so sufficiently near initial states stay there and converge at least as e^(−mt). This is a local argument; no invariance or attraction of the entire cohesive domain has been assumed.

Differentiate arcsin(f_e/K_e) while holding K_e fixed to get $\partial\delta_e/\partial f_e=1/\sqrt{(K_e^2-f_e^2)}=1/w_e$. Because f_e is a linear cut sum of b, this also gives exact frequency sensitivities. Holding the other flows fixed and changing only f_e shifts the phases by a constant on each component of the cut. In the mean-zero gauge, the two constants differ by dδ_e; therefore $||\partial\theta/\partial f_e||_2=\sqrt{(s(n-s)/n)}/w_e$. Thus sensitivity diverges as an edge saturates, even though the equilibrium remains locally stable before the boundary.

The rate estimates quantify the same boundary. The inequality $L_w\succcurlyeq w_{\min} BB^{T}$ gives $\lambda_2(L_w)\ge w_{\min}\lambda_2(BB^{T})$ by the Rayleigh principle on $1^{\perp}$. For the upper bound, use a test vector equal to (n−s)/n on a component of size s and −s/n on the other. Its only nonzero edge difference is one across the removed edge, its squared norm is s(n−s)/n and its energy is w_e. Its Rayleigh quotient is $n\,w_e/[s(n-s)]$, proving the bound. If one bridge weight tends to zero, the convergence rate therefore tends to zero; the tree is dynamically approaching disconnection in its linearized restoring force.

For observability, let $A=-L_w$ and $C=B^{T}$. Then ker C=span{1}, A1=0, and the unobservable linear subspace $\bigcap_{j\ge0}\ker(CA^j)$ is exactly span{1}: the j=0 kernel already gives an upper bound, while common phase survives every power. For sine measurements the local derivative is $\mathrm{diag}(\cos\delta^*)B^{T}$, whose diagonal factor is invertible under strict cohesion, so the same conclusion holds. Adding an absolute phase measurement e_r^{T} makes the stacked instantaneous observation injective. Separately, replacing ω by ω+c1 changes a solution to θ(t)+ct1 and leaves every relative-phase output unchanged. Relative data can identify b but cannot determine Ω without a time or phase reference.

A stronger blindness example does not rely on common phase. Take a star with center 1 and identical coupling k>0, equal natural frequencies removed by a rotating frame, and at least two leaves. Put $\theta_1(t)=0$, $\theta_0(t)=a(t)$, $\theta_2(t)=-a(t)$, and all other leaf phases zero, where $\dot a=-k\sin a$. The center receives sin a+sin(−a)=0, and each leaf equation is satisfied. For $|a_0|<\pi$, $a(t)=2\arctan(\tan(a_0/2)e^{(-kt)})$ is an exact solution converging to synchronization. Its center output is identically zero, just as for the equilibrium, so arbitrarily close nonequilibrium states are not identifiable from that output. At linear order, leaf vectors with zero center and zero leaf sum form an (n−2)-dimensional invisible eigenspace with eigenvalue −k. Adding a measurement of one displaced leaf or one of its incident edges detects this particular pair motion, while measuring every edge difference detects all mean-zero states. Thus local convergence and local observability are independent properties even within an exactly solvable stable family.

Worked example

Use vertices 0,1,2,3 with edges oriented 0→1, 1→2, 1→3. Let Ω=2, b=(−1/2,1/6,1/3,0), ω=(3/2,13/6,7/3,2), couplings K=(5/6,13/15,1/2), and cut flows f=(1/2,1/3,0). These are rational input data.

The incidence equation is Bf=$(-f_0, f_0-f_1-f_2, f_1, f_2)$=b. The edge sines are s=(3/5,5/13,0), and choose positive cosines c=(4/5,12/13,1). The exact unit phasors are $z_0=1$, $z_1=(4+3i)/5$, $z_2=(33+56i)/65$ and $z_3=(4+3i)/5$. Their head-tail ratios have precisely those rational real and imaginary parts. Equivalently, choose phases (0,α,α+β,α), where α=arctan(3/4) and β=arctan(5/12), before subtracting the mean. All edge differences are principal and even satisfy $|\sin\delta_e|<4/5$, a strict certificate for the cohesive bound γ=arcsin(4/5).

The Hessian weights are w=Kc=(2/3,4/5,1/2). Every principal cofactor of the stable weighted tree Laplacian equals their product 4/15, and its kernel is exactly the common-phase line. The unweighted four-node star has $\lambda_2=1$, so the transverse rate obeys $1/2\le\lambda_2(L_w)\le2/3$, the upper bound using the leaf-3 cut. These bounds and the rational Laplacian entries can be checked without evaluating any inverse trigonometric function. Flipping any subset of the three edge cosine signs, while leaving the sine values fixed, constructs all eight phase-locked branches with rational phasors. Their unstable dimensions have counts 1,3,3,1 for indices 0,1,2,3; the stable branch is the sole index-zero case.

Now retain K and the first two flows but replace f_2 by $q\in(-1/2,1/2)$. This changes b to $(-1/2,1/6-q,1/3,q)$. The third weight is $w_2(q)=\sqrt{(1/4-q^2)}$, which is always the minimum weight. Consequently $w_2(q)\le\lambda_2(L_w(q))\le(4/3)w_2(q)$, while $d\delta_2/dq=1/w_2(q)$ and the mean-zero phase sensitivity norm is $\sqrt{3}/(2w_2(q))$. Thus λ_2 times the edge-phase sensitivity stays between 1 and 4/3, while the rate vanishes and sensitivity diverges at either saturation boundary. The rational parametrization $q=t/(1+t^2)$, $w_2=(1-t^2)/(2(1+t^2))$ for $0\le t<1$ supplies exact examples arbitrarily close to the boundary.

The nonlinear equilibrium, its positive Hessian and its cut-flow certificate are explicit. The parameter family proves a quantitative relation between critical slowing and ill-conditioned phase reconstruction. The separate exact star trajectory proves that a stable network can still be invisible to an inadequate observation panel.

Scope of the result

The theorem covers symmetric positive couplings on a finite tree and a specified principal cohesive lift. It does not assert global convergence, attraction from arbitrary phases, robustness to noise or delays, or sufficiency of edge bounds on general cycles. At a saturated edge $|f_e|=K_e$ the positive-weight and exponential-stability proof fails; existence at the closed π/2 boundary does not restore its missing rate. The rational verifier certifies finite examples and exact identities, not the general theorem in a proof assistant. The tree-locking criterion is established literature; no novelty claim is made for it.

Further development

The completed operational consequence is a two-stage certificate: first verify Bf=b and edge feasibility to certify a cohesive equilibrium; then verify the rank of the actual observation map on the mean-zero state space. For a full edge panel the residual blindness is exactly common phase, whereas a center-only symmetric-star panel fails even for exact nonlinear trajectories. If a graph contains cycles, an additional cycle-compatibility problem must be solved before reusing this certificate design.

References: R008, R009

S010 The complete one-magnon Gaudin family and its scalar equations

Classical rank-one structure developed into a proved finite-sector theorem with exact rational and exceptional certificates

The two-site calculation extends to an explicit theorem for every number of spin-1/2 sites in the one-magnon sector. Simple critical points of one polynomial give a complete joint eigenbasis. For distinct real sites these critical points necessarily interlace and are simple, so this sector has no missing regular Bethe states apart from the explicitly retained uniform descendant. A three-site rational example is solved on the entire eight-dimensional tensor space, and the complex discriminant-zero locus exhibits a genuine Jordan degeneration.

Definitions and assumptions

Let $n\geq 2$, take pairwise distinct complex sites $z_1,\dots,z_n$ and put $p(x)=\prod_i(x-z_i)$. The one-magnon space has basis e_j, meaning that only spin j is down.

Use $\Omega_{ij}=(1/2)h_i h_j+e_i f_j+f_i e_j$ and $H_i=\Sigma_{(j\neq i)}\Omega_{ij}/(z_i-z_j)$. Define $E_i^0=(1/2)\Sigma_{(j\neq i)}1/(z_i-z_j)$ and the uniform vector $u=\Sigma_j e_j$.

For w not equal to a site, define $a_j(w)=1/(w-z_j)$, $B(w)=\Sigma_j a_j(w)=p'(w)/p(w)$, and $v(w)=\Sigma_j a_j(w)e_j$. The scalar equation uses $u_w(x)=p'(x)/(2p(x))-1/(x-w)$ and $V_w=u_w^2-u_w'$. The symbol u_w is a scalar connection coefficient and is distinct from the uniform vector u.

Critical points give a complete spectral decomposition under the stated simple-root condition

If p' has $n-1$ distinct roots $w_1,\dots,w_{n-1}$, the vectors $u,v(w_1),\dots,v(w_{n-1})$ are a basis of the one-magnon space. Their H_i eigenvalues are E_i^0 and $E_i(w)=E_i^0+a_i(w)$, respectively. For distinct real sites, the hypothesis always holds: exactly one critical point lies in each consecutive-site interval. For every critical w, $V_w=(3/4)\Sigma_i(x-z_i)^{-2}+\Sigma_i E_i(w)/(x-z_i)$, with no pole at w. Its scalar equation has a rational-over-sqrt(p) fundamental pair of Wronskian one.

Derivation or proof

The action on a pair of sites is explicit: $\Omega_{ij}\,e_i=-e_i/2+e_j$, $\Omega_{ij}\,e_j=e_i-e_j/2$, while $\Omega_{ij}\,e_k=e_k/2$ if k differs from both. Therefore, for $k\neq i$, the kth component of $H_i\,v(w)$ is $E_i^0a_k+(a_i-a_k)/(z_i-z_k)=(E_i^0+a_i)a_k$. Its ith component equals $E_i^0a_i-a_i\Sigma_{(j\neq i)}a_j=(E_i^0+a_i)a_i-a_iB(w)$. At a critical point $B=0$, this is the claimed eigenvector. Directly $H_i\,u=E_i^0u$. For any fixed i, the n displayed eigenvalues are distinct because the map $w\mapsto 1/(w-z_i)$ is injective and never zero. They therefore give a basis, proving completeness within this finite sector without assuming it. In particular the H_i are simultaneously diagonal in that basis. For real ordered sites, Rolle's theorem supplies at least one root of p' in each of $n-1$ disjoint gaps. Since $\deg p'=n-1$, these are all its roots, each simple. The vectors are orthogonal: $u^Tv(w)=B(w)=0$ and $v(w)^Tv(v)=[B(w)-B(v)]/(v-w)=0$ for distinct critical roots. Their squared norm is $-p''(w)/p(w)$, which is positive for real sites and nonzero whenever the critical root is simple. Thus $P_w=v(w)v(w)^T/[-p''(w)/p(w)]$ and $P_0=uu^T/n$ give explicit spectral projectors.

For the scalar equation, a physical site z_i has $u_w=1/[2(x-z_i)]+E_i(w)+O(x-z_i)$, so V_w has double coefficient $3/4$ and simple residue $E_i(w)$. At $x=w$, write $u_w=-1/(x-w)+B(w)/2+O(x-w)$; the

double pole cancels and the simple residue is $-B(w)$, hence zero at a Bethe root. Since V_w decays at infinity, its partial fractions are exactly the asserted expression. One solution is $y_1=(x-w)/\mathrm{sqrt}(p(x))$. Reduction of order gives $y_2=y_1\int p(x)/(x-w)^2\,dx$. Taylor expansion at w has no linear term because $p'(w)=0$, so an explicit numerator is $q_2(x)=-p(w)+\Sigma_{m=2}^{n} p^{(m)}(w)(x-w)^m/[m!(m-1)]$. Then $y_2=q_2/\mathrm{sqrt}(p)$, and $y_1y_2'-y_1'y_2=1$. Both solutions are regular at the apparent point w. Their only multivalued factor is a common square root, so the resulting monodromy is scalar and projectively trivial. This is a concrete scalar-oper statement with all normalizations fixed.

Worked example

Take three sites $z=(0,1,8/3)$, and separately study $z=(0,1,t)$ as a complex parameter family.

For $z=(0,1,8/3)$, p' has roots 2 and$4/9$. Their highest one-magnon coefficient vectors are $(1/2,1,-3/2)$ and $(9/4,-9/5,-9/20)$, giving Hamiltonian triples $(-3/16,6/5,-81/80)$ and $(25/16,-8/5,3/80)$. Each generates a two-dimensional spin doublet under total lowering. The vacuum generates a quartet with triple $(-11/16,1/5,39/80)$. The verifier constructs all eight rational eigenvectors and all three full 8×8 Hamiltonians, verifies joint eigenvector relations and rank eight, and checks both scalar differential equations through identically zero residual polynomials and Wronskians. As another independent exact example, four sites $(-7,-1,1,7)$ have critical roots $(-5,0,5)$, so their complete four-dimensional one-magnon diagonalization and three scalar solution pairs remain rational.

For $z=(0,1,t)$, $p'=3x^2-2(1+t)x+t$ and its root discriminant is $4(t^2-t+1)$. The exceptional sites are $t=(1\pm i\sqrt{3})/2$. They are distinct from 0 and1, but the two Bethe roots coalesce at $w=(1+t)/3$. Here $B(w)=B'(w)=0$. Differentiating the exact off-shell relation gives $(H_i-E_i(w))v'(w)=-a_i(w)^2v(w)$. The right side is nonzero, and v' is independent of v because distinct sites prevent all a_j from being equal. Together with u these vectors span the three-dimensional sector, so each H_i has a size-two Jordan block there. The checker works exactly in Q(t) with $t^2-t+1=0$ rather than relying on near-coincident numerical eigenvalues.

The finite real-site completeness theorem is closed under its stated assumptions. The three-site complex exceptional locus is also explicit, and its loss of diagonalizability is certified by generalized eigenvectors. The scalar-oper formula survives there, but it cannot by itself distinguish the generalized direction in a nonreduced spectral problem.

Scope of the result

The general theorem covers one-magnon states and the uniform descendant, not every magnetization sector for arbitrary n. The full eight-dimensional completeness statement applies only to the explicitly checked three-site example. General higher-rank Gaudin opers, arbitrary representations and a global Langlands correspondence require additional theory. Projectively trivial scalar monodromy here follows from the explicit fundamental pair, not from a broad assertion that all ODEs associated with a spin system have trivial monodromy.

Further development

S014 completes the distinct-complex-site one-magnon Jordan classification, including arbitrary critical multiplicities and a closed characteristic formula. For multi-magnon sectors the Bethe equations include root-root interactions. A disciplined next step is to compute a finite highest-weight multiplicity space independently, then compare regular, singular and generalized Bethe vectors with its exact joint spectrum. The exceptional-point calculation shows why a scheme-level spectrum may contain information that a set of distinct eigenvalues loses.

Related entries: M104, M105, M016, M038, S001, S014

References: R010

S011 Minimum induced subgraph order and the sharp heat observation margin

Exact finite classification and analytic margin proved

The rook-Shrikhande collision can be pushed beyond merely producing two graphs with the same spectrum. For this pair, every permutation-invariant induced-subgraph count through order three agrees, all eleven four-vertex pattern counts are classified, and a single labeled off-diagonal heat observation has a sharp optimal time and a sharp worst-case error threshold. These are different observation classes and must not be conflated.

Definitions and assumptions

For a finite simple graph G and an unlabeled k-vertex graph H, N_H(G) is the number of k-element subsets of V(G) whose induced graph is isomorphic to H. Dividing by C(|V(G)|,k) gives the distribution obtained by uniform vertex-subset sampling.

The feature panel through order d is the vector of all N_H(G) for all unlabeled H with at most d vertices. This vector is invariant under relabeling G.

The two candidate graphs R and S are the sixteen-vertex rook and Shrikhande graphs from M020. Their vertices are assigned the known coordinate labels in Z_4^2.

For the separate labeled observation problem, observe y=(exp(−tL))_{uv}+η at t>0, with known u=(0,0), v=(0,2) and adversarial error |η|≤ε.

A feature-class lower bound and an optimal labeled experiment

The feature panels through order three coincide for R and S; the order-four panels differ, so four is the minimum distinguishing order within this feature class. Their K_4 counts are eight and zero. For the known labels u,v, the heat-entry difference is d(t)=(e^(−4t)−e^(−8t))/4. It has unique maximum 1/16 at t*=log(2)/4. At that time the candidate observations are 5/64 and 1/64. Uniform identification under the stated bounded-error model is possible exactly when ε<1/32.

Derivation or proof

For any strongly regular graph with parameters (v,k,λ,μ), the number of edges is vk/2 and the triangle count is vkλ/6. Each vertex contributes C(k,2) neighbor pairs; subtracting its kλ/2 neighbor-neighbor edges counts induced three-vertex paths centered there, so their total is v[C(k,2)−kλ/2]. For an edge, the vertices adjacent to neither endpoint number v−2k+λ, giving (vk/2)(v−2k+λ) induced single-edge triples. Subtract these three classes from C(v,3) to obtain the independent triples. With (16,6,2,2), these counts are respectively triangles32, paths144, single-edge triples288 and independent triples96. The vertex, edge, nonedge and empty-set counts also agree. Therefore every function of the specified feature panel of order at most three agrees; no decision rule using only that panel can separate the pair. A four-clique occurs on each row and each column of the rook graph, giving eight. The exhaustive canonical certificate finds none in Shrikhande, proving that order four suffices. It classifies all 1,820 four-subsets in each graph rather than checking only one witness.

For the heat formula, both adjacency matrices satisfy $A^2=4I+2J$. On the constant vector the heat operator acts by one; on the adjacency eigenvalues +2 and −2 it acts by a=e^(−4t) and b=e^(−8t).

Solving for its coefficients in span{I,A,J} gives K_t=((a+b)/2)I+((a−b)/4)A+((1−2a+b)/16)J. The aligned pair is an edge of R and a nonedge of S, so their difference is (a−b)/4. Put x=e^(−4t)∈(0,1): d=x(1−x)/4=1/16−(x−1/2)²/4. This proves the unique maximum and its time without numerical optimization. At x=1/2, the common nonedge value is 1/64 and adding the edge coefficient gives 5/64.

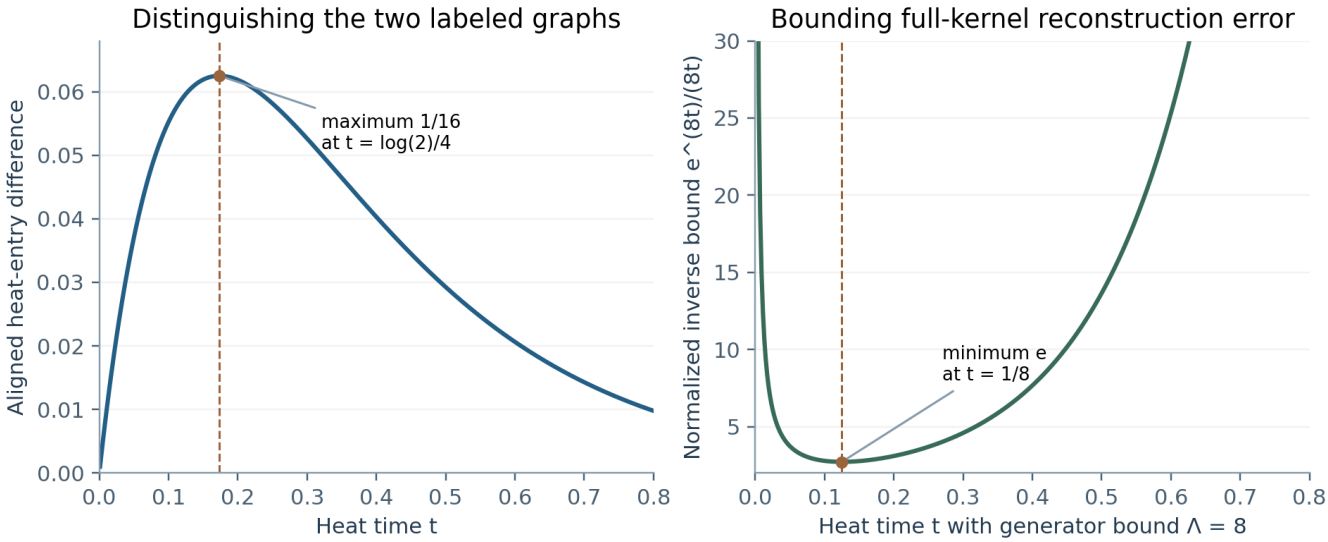
Under either hypothesis the possible noisy values form a closed interval of radius ε around its candidate value. These intervals are disjoint exactly when 2ε<1/16. The midpoint threshold 3/64 then separates them for every permitted error. At ε=1/32 both hypotheses can produce the midpoint: subtract 1/32 from 5/64 or add 1/32 to 1/64. No deterministic or randomized classifier can guarantee correct identification from that indistinguishable observation. Larger error budgets also contain this ambiguity.

Worked example

Canonicalize a labeled four-vertex graph by enumerating all twenty-four vertex permutations and taking the smallest six-edge bit mask. First enumerate all sixty-four labeled four-vertex graphs to obtain the complete set of eleven canonical classes; then classify every four-subset of R and S.

The counts in canonical-mask order are empty24/16, one-edge-plus-two-isolates288/336, path-plus-isolate576/480, star0/32, triangle-plus-isolate96/128, two-disjoint-edges216/192, four-path288/384, triangle-with-pendant288/192, four-cycle36/12, diamond0/48 and four-clique8/0, where each pair is rook/Shrikhande. Each column sums to 1,820. These exact distributions are stored in supplement_checks.json. At the optimal heat time the shared diagonal is 25/64, the edge value5/64 and nonedge value1/64; 25/64+6(5/64)+9(1/64)=1 verifies the row-sum normalization.

The pair is completely indistinguishable by induced pattern counts of orders zero through three, yet every four-vertex class count differs. A labeled heat entry is a different feature and can separate the same aligned candidates with one scalar observation.



The optimal time for separating one aligned graph pair differs from the time minimizing a full-kernel inverse-error bound. The left margin has exact maximum 1/16; the right normalized bound has exact minimum e. These are distinct objectives.

Scope of the result

The minimum-order statement is restricted to the declared induced-subgraph count panels; it is not a universal lower bound on all graph invariants, sensors, computational algorithms or sample counts. The heat threshold assumes exact time control, fixed known vertex correspondence, the two stated candidate graphs and the bounded-error model. It proves neither recovery of an unknown unlabeled graph nor uniform stability of taking a matrix logarithm. An unknown vertex permutation can change which pair is an edge and must be included as a nuisance parameter in a different problem. The continuous optimum t*=log(2)/4 is exact; numerical time selection incurs a separate margin loss.

Further development

The exact loss from a known mistiming t=t*+h is 1/16−d(t)=(e^(−4h)−1)²/16. If the actual time is observed but the design must tolerate |h|≤δ<t*, the worst same-time gap is [1−(e^(4δ)−1)²]/16, and a threshold adjusted to the observed time works for errors less than half that gap. If the actual time is unobserved, that bound alone is insufficient: each hypothesis ranges over different possible times. Put l=e^(−4δ)/2 and u=e^(4δ)/2. Over that interval, the least rook entry is (1+2u−3u²)/16, while the greatest Shrikhande entry is (1−l)²/16. The uniformly separating interval gap is therefore Δ=[2(u+l)−3u²−l²]/16, when positive, and the admissible bounded error is ε<Δ/2. These extrema follow because the Shrikhande entry decreases in x and the rook entry is concave; the difference of its endpoint values has sign (u−l)[2−3(u+l)]<0. For δ=log(3/2)/4, the exact candidate-set endpoints are 13/256 and 1/36, giving Δ=53/2304 and ε<53/4608. Thus calibrated deviation and an unobserved timing nuisance are genuinely different robustness problems. No version of these bounds establishes uniform unlabeled graph recovery.

References: R011

S012 Singular quadratic duality requires an observation quotient

Classical convex duality proved with a complete finite-dimensional classification and exact certificates

A pseudoinverse cannot simply replace an inverse inside every constrained-energy formula. Zero-energy input directions can generate nonzero observed outputs, and these free outputs must be quotiented out before applying the usual minimum-energy expression. The same geometry explains the finite domain of the Fenchel conjugate, the divergent part of ridge regularization, and changes in rank or active constraints. This supplies an algebraic complement to the positive-definite projection calculation in S006; it is a classical result developed here, not a claim of a new duality theorem.

Definitions and assumptions

Work in finite-dimensional real Euclidean spaces. Let $Q=Q^{T}\geq 0$ be $n\times n$, $f(x)=x^{T}Qx/2$, $N=\ker Q$ and $R=N^{\perp}=\operatorname{range} Q$. The Moore-Penrose inverse Q^{+} inverts Q on R and is zero on N.

For $B:R^{n}\to R^{m}$ and output $z\in R^{m}$, define the output space $W=BN$ generated at zero energy. Let P be the Euclidean orthogonal projection onto $W^{\perp}$ and put $H=PBQ^{+}B^{T}P$. This H is symmetric positive semidefinite.

The equality-constrained value is $v(z)=\inf\{f(x):Bx=z\}$, with value $+\infty$ if infeasible. Inequalities are not part of this general formula; the separately stated parameter example includes $x\geq 0$ and has its own KKT certificates.

The Fenchel conjugate is $f^*(y)=\sup_x(y^Tx-f(x))$. For $\varepsilon>0$, the ridge energy is $f_\varepsilon(x)=x^T(Q+\varepsilon I)x/2$. Its conjugate is finite everywhere.

Conjugate domain, ridge limit and complete equality-constrained value

The conjugate is $f^*(y)=y^TQ^+y/2$ for $y\in R$, and $+\infty$ otherwise. Writing $y=y_R+y_N$, the ridge conjugate is $f_\varepsilon^*(y)=y_R^T(Q|_R+\varepsilon I_R)^{-1}y_R/2+||y_N||^2/(2\varepsilon)$. Thus it tends to $f^*(y)$, with an explicit $1/\varepsilon$ divergence off R. The equation $Bx=z$ is feasible exactly when $Pz\in rangeH$, equivalently $z\in rangeB$. When feasible, $v(z)=(Pz)^TH^+(Pz)/2$. Its unique minimum-energy R-component is $r^*=Q^+B^TP\ H^+Pz$. Choose $n^*\in N$ with $Bn^*=z-Br^*$. All minimizers are $r^*+n^*+(N\cap kerB)$. A dual optimizer is $\lambda^*=H^+Pz$; the dual admissibility condition is $B^T\lambda\in R$, equivalently $\lambda\in W^\perp$. If feasibility fails, the dual is unbounded above along an explicitly separating direction.

Derivation or proof

First split $x=r+n$ orthogonally with $r\in R$ and $n\in N$. Since $Qn=0$, $y^Tx-f(x)=y_R^Tr-r^TQr/2+y_N^Tn$. If $y_N\neq0$, choosing $n=t\,y_N$ and sending $t\to+\infty$ proves that the supremum is infinite. If $y_N=0$, Q is positive definite on R, and completing the square there gives the unique maximizing $r=Q^+y$ and value $y^TQ^+y/2$; any null component of x is allowed. Diagonalizing Q separately on R and N yields the ridge formula. If Q has smallest positive eigenvalue a, then on R the finite ridge error obeys $0\le f^*(y_R)-f_\varepsilon^*(y_R)\le\varepsilon||y_R||^2/(2a^2)$. No such finite uniform limit exists for a fixed nonzero y_N because its contribution is exactly $||y_N||^2/(2\varepsilon)$.

Next write $Bx=Br+Bn$. A null component can supply precisely W, so the constraint is equivalent to $PBr=Pz$. On R set $u=(Q|_R)^{(1/2)}r$ and $C=PB(Q|_R)^{(-1/2)}$, interpreted as a map from R to $W^\perp$. The energy is $||u||^2/2$ and the quotient constraint is $Cu=Pz$. Since $H=CC^T$, $rangeH=rangeC$: their orthogonal complements both equal $\ker C^T$, because $a^THa=||C^Ta||^2$. Thus feasibility is exactly the stated range test. For a feasible right side, the least-norm solution is $u^*=C^TH^+Pz$. Indeed $Cu^*=HH^+Pz=Pz$, and u^* lies in range C^T, orthogonal to every feasible increment in $\ker C$. Pythagoras proves uniqueness and the value $||u^*||^2/2=(Pz)^TH^+Pz/2$. Returning to r gives the displayed formula. The residual $z-Br^*$ lies in $\ker P=W$, so a suitable n^* exists. Strict convexity on R fixes r^*, while all permissible null components differ by $N\cap kerB$.

For the dual, use $L(x,\lambda)=f(x)+\lambda^T(z-Bx)$. Taking the infimum over x gives $d(\lambda)=\lambda^Tz-f^*(B^T\lambda)$. The finite-domain condition says λ annihilates BN: $n^TB^T\lambda=0$ for every $n\in N$. Therefore $\lambda=P\lambda$ and $d(\lambda)=\lambda^TPz-\lambda^TH\lambda/2$. At $\lambda^*=H^+Pz$, stationarity and feasibility give $Qx^*=B^T\lambda^*$, $Bx^*=z$, and primal and dual values agree. This proves strong duality directly without relying on an unstated constraint qualification. If Pz is outside $rangeH$, let a be its nonzero projection onto $\ker H\cap W^\perp$. Then B^Ta lies in R and $a^THa=0$ forces $B^Ta=0$. The dual ray $\lambda=ta$ has $d(ta)=t||a||^2\to+\infty$, while $a^Tz\neq0$ proves $Bx=z$ impossible. These are both an infeasibility certificate and a dual-geometry explanation.

The simpler expression $z^T(BQ^+B^T)^+z/2$ is valid without the quotient when $BN=\{0\}$, together with the feasibility condition $z\in range(BQ^+B^T)$. Dropping either condition can give a wrong finite value. The case $Q=0$ is covered: every feasible output is free, $H=0$ and the value is zero, while infeasible outputs have value $+\infty$.

Worked example

First use $Q=diag(2,0,3)$, $B=[[1,1,0],[0,1,1]]$ and $z=(1,0)$. Then examine the parameter family $Q_t=diag(1,t,0)$, $b_a=(1,1,a)$, equality $b_a\,x=1$, and, as a distinct program, the extra constraint $x\ge0$, for $t\ge0$ and $a\in R$.

In the two-output example $N=\operatorname{span}(e_2)$, $W=\operatorname{span}((1,1))$ and $P=[[1,-1],[-1,1]]/2$. Direct exact multiplication gives $H=(5/24)[[1,-1],[-1,1]]$ and $H^{+}=(6/5)[[1,-1],[-1,1]]$. Therefore $\lambda^*=(6/5,-6/5)$, $r^*=(3/5,0,-2/5)$, $n^*=(0,2/5,0)$, and $x^*=(3/5,2/5,-2/5)$. Its energy and dual value are both 3/5. The raw matrix $BQ^{+}B^{T}=\operatorname{diag}(1/2,1/3)$ would instead give the incorrect naive value one. An independent elimination sets $x=(1-s,s,-s)$, giving energy $1-2s+(5/2)s^2$, whose unique minimizer is $s=2/5$. This verifies both the quotient formula and the counterexample without using pseudoinverse numerical software.

For the parameter family with equality only, Q_t has rank two for $t>0$ and rank one at $t=0$. If $t>0$ and $a\neq 0$, $x=(0,0,1/a)$ is feasible and has zero energy. If $t>0$ and $a=0$, the minimum is $t/[2(1+t)]$, attained at $x_1=t/(1+t)$, $x_2=1/(1+t)$, with arbitrary x_3. At $t=0$ the free second coordinate makes the minimum zero for every a. The raw pseudoinverse expression gives $t/[2(1+t)]$ for every a when $t>0$ and gives $1/2$ at $t=0$. Thus it is wrong on entire parameter strata, not merely at an isolated numerical accident.

Now impose $x\geq 0$. For $t>0,a>0$, the unique minimizer is $(0,0,1/a)$, with energy zero. For $t>0,a=0$, all minimizers are $(t/(1+t),1/(1+t),s)$, $s\geq 0$, with value $t/[2(1+t)]$. For $t>0,a<0$, the unique minimizer is $(t/(1+t),1/(1+t),0)$, with the same positive value. To certify the latter two branches choose $\lambda=t/(1+t)$, $\nu=(0,0,-a\lambda)$ and stationarity $Q_t\,x-\lambda b_a-\nu=0$. For $a\leq 0$, $\nu\geq 0$, and complementary slackness holds; at $a=0$ its third multiplier vanishes. For $a>0$ choose $\lambda=0,\nu=0$ at the zero-energy minimizer. At $t=0$ every feasible vector with $x_1=0$ is optimal: if $a>0$ this is the segment $x_3\in[0,1/a]$, $x_2=1-ax_3$; if $a\leq 0$ it is the ray $x_3\geq 0$, $x_2=1-ax_3$. These zero-energy branches also have $\lambda=\nu=0$ certificates. For example $t=2,a=-1$ has constrained optimum $(2/3,1/3,0)$, value $1/3$ and $\nu_3=2/3$, whereas its equality-only optimum $(0,0,-1)$ has value zero.

For every listed nonnegative candidate, convexity makes these KKT certificates globally sufficient: for any other feasible $x=x^*+h$, the objective difference equals $h^{T}Q_t h/2+\nu^{T}x\geq 0$. This is a proof for all real parameters in the declared strata; the verifier replays representative rational points and checks feasibility, stationarity, nonnegative multipliers, complementarity and primal-dual agreement exactly. As $a\to 0$ from above at fixed $t>0$, the zero-energy minimizer escapes to infinity while the value jumps upward at $a=0$. The discontinuity comes from loss of uniform control of null directions, not from a failure of the theorem.

The dual must discard observed outputs that can be supplied without energy, by restricting multipliers to their annihilator. Inequalities can prevent use of those directions, which is why a linear nullspace quotient alone is not a formula for an inequality-constrained problem. Rank, output compatibility and active constraints form separate strata. This makes the relation to M062 and M093 precise and complements S006 without conflating zero energy, zero variance and missing statistical information.

Scope of the result

Everything here is finite-dimensional with a symmetric positive-semidefinite quadratic form and exact equality constraints. In an infinite-dimensional Hilbert space the range need not be closed and minimizers or bounded pseudoinverses may fail to exist. An indefinite Q can make the primal unbounded below. The nonnegative parameter family is solved separately by KKT; the quotient formula does not automatically incorporate general inequalities. A ridge limit is a declared regularization, not a justification for deleting divergent terms or for replacing inverses by pseudoinverses without range tests.

Further development

For rational parameter-dependent Q and B, produce exact rank strata, bases for kerQ and BkerQ, compatibility equations for Pz and KKT active-set certificates. Preserve both the domain conditions and the optimizer set at boundary parameters. In a statistical application specify whether Q describes energy, precision or covariance before transferring these algebraic formulas into a measurement claim.

Related entries: M062, M093, S006, M017, M068

References: R012

S013 An exact exceptional point and its two-sheeted spectral cover

Finite parameter-family classification and exact Jordan/projector certificates; no novelty claim

The three-site sl_2 Gaudin family has an explicit exceptional locus where two highest-weight eigenlines coalesce. At that locus a genuine Jordan block replaces the two eigenvectors. Nearby spectral projectors diverge although their sum stays constant. Analytic continuation of the eigenvalue labels exchanges the two lines. This is spectral-cover monodromy in a parameter, not monodromy of a KZ connection or of the scalar differential equation in its spatial variable.

Definitions and assumptions

Use the one-magnon sector and normalization of S010, with sites $z=(0,1,t)$, $t\neq0,1$. Put $p_t(x)=x(x-1)(x-t)$, $D(t)=t^2-t+1$ and $w_\pm=(1+t\pm\sqrt{D})/3$.

For $D\neq0$, let $v_\pm$ have components $1/(w_\pm-z_j)$, and $E_{i\pm}=E_i^0+1/(w_\pm-z_i)$. The uniform vector is $u=(1,1,1)^T$ and the highest-weight projection is $Q=I-uu^T/3$.

Transpose is the complex bilinear transpose, not the Hermitian adjoint. For complex t the resulting spectral projectors need not be orthogonal for the standard Hermitian norm. Their divergence is compatible with Q remaining bounded.

Jordan degeneration, projector cancellation and spectral-label exchange

The only critical-root collisions in this family occur at $t_0=(1\pm i\sqrt{3})/2$, the simple zeros of D. At $t=t_0$, $w_0=(1+t_0)/3$ and $v_0=v(w_0)$ satisfy $(H_i-E_i)v_0=0$ and $(H_i-E_i)v'_0=-a_i^2v_0$, where $a_i=1/(w_0-z_i)$. Thus each H_i has a size-two Jordan block on im Q. For $D\neq0$, $P_\pm=\mp p_t(w_\pm)v_\pm v_\pm^T/(2\sqrt{D})$, and $P_+ + P_- = Q$ exactly. As t approaches t_0 on either local sheet, $\sqrt{D}P_+ \to -p_(t_0)(w_0)v_0v_0^T/2$ and $\sqrt{D}P_-$ has the opposite limit. A loop encircling just one zero of D exchanges P_+ and P_-.

Derivation or proof

Differentiating p_t gives $p'=3x^2-2(1+t)x+t$, so the critical roots have the displayed quadratic formula. Its discriminant vanishes precisely at the two stated sites, and $D'(t_0)=2t_0-1\neq0$. At the collision $B=p'/p$ and B' both vanish at w_0, while $p(w_0)\neq0$ because the physical sites are still distinct. The off-shell identity from S010 is $H_i\, v(w)=E_i(w)v(w)-a_i(w)B(w)e_i$. Differentiating in w at fixed t_0 eliminates both residual terms and gives the generalized-eigenvector relation, with $E_i'=-a_i^2\neq0$. If v' were proportional to v, all a_j would have to be equal, contrary to distinct sites. Both vectors have coordinate sum zero; with u they span the full sector. Therefore the block is genuinely Jordan, not a repeated diagonal eigenvalue.

At a simple critical root the bilinear square is $v(w)^{T}v(w)=-p''(w)/p(w)$. Since $p''(w_{\pm})=\pm 2\sqrt{D}$, the projector formula follows. The eigenvectors u, v_{+}, v_{-} form a bilinear-orthogonal basis when $D\neq 0$, so their projectors sum to I and $P_{+}+P_{-}=Q$. The numerator of P_{+} tends to a nonzero rank-one matrix, giving divergence proportional to $|D|^{-1/2}$ in any fixed finite-dimensional matrix norm. With $\varepsilon=\sqrt{D}$, one has $t-t_0=O(\varepsilon^2)$, $w_{\pm}=w_0\pm\varepsilon/3+O(\varepsilon^2)$, and $p_t(w_{\pm})=p_t(t_0)(w_0)+O(\varepsilon^2)$. Thus $P_{\pm}=\mp p_0 v_0 v_0^{T}/(2\varepsilon)-p_0(v'_0 v_0^{T}+v_0 v'^{T}_0)/6+O(\varepsilon)$. The divergent terms cancel. In fact $Q=-p_0(v'_0 v_0^{T}+v_0 v'^{T}_0)/3$ at the exceptional point. The finite surviving nilpotent is $N_i=(H_i-E_i)Q=(p_0 a_i^2/3)v_0 v_0^{T}$. It is nonzero and $N_i^2=0$ because $v_0^{T}v_0=0$.

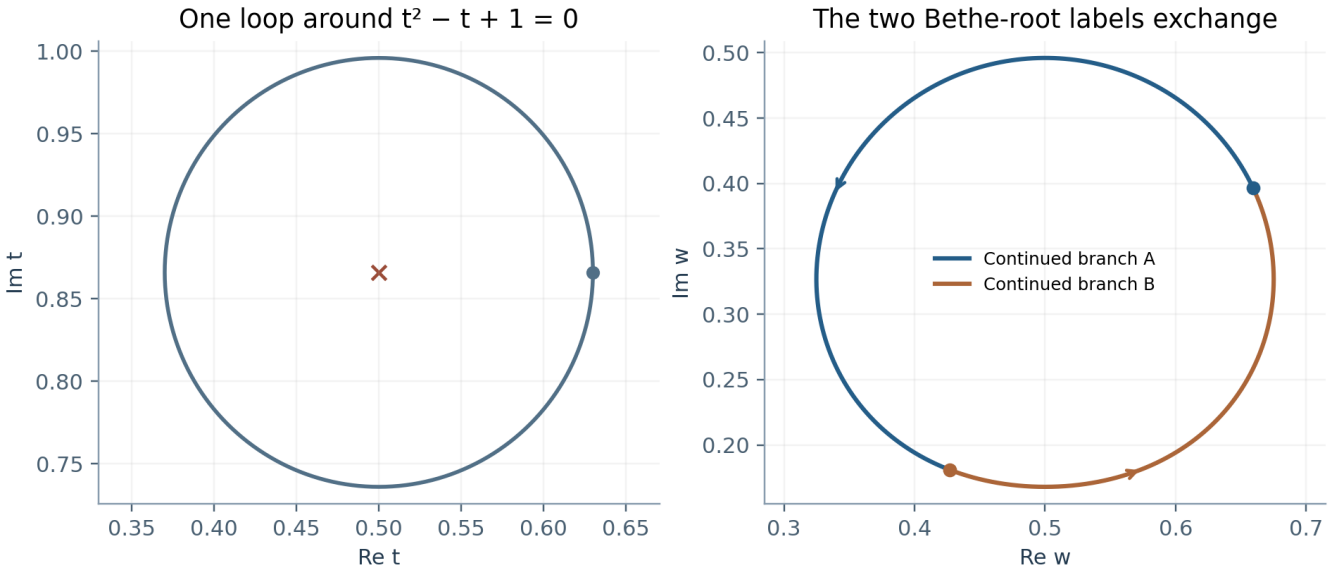
Finally $D=(t-t_0)(t-t_1)$. A sufficiently small loop around t_0 winds D once around zero because t_1 lies outside. Analytic continuation of $\sqrt{D}$ therefore changes sign, exchanging the two roots, eigenvalues and projectors. This proof concerns the two-sheeted algebraic spectral cover; it introduces no connection or parallel-transport operator.

Worked example

Perform the exceptional calculation exactly in $K=Q(\tau)$, $\tau^2-\tau+1=0$, taking $t_0=\tau$ and $w_0=(1+\tau)/3$.

The supplemental verifier constructs all three 3×3 Hamiltonians over K. It checks their commutators, the eigenvector and generalized-eigenvector identities, $\operatorname{rank}(u, v_0, v'_0)=3$, the constant-projector formula and $N_i=(p_0 a_i^2/3)v_0 v_0^{T}$ with $N_i\neq 0$ and $N_i^2=0$. These are exact algebraic calculations, independent of numerical eigenvalue separation. Additional complex-number evaluations approach the exceptional point along $D=\varepsilon^2$ and check the leading projector coefficient and cancellation. A separately tracked square root around a small parameter loop exchanges the two labels, agreeing with the analytic winding argument.

This closes the finite three-site exceptional-point question: location, Jordan structure, projector blowup, cancellation and label exchange are explicit. The exact field certificate carries the algebraic result; numerical paths illustrate its limiting and continuation behavior.



Numerical continuation of the exact quadratic Bethe-root formula around a simple discriminant point exchanges the two root labels. This illustrates spectral-cover continuation, not KZ connection holonomy. The source script checks endpoint exchange and the polynomial residual to 1e-12.

Scope of the result

For real distinct sites, $D=(t-1/2)^2+3/4$ is positive and cannot reach this exceptional locus; the real-site orthogonal projectors therefore do not develop this singularity. The result does not classify every higher-site exceptional locus. Spectral-cover label exchange, spatial scalar-ODE monodromy and KZ connection monodromy are three different constructions and must not be identified.

Further development

S014 completes the distinct-site one-magnon answer: each multiplicity-m root of p' gives exactly one size-m Jordan block, and the commuting spectral algebra retains its nilpotent direction. Further work should address multi-magnon sectors or collisions of the physical sites, where the present hypotheses and one-coordinate construction do not apply unchanged.

Related entries: S010, M104, M105, M042, S014

References: R010

S014 Complete complex one-magnon Jordan classification

Proved finite-sector classification with exact Gaussian-rational examples; classical framework, no global novelty claim

The simple-critical-point qualification in S010 can be removed. For arbitrary distinct complex spin-1/2 sites, multiplicities of the roots of p' determine all Jordan block sizes in the one-magnon sector. One explicit local coordinate turns the Bethe vector's derivatives into a Jordan chain of the correct length. The chains exhaust the sector, and every Hamiltonian generates the same finite commuting algebra after polynomial interpolation.

Definitions and assumptions

Retain $p(x)=\prod_{j=1}^{n}(x-z_j)$, H_i and E_i^0 from S010, with all physical sites distinct. A critical root w_r has multiplicity m_r as a root of p'. Since p has simple roots, $p(w_r)\neq 0$; $\sum_r m_r=n-1$.

Fix any Hamiltonian index i. At w_r write $a=1/(w_r-z_i)$, use local coordinate $q=1/(w-z_i)-a$, and define $w(q)=z_i+1/(a+q)$. Expand $v(w(q))=\sum_{k\geq 0}V_{r,k}q^k$ near $q=0$.

A length-m Jordan chain for $H-E$ is a list $V_0,\dots,V_{m-1}$ with $(H-E)V_0=0$ and $(H-E)V_k=V_{k-1}$ for $k\geq 1$. No factorial is missing: V_k is the Taylor coefficient, equivalently the kth derivative divided by $k!$.

Critical multiplicity equals Jordan length for every H_i

On the n-dimensional one-magnon space, each H_i has one uniform-descendant block of size one and eigenvalue E_i^0, together with exactly one block of size m_r and eigenvalue $E_i(r)=E_i^0+1/(w_r-z_i)$ for each distinct critical root w_r. Consequently H_i is diagonalizable exactly when p' is squarefree, and its minimal polynomial equals its characteristic polynomial of degree n. With $A_i=H_i-E_i^0 I$, one has $A_j=A_i[I+(z_i-z_j)A_i]^{-1}$. The characteristic polynomial is $\chi_{H_i}(\lambda)=u^n\,p'(z_i+1/u)/p'(z_i)$, $u=\lambda-E_i^0$, interpreted by polynomial expansion. The unital algebra generated by the H_j is $\mathbb{C}[H_i]$, isomorphic to $\mathbb{C}\times\prod_r \mathbb{C}[q]/(q^{m_r})$.

Derivation or proof

The exact off-shell identity is $H_i\,v(w)=(E_i^0+a_i(w))v(w)-a_i(w)B(w)e_i$, with $B=p'/p$. At a multiplicity-m critical root, $B(w)$ vanishes to order m. The local coordinate q is valid because $a_i'(w_r)=-a^2\neq0$, and its inverse is the displayed rational function. Moreover $E_i(w(q))=E_i(r)+q$ exactly. Thus $(H_i-E_i(r))v(w(q))=q\,v(w(q))+O(q^m)$. Equating Taylor coefficients through q^{m-1} gives the claimed chain. It is independent: if a linear relation has highest nonzero coefficient at V_k, applying $(H_i-E_i(r))^k$ leaves that coefficient times V_0, a contradiction because $V_0\neq0$. Distinct critical roots have distinct $E_i(r)$, since $w\mapsto1/(w-z_i)$ is injective; none equals E_i^0. Their generalized eigenspaces and the uniform vector are therefore independent. Their dimensions total $1+\Sigma m_r=n$, so they form a complete basis and no larger or additional block can exist.

For any other j, put $d=z_i-z_j$ and $A=1+da$. This A is nonzero because w_r is not a physical site. In the same coordinate, $a_j(w(q))=(a+q)/(A+dq)=a_j(w_r)+\Sigma_{\ell\geq1}c_{j\ell}\,q^\ell$, where $c_{j\ell}=(-d)^{\ell-1}/A^{\ell+1}$. The off-shell remainder still has order m. Hence H_j on the chain is $E_j(r)I+\Sigma_{\ell=1}^{m-1}c_{j\ell}\,N^\ell$, with $NV_k=V_{k-1}$. Its linear coefficient is $c_{j1}=A^{-2}\neq0$, so its nilpotent part is N times an invertible polynomial in N and has the same Jordan length and generalized-eigenspace flags. Equivalently these jets give $A_j=A_i(I+dA_i)^{-1}$ on each block and on the uniform vector. The denominator is invertible: its critical eigenvalues are $(w_r-z_j)/(w_r-z_i)$, all nonzero, and its uniform eigenvalue is one. An inverse of this finite matrix is polynomial in A_i, giving $\mathbb{C}[H_i]$ as the entire generated algebra. Finally $\chi=u\prod_r(u-1/(w_r-z_i))^{m_r}$; substituting the factorization $p'(x)=n\prod_r(x-w_r)^{m_r}$ gives the displayed polynomial formula. The degree-n minimal polynomial and the Chinese remainder theorem yield the algebra decomposition. This proves the entire finite-sector classification, including its nonsemisimple cases.

Worked example

Take four physical sites $(1,-1,i,-i)$, so $p=x^4-1$ and $p'=4x^3$. Fix i to be the site $z_i=1$.

There is one critical root $w=0$ of multiplicity three. Its chain coefficients are $V_0=(-1,1,i,-i)$, $V_1=(1,1,-1,-1)$, $V_2=(0,2,-1-i,-1+i)$. The uniform vector and these three vectors have rank four. The eigenvalues are $E_j^0=3/(4z_j)$ and $E_j=-1/(4z_j)$. In particular H_i has characteristic and minimal polynomial $(\lambda-3/4)(\lambda+1/4)^3$. On $Q=I-uu^{\mathsf T}/4$, $N=(H_i+I/4)Q$ obeys $N^3=0$ and $N^2\neq0$. All equations are checked exactly in $\mathbb{Q}(i)$.

A mixed-multiplicity example uses eight distinct sites $\pm1,\pm i,\pm2,\pm2i$, giving $p=(x^4-1)(x^4-16)=x^8-17x^4+16$ and $p'=4x^3(2x^4-17)$. The critical multiplicities are three at zero and one at each of four nonzero roots. Thus each H_i has block sizes $3,1,1,1,1,1$, including the uniform block. At base site1, $E_i^0=37/60$ and $E_i(0)=-23/60$. If $q=\lambda-37/60$, its characteristic polynomial is $(\lambda-37/60)(\lambda+23/60)^3[17q^4-2(q+1)^4]/15$. The verifier independently calculates the full 8×8 characteristic polynomial by trace/Newton identities over $\mathbb{Q}(i)$, checks this factorization and the length-three chain. The four remaining factors have distinct roots because the map $w\mapsto1/(w-1)$ is injective on the four simple roots of $2w^4-17$.

The multiplicity-three and mixed examples verify higher Jordan lengths and coexistence of simple and nonsimple blocks. The arbitrary-n result is supplied by the proof, not extrapolated from those matrices.

Scope of the result

This classification concerns the one-magnon sector for distinct physical sites and the stated spin-1/2 quadratic Gaudin Hamiltonians. It does not settle multi-magnon completeness, higher representations, collisions of physical sites, or a global Langlands correspondence. The direct-product algebra is a finite spectral algebra, not a claim that a general moduli space has been constructed.

Further development

In multi-magnon sectors, root-root interactions and admissibility conditions prevent this one-coordinate argument from being transferred unchanged. An appropriate next problem is to seek an equally explicit off-shell jet construction on a fixed small multiplicity space and compare it against an independently computed joint Jordan form.

Related entries: S010, S013, M105, S001

References: R010

A Arithmetic nonassociative algebra and exact wave dynamics

M001 Separate the meanings of Q

2023-01-23 · historical evidence A · Definitions resolved with a structural obstruction

The original notation can be repaired completely at the level of objects. Robinson arithmetic Q is a first-order theory, Q_p is a valued field of characteristic zero, F_p[[q]] is a formal power-series ring of characteristic p, and D_q is an operator depending on a numerical or formal parameter. A proposed connection must name the relevant category and map; the shared letter Q supplies neither.

Definitions and assumptions

Robinson Q consists of axioms in the language of zero, successor, addition and multiplication; it is not a number field.

Q_p is the completion of Q for |x|_p=p^(−v_p(x)); Z_p is its valuation ring.

F_p[[q]] consists of formal series Σa_n q^n. Its q-adic topology measures vanishing order, not numerical size.

D_q f(x)=(f(qx)−f(x))/((q−1)x), initially for q≠1 and x≠0.

Characteristic obstructs a claimed field identification

> There is a canonical dense embedding Q→Q_p and a compatible reduction system Z_p→Z/p^nZ. There is no unital ring homomorphism Q_p→F_p[[q]], or in the reverse direction. In particular these objects cannot be identified by renaming their parameters.

Derivation or proof

A unital homomorphism sends an integer n·1 to the same n·1. In characteristic p, p·1=0, whereas p is nonzero and invertible in Q_p. If φ:Q_p→F_p[[q]] were unital, then 1=φ(p p^(−1))=φ(p)φ(p^(−1))=0, a contradiction. In the reverse direction, applying φ to p·1=0 would give p·1=0 in Q_p, also impossible. The dense embedding Q→Q_p instead preserves characteristic and the field operations. The reductions from Z_p preserve ring operations but have nontrivial kernel p^nZ_p, so they are quotients rather than embeddings. This distinction already eliminates several possible informal identifications without requiring a difficult theorem.

Worked example

Compare a rational number, a p-adic approximation and a formal series.

The rational 1/3 maps into Q_7 and can be approximated by compatible residues modulo 7^n. The formal series (1−q)^(−1)=1+q+q²+⋯ belongs to F_7[[q]] because its coefficients satisfy a formal multiplication identity. Its formal variable q is not automatically the prime 7 or a neural-network parameter.

A corrected dictionary gives legitimate arrows and a proof that one tempting arrow cannot exist.

Scope of the result

An arithmetic theory can be interpreted in structures, and equal-characteristic local fields can resemble mixed-characteristic fields in selected respects. The characteristic obstruction does not rule out every model-theoretic or categorical comparison; it rules out the specified unital ring identifications.

Further development

For every later use of q or Q, record its type, admissible values and morphisms. A meaningful bridge should transport a named theorem, not merely an expression.

References: R013, R014

M002 p-adic completion with certified digits

2023-01-23 · historical evidence A · Simple-root lifting proved and calculated

A useful p-adic computation is a compatible sequence of exact congruences with a precision guarantee. The essential advantage in this example is that a simple solution modulo p determines one branch at every higher precision; decimal proximity has no role in the argument.

Definitions and assumptions

v_p(x) is the exponent of p in a nonzero rational x, and |0|_p=0.

An approximation x_n modulo p^n certifies a p-adic ball x_n+p^nZ_p.

A simple root modulo p satisfies f(x_1)≡0 and f′(x_1) not≡0 modulo p.

Unique next digit in a Hensel lift

> For f∈Z_p[x], a simple root x_n modulo p^n has exactly one lift x_n+t p^n modulo p^(n+1), with t∈{0,...,p−1}. The compatible lifts converge to a unique root in x_1+pZ_p.

Derivation or proof

Taylor expansion gives f(x_n+t p^n)≡f(x_n)+t p^n f′(x_n) modulo p^(n+1); all terms of degree at least two are divisible by p^(2n), hence by p^(n+1). Dividing by p^n reduces the lift equation to f(x_n)/p^n+t f′(x_n)≡0 modulo p. The derivative is a unit, so this linear equation has one digit t. Repeating constructs a compatible inverse-limit element. Completeness gives a limit α; polynomial continuity gives f(α)=0. If two roots share the residue x_1, the same unique-digit argument forces agreement modulo every p^n and therefore equality. A finite calculation consequently certifies a ball containing the unique branch, not an approximate real root.

Worked example

Lift x^2-2 from the root 3 modulo 7.

At precision 7^2, write x=3+7t. Dividing f(x) by 7 gives 1+6t≡0 modulo 7, hence t=1 and x=10 modulo 49. Next x=10+49t gives 2+6t≡0, hence t=2 and x=108 modulo 343. Directly, $108^2-2=34\cdot343$. The verifier continues the uniquely selected digits through precision 7^6.

The result is a root α∈Z_7 with α≡108 modulo 343, with a proof that no other root in 3+7Z_7 has those initial conditions.

Scope of the result

If f′ vanishes modulo p, uniqueness and existence require a different argument; the simple-root calculation cannot be recycled unchanged. This example establishes an arithmetic algorithm, not a general computational speedup over real methods.

Further development

Develop Newton lifting with precision doubling and an independent residual valuation; compare digit lifting and Newton lifting on the same branch.

References: R015

M003 Affine dynamics on Z_p

2023-01-23 · historical evidence A · Complete fixed-point and attraction classification for the stated family

The affine family can be settled exactly. The hypotheses a,b∈Z_p ensure that f(x)=ax+b maps Z_p into itself and never expands p-adic distance. Attracting, neutral and translational behavior can then be distinguished without metaphor.

Definitions and assumptions

A fixed point solves f(x*)=x*.

Attraction means f^n(x)→x* in the p-adic metric for nearby initial states.

Congruence modulo p^N means two states cannot be distinguished at N-digit precision.

Affine fixed points and orbit differences

If a≠1, a fixed point in Z_p exists exactly when b/(1−a)∈Z_p, equivalently b=0 or v_p(b)≥v_p(1−a), and is then unique. If a=1, b=0 makes every point fixed and b≠0 gives none. For |a|_p<1 the unique point attracts every state; for |a|_p=1 no distinct state converges to a fixed point.

Derivation or proof

The fixed-point equation is (1−a)x=b, which gives the classification because Q_p is a field and membership in Z_p is exactly nonnegative valuation. Subtract two trajectories: f^n(x)−f^n(y)=a^n(x−y). Taking norms gives |a|_p^n|x−y|_p. This converges to zero precisely in the nonunit case, while for a unit it remains constant whenever x≠y. When |a|_p<1, 1−a is a unit, so every b supplies an integral fixed point. For a≠1, direct induction gives f^n(x)=a^n x+b(1−a^n)/(1−a). Congruent initial states remain congruent at every step, since multiplication by a∈Z_p cannot lower their agreement precision.

Worked example

Take p=7, a=7 and b=1.

The fixed point is −1/6∈Z_7. Starting from x_0=0 gives x_n=1+7+⋯+7^(n−1), and x_n−x*=7^n/6. Its valuation is exactly n. Thus n iterations supply n correct base-seven digits. In contrast a=8,b=1 has no fixed point in Z_7 because (1−8)x=1 would require v_7(x)=−1.

The same affine notation yields global contraction in one case and a provable integral fixed-point obstruction in the other.

Scope of the result

Neutral maps may have rich periodic structure modulo p^N; constant separation does not mean every orbit is constant. This classification is for affine maps on Z_p, not arbitrary nonlinear p-adic dynamics.

Further development

Classify neutral affine cycles modulo p^N, then identify which cycle properties stabilize as N grows. Use gcd divisibility as an independent finite checker.

References: R014

M004 A valid probability model and identifiable parameters

2023-01-23 · historical evidence A · Range failure proved and a finite-design model specified

The original product can be analyzed before any modeling claim is made. A formula is a probability only on an admissible domain, and fitted coefficients are meaningful only when the observation design can distinguish them.

Definitions and assumptions

For the original expression $P(x,y)=(1+x)(1-y)$, assume $x,y\in[0,1]$.

For the alternative logistic model let $z=(1,x,y)$, $\eta=z\cdot\beta$ and $\sigma(\eta)=1/(1+e^{-\eta})$.

A finite design matrix X has rows z_i and identifies β when distinct coefficients give distinct modeled probability vectors.

Exact admissible domain and logistic identifiability

On $[0,1]^2$ the product lies in $[0,1]$ exactly when $y\geq x/(1+x)$. A logistic model yields probabilities strictly between zero and one for every finite β. At a fixed design, its coefficient vector is identifiable exactly when X has full column rank.

Derivation or proof

Nonnegativity of the product is automatic on the square. The upper bound is $(1+x)(1-y)\leq 1$, equivalent to $y\geq x/(1+x)$ because $1+x>0$. The point $(1,0)$ violates it with $P=2$. For the logistic family, σ is a strictly increasing bijection $\mathbb{R}\to(0,1)$. Thus equal probability vectors are equivalent, after applying logit, to $X\beta=X\beta'$. Full column rank makes this imply $\beta=\beta'$; a nonzero null vector otherwise supplies indistinguishable coefficients. For independent Bernoulli observations the log likelihood is $\Sigma[y_i\eta_i-\log(1+e^{\eta_i})]$. Its Hessian is $-X^{T}WX$ with W diagonal and positive at finite β. Full rank gives strict concavity, but does not guarantee that a finite maximizer exists: complete separation can send the optimum to infinity.

Worked example

Use the three design points $(0,0),(1,0),(0,1)$.

Their augmented rows form an invertible 3×3 matrix. $\beta=(0,\log3,-\log3)$ gives probabilities $(1/2,3/4,1/4)$. Taking logits yields $(0,\log3,-\log3)$, from which β is uniquely recovered by solving the

linear system. Repeating only the first point would identify β_0 but leave the other two coefficients free.

The example separates probability range, structural identifiability and existence of an estimator-three questions the original toy formula left unresolved.

Scope of the result

Choosing a logistic family does not establish empirical truth, independence or adequate calibration. Identifiability is conditional on the design and model; it is not a performance guarantee.

Further development

Add a specified observation model and investigate design conditioning or separation before optimizing the likelihood.

M005 Specify a hard problem before using arithmetic complexity

2023-01-23 · historical evidence A · A toy encryption proposal is explicitly broken

Sophisticated number systems do not themselves supply one-wayness. A mathematically assessable proposal must say what an adversary receives, what recovery problem is being claimed difficult, and how problem instances are distributed.

Definitions and assumptions

A toy modular scheme publishes a prime p and nonzero multiplier a and maps a message x to y=ax modulo p.

Correctness of a private recovery algorithm is different from hardness of public recovery.

An instance distribution specifies which public parameters, secrets and observations occur; worst-case difficulty alone need not control that distribution.

Public linear arithmetic is publicly invertible

> For the stated toy map, every message is recovered from public data using x=a^(−1)y modulo p. Extended Euclid computes a^(−1), so adding large notation or expressing the same arithmetic p-adically cannot turn this construction into a secret inversion problem.

Derivation or proof

Since p is prime and a is nonzero modulo p, gcd(a,p)=1. The extended Euclidean algorithm produces integers u,v with ua+vp=1. Reducing modulo p gives ua≡1, and multiplying y=ax by u recovers x≡uy. This is a deterministic polynomial-time attack in the bit length of p. More generally, a public invertible linear transformation over a field is inverted by linear algebra. If its matrix is hidden but an oracle permits arbitrary message queries, querying the coordinate basis recovers its columns. Neither attack needs to search the message space or exploit implementation defects. They attack the mathematical specification itself.

Worked example

Take p=101, a=7 and observed y=35.

Euclid gives $7^{-1}\equiv 29 \pmod{101}$, since $7\cdot 29=203\equiv 1$. Therefore $x\equiv 29\cdot 35\equiv 5$. The verifier recovers x using only these public quantities. A larger prime changes the input size but not the kind of attack.

The construction is correct as a reversible encoding and fails as a claimed secret arithmetic trapdoor.

Scope of the result

This example does not establish security or insecurity of an unrelated fully specified cryptosystem. A different map might pose a genuinely different problem, but its hardness must be argued independently of the prestige of its geometric or arithmetic vocabulary.

Further development

For any proposed replacement, give generation, evaluation, recovery, correctness and an explicit public attack model. Test algebraic simplifications on small instances before attaching a security claim.

M006 Zeno through a convergent series and a time law

2023-02-02 · historical evidence A · Convergence and endpoint behavior proved

Infinitely many subdivisions do not determine elapsed time. Once both the distance increments and the time increments are stated, the mathematical ambiguity disappears: an infinite event sequence can accumulate at a finite time, or it can continue for an infinite duration.

Definitions and assumptions

Let distance increments be $d_n=2^{-n}$, with milestones $x_N=\Sigma_{n=1}^{N} d_n$.

Let positive time increments be τ_n and milestone times $t_N=\Sigma_{n=1}^{N} \tau_n$.

A finite accumulation time exists exactly when the positive time series converges.

Finite distance does not fix finite elapsed time

> The distances sum to one. Choosing $\tau_n=2^{-n}$ reaches milestones at $t_N=1-2^{-N}$ and admits the continuous extension $x(1)=1$. Choosing $\tau_n=1/n$ instead takes infinite total time while approaching the same spatial limit.

Derivation or proof

For the geometric sum, multiply $S_N=1/2+\cdots+2^{-N}$ by two and subtract to obtain $S_N=1-2^{-N}$. Given $\varepsilon>0$, choose $N>\log_2(1/\varepsilon)$; then $0<1-S_N<\varepsilon$, proving convergence. With equal distance and time increments, the piecewise linear motion is simply $x(t)=t$ on $[0,1)$, so continuity supplies the endpoint $x(1)=1$. In the harmonic-time model, the subsequence of times diverges because each block from 2^k+1 through 2^{k+1} contributes at least $1/2$. A continuous piecewise linear interpolation still approaches spatial position one, but only as $t\to\infty$. The number of labeled milestones is identical in both descriptions; its cardinality does not decide the total time.

Worked example

Compare the first N milestones in the two parametrizations.

Both have spatial coordinate 1−2^(−N). In the first model the remaining time is 2^(−N). In the second, any finite milestone leaves an infinite tail Σ_(n>N)1/n. Their different conclusions follow from different time assignments, not from a contradiction in summing the same distances.

The revised question separates convergence, continuity and parametrization, each of which can be proved with elementary estimates.

Scope of the result

A mathematical parametrization does not assert that a physical device can execute an unbounded rate of discrete operations. The continuous trajectory requires no such discrete mechanism; a physical implementation would add its own constraints.

Further development

Study rectifiable curves with infinitely many segments and ask separately whether a chosen parametrization has bounded speed, finite length and a continuous endpoint.

M007 Octonions with a declared multiplication convention

2023-02-20 · historical evidence A · Nonassociativity exhibited and Moufang law certified by all coefficients

An analogy involving octonions becomes mathematics once multiplication, regrouping and the surviving identities are explicit. The real octonions are a division algebra but not an associative algebra; consequently their nonzero elements form a Moufang loop rather than a group.

Definitions and assumptions

Represent an octonion as a pair (a,b) of quaternions with product (a,b)(c,d)=(ac−conj(d)b, da+b conj(c)).

Use the quaternion convention ij=k, jk=i, ki=j. Set e1=(i,0), e2=(j,0), e3=(k,0), e4=(0,1), e5=(0,i), e6=(0,j), e7=(0,k).

The associator is [a,b,c]=(ab)c−a(bc). The Euclidean norm on R^8 is multiplicative.

A nonzero associator with an exact surviving identity

> Under this convention [e1,e2,e4]=2e7. Nevertheless the associator is alternating, and the Moufang identity (ab)(ca)=a((bc)a) holds for all real octonions. In particular any two elements generate an associative subalgebra, the classical alternative-algebra theorem.

Derivation or proof

The displayed associator follows directly from e1e2=e3, e3e4=e7, e2e4=e6 and e1e6=−e7. For an independent finite certificate of alternation, expand the trilinear associator in the basis 1,e1,…,e7. The verifier checks all 512 triples and confirms [a,b,c]+[b,a,c]=0 and [a,b,c]+[a,c,b]=0 at every coefficient. Bilinearity of multiplication then proves alternation for arbitrary real coordinates. The Moufang residual is quadratic in a and linear in b and c. It is therefore enough to check a on each basis vector and each pairwise sum of basis vectors, with b and c on basis vectors: polarization recovers all squared and mixed coefficients. All 2304 evaluations vanish exactly. This finite coefficient certificate verifies the full identity, not merely a random sample.

Worked example

Regroup e1,e2,e4 and compare the two results.

The left grouping gives (e1e2)e4=e7 and the right gives e1(e2e4)=−e7. Their difference has norm two. If the third factor instead belongs to the associative quaternion algebra generated by e1,e2, regrouping makes no difference.

The example demonstrates both a genuine failure and a precise replacement law. No philosophical interpretation is needed for either statement.

Scope of the result

Other octonion tables may change basis signs and the displayed associator sign. The convention must travel with every numerical example. Alternativity does not authorize arbitrary reassociation of three unrelated factors.

Further development

Use the explicit coefficient tensor as an exact algebraic input when investigating invariant subalgebras, automorphisms or defect loci.

References: R016, R017

M008 The sharp octonion regrouping bound

2023-02-20 · historical evidence A · Sharp inequality proved with an equality example

The earlier proposal to measure a regrouping defect can be carried to a sharp result. The defect is not just a qualitative marker: its exact universal scale follows from norm multiplicativity, and a basis triple attains that scale.

Definitions and assumptions

For octonions define d(a,b,c)=||(ab)c−a(bc)||.

Let n=||a||||b||||c||. Norm multiplicativity gives ||(ab)c||=||a(bc)||=n.

For unit factors the squared defect is 2−2⟨(ab)c,a(bc)⟩, using the Euclidean inner product.

Sharp norm bound and equality criterion

> For all real octonions, 0≤d(a,b,c)≤2||a||||b||||c||. If n>0, equality holds exactly when (ab)c=−a(bc). The constant two is optimal.

Derivation or proof

Write u=(ab)c and v=a(bc). The triangle inequality gives ||u−v||≤||u||+||v||=2n. Alternatively, expand the squared norm: $||u-v||^2=2n^2-2\langle u,v\rangle$. Cauchy-Schwarz gives $\langle u,v\rangle\geq -n^2$. Equality requires u and v to be oppositely directed; because their norms are equal and nonzero, this is exactly u=−v. The declared convention in M007 gives u=e7 and v=−e7 for a=e1,b=e2,c=e4, so the bound is attained at unit norm. Trilinearity also gives d(ra,sb,tc)=|rst|d(a,b,c) for real scalars. A norm-preserving algebra automorphism transports the associator and therefore preserves the defect.

Worked example

Evaluate the normalized defect on a maximally nonassociative basis triple and an associative quaternion triple.

For (e1,e2,e4), d=2. For (e1,e2,e3), both products equal −1, so d=0. The two calculations sit at opposite endpoints of the proved interval. The normalized quantity δ=d/(||a||||b||||c||) is consequently dimensionless on nonzero triples and ranges through values in [0,2].

The revised concept now has a universal bound, a certified sharpness witness, scale behavior and an exact criterion for attaining the upper endpoint.

Scope of the result

This does not classify every zero-defect triple or every orbit of the automorphism group. A zero associator for one triple alone should not be promoted to a statement that an entire generated algebra is associative.

Further development

Study the polynomial zero locus [a,b,c]=0 on three unit spheres and its orbit structure. Distinguish this classification problem from the sharp inequality already proved.

References: R016

M009 Hyperbolic identities that actually verify a wave

2023-03-11 · historical evidence U · Exact derivative and travelling-wave calculation

The corrected sech² identity supplies enough structure to verify a whole travelling-wave family. Instead of recognizing a familiar shape, we derive the amplitude-speed relation required by a specified nonlinear PDE.

Definitions and assumptions

Let T=tanh z and f(z)=sech²z=1−T².

The derivatives satisfy T′=f, f′=−2fT and d/dz=(1−T²)d/dT when applied to polynomials in T.

For a wave u(x,t)=A f(κ(x−ct−x0)), primes on u below denote spatial derivatives.

Amplitude and velocity are fixed by substitution

For nonzero κ and the equation u_t+6uu_x+u_xxx=0, the sech² ansatz is a solution when A=2κ² and c=4κ². These constants are linked by the equation; they cannot be chosen independently.

Derivation or proof

Differentiate $f'=-2fT$ to obtain $f''=-2f'T-2fT'=4fT^2-2f^2=4f-6f^2$. Differentiating again gives $f'''=(4-12f)f'$. After rescaling, $u_{xxx}=4\kappa^2 u_x-(12\kappa^2/A)u\,u_x$. The travelling-wave time derivative is $u_t=-c\,u_x$. The PDE residual is therefore $(4\kappa^2-c)u_x+(6-12\kappa^2/A)u\,u_x$. The stated amplitude and speed annihilate both coefficients. Conversely, for a nonconstant profile with $A\neq0$, vanishing for the whole wave forces those coefficients to vanish because u takes a nontrivial interval of values wherever u_x is nonzero. The verifier independently represents $u=2(1-T^2)$ as a polynomial and obtains an identically zero residual using exact integer coefficients.

Worked example

Use $\kappa=1$, $x_0=0$.

Then $u=2\operatorname{sech}^2(x-4t)$. Its maximum is two at $x=4t$ and its spatial integral is four, since $\int \operatorname{sech}^2 z\,dz=\tanh z$. The profile is localized and translates with speed four. A version with amplitude six and the same width would not solve this normalization of KdV.

The identity repair now produces a complete exact check, conserved profile mass and a falsifiable relationship among width, height and speed.

Scope of the result

Changing the sign or coefficient of the nonlinear term changes the amplitude relation. This calculation verifies an explicit family and is not a classification of all solutions or a stability proof.

Further development

Use the first integrals of the travelling-wave ODE to classify decaying profiles under explicit smoothness and boundary assumptions.

M010 Gudermannian coordinates with the measure retained

2023-03-11 · historical evidence U · An exact coordinate transformation and endpoint asymptotic

The Gudermannian is a genuine bridge between hyperbolic and circular expressions. Its advantage is algebraic simplification on a bounded angular interval; its cost is a singular inverse-coordinate derivative near that interval’s endpoints.

Definitions and assumptions

Set $\theta=\operatorname{gd}(x)=\arctan(\sinh x)$, mapping $\mathbb{R}$ bijectively to $(-\pi/2,\pi/2)$.

Then $\sin\theta=\tanh x$, $\cos\theta=\operatorname{sech} x$ and $\theta'=\cos\theta$.

The inverse is $x=\operatorname{asinh}(\tan\theta)=\operatorname{artanh}(\sin\theta)$, with $dx/d\theta=\sec\theta$.

Transforming derivatives and integrals consistently

For a smooth function $F(\theta)$, $\partial_x F=\cos\theta\,F_\theta$ and $\partial_x^2 F=\cos^2\theta\,F_{\theta\theta}-\sin\theta\cos\theta\,F_\theta$. For an integrable profile $u(x)=U(\operatorname{gd}(x))$, $\int_{\mathbb{R}} u(x)dx=\int_{(-\pi/2)}^{(\pi/2)}U(\theta)\sec\theta\,d\theta$. The angular compression does not discard this Jacobian.

Derivation or proof

Differentiate θ=arctan(sinh x): θ′=cosh x/(1+sinh²x)=sech x. Its range and monotonicity show it is a bijection. Since cosθ is positive in that range, the elementary identities follow by resolving a right triangle with tangent sinh x. The differential operator is ∂_x=cosθ∂_θ; applying it twice gives the stated second derivative. The ordinary change-of-variables theorem supplies dx=secθdθ. For x→+∞, π/2−gd(x)=2arctan(e^(−x))~2e^(−x). Hence the inverse sensitivity secθ grows like 1/(π/2−θ). Bounded angular coordinates therefore do not guarantee bounded recovery error in x.

Worked example

Transform u(x)=Asech²x.

The angular profile is U(θ)=Acos²θ. Retaining the measure gives ∫u dx=A∫cosθdθ=2A, matching the direct calculation A[tanh x]_(−∞)^∞. If one integrated Acos²θ against dθ without the Jacobian, the answer would instead be Aπ/2 and would describe a different measure.

The transformation simplifies a profile while preserving its integral only when the derivative and measure factors are carried along.

Scope of the result

For a moving coordinate θ=gd(κ(x−ct)), time differentiation also contributes θ_t=−cκcosθ. A stationary spatial formula cannot simply be inserted into a time-dependent PDE without this extra chain rule.

Further development

Transform an entire variational problem or PDE, including boundary conditions, noise and observation measure; assess conditioning in the transformed metric.

References: R002

M011 KdV and KP with matched conventions

2023-03-11 · historical evidence U · One-dimensional family and oblique KP line wave verified

KdV is an evolution equation in one spatial coordinate. KP adds a second spatial coordinate and a sign that changes its transverse behavior. An exact KdV solution embeds in either stated KP sign when it is independent of the second coordinate, but this elementary inclusion does not solve the full two-dimensional problem.

Definitions and assumptions

Fix KdV as u_t+6uu_x+u_xxx=0.

Fix KP as (u_t+6uu_x+u_xxx)_x+3σu_yy=0 with σ=±1.

A line-wave phase is η=kx+ly−ωt+δ with real k≠0 and l,δ; set U(η)=(k²/2)sech²(η/2).

An explicit KP line-wave dispersion relation

The profile u(x,y,t)=U(kx+ly−ωt+δ) solves the stated KP equation when ω=k³+3σl²/k. Taking l=0 recovers the KdV family of M009 after k=2κ.

Derivation or proof

The travelling-wave derivative identity in the phase variable is $U'''=U'-(6/k^2)UU'$. Thus the expression inside the outer x derivative is $-\omega U'+6kUU'+k^3U'''=(k^3-\omega)U'$. Its x derivative is $k(k^3-\omega)U''$, while $u_{yy}=l^2U''$. The KP residual is $[k(k^3-\omega)+3\sigma l^2]U''$ and vanishes under the displayed relation. With $l=0$, $\eta=2\kappa(x-4\kappa^2t-x0)$, giving precisely $2\kappa^2\operatorname{sech}^2(\kappa(x-4\kappa^2t-x0))$. This calculation fixes every coefficient and sign directly rather than inferring them from a remembered formula.

Worked example

Choose $\sigma=+1$, $k=2$, $l=1$ and $\delta=0$.

Then $\omega=8+3/2=19/2$, and $u=2\operatorname{sech}^2((2x+y-(19/2)t)/2)$. The crest is the straight line $2x+y=(19/2)t$. It is localized in the normal direction but extends along the line, so its total integral over the whole plane is not finite.

The line wave is an exactly verified two-dimensional solution with a clear localization limitation.

Scope of the result

Neither transverse stability nor general KP multi-soliton classification follows from this substitution. The signs $\sigma=\pm1$ must remain explicit, and conventions in external references may use time or field rescalings.

Further development

Introduce a tau-function compatible with the same convention and study a genuine interaction of nonparallel line solitons rather than adding individual profiles.

References: R018

M012 A complete two-soliton construction in a fixed KdV convention

2023-03-11 · historical evidence U · Tau-function interaction and phase shifts derived

The interaction term can be determined rather than guessed. A positive tau-function supplies a nonsingular exact solution and gives a clean explanation of why the answer is not the sum of two single waves.

Definitions and assumptions

Take $k1>k2>0$, $\eta_i=k_i\,x-k_i^3t+\delta_i$ and $A=((k1-k2)/(k1+k2))^2$.

Set $\tau=1+e^{\eta1}+e^{\eta2}+A\,e^{(\eta1+\eta2)}$, and $u=2\partial_x^2\log\tau$.

For exponentials, a Hirota bilinear operator acts by parameter differences; write $P(k,\omega)=k\omega+k^4$.

Interaction coefficient and asymptotic shifts

This τ obeys $(D_xD_t+D_x^4)\tau\cdot\tau=0$ and u solves $u_t+6uu_x+u_{xxx}=0$. The fast wave’s spatial shift is $-\log(A)/k1>0$ and the slow wave’s is $\log(A)/k2<0$. The profile mass is $2(k1+k2)$.

Derivation or proof

Each one-wave term satisfies $P(k_i,-k_i^3)=0$. The only new mixed exponential coefficient is proportional to $A\,P(k1+k2,-k1^3-k2^3)+P(k1-k2,-k1^3+k2^3)$. These two polynomial values are $3k1k2(k1+k2)^2$ and $-3k1k2(k1-k2)^2$, so their cancellation forces the stated A. Other coefficients reduce to the one-wave dispersion equations. Dividing the bilinear identity by τ^2 gives $(\log\tau)_{xt}+(\log\tau)_{xxxx}+6((\log\tau)_{xx})^2=0$; differentiating in x yields KdV for u. Because all four tau coefficients are positive, $\tau>0$ for real x,t, eliminating zeros and singularities. In the fast moving frame, the effective one-wave factor changes from $1+e^{\eta 1}$ to $1+A\,e^{\eta 1}$ between early and late times, giving the fast shift. In the slow frame the factor changes in the reverse order, giving the slow shift. Finally $\int u\,dx=2[(\log\tau)_x]_{-\infty}^{\infty}=2(k1+k2)$.

Worked example

At $x=t=\delta 1=\delta 2=0$ choose $k1=2$ and $k2=1$.

Here $A=1/9$. Direct rational differentiation gives $u(0,0)=153/98\approx 1.56122$, whereas adding the two isolated wave values gives 5/2. The verifier checks every bilinear exponential coefficient exactly at three distinct rational parameter pairs; the preceding symbolic coefficient calculation proves the full parameter family.

The interaction coefficient, nonsingularity, mass and phase shifts are all specified and derived within one consistent normalization.

Scope of the result

This is the classical KdV two-soliton family, also a y-independent KP solution. It is not a construction of every KP interaction. Coincident or zero k values require a separate limiting treatment.

Further development

Generalize to N waves with pair interaction factors and investigate which positivity assumptions guarantee a nonsingular tau-function.

References: R018, R019

M013 Radial conservation and stereographic geometry

2023-03-11 · historical evidence U · Two distinct inverse-power mechanisms calculated

A coordinate projection and a conserved flux are mathematically different constructions. Each can generate a position-dependent factor, but the factors belong to different measures and have different meanings. Writing both down makes the proposed identification testable.

Definitions and assumptions

In Euclidean R^d with $d\geq 2$, a radial field has the form $J(x)=F(r)e_r$ for $r=||x||>0$.

Conserved outward flux across a sphere means $F(r)$ times its area is independent of r.

Stereographic coordinates (u,v) on the unit sphere use $r^2=u^2+v^2$ and $X=(2u,2v,r^2-1)/(1+r^2)$.

Flux power law versus a projection Jacobian

A rotationally symmetric conserved flux in dimension d has F(r)=C/r^(d−1), giving inverse-square magnitude in d=3. The unit-sphere stereographic metric is ds²=4(du²+dv²)/(1+u²+v²)² and its area measure is 4du dv/(1+u²+v²)². The latter is not a derivation of the former.

Derivation or proof

The sphere of radius r has area ω_(d−1)r^(d−1). If its flux is Φ, then Φ=ω_(d−1)r^(d−1)F(r), yielding F(r)=Φ/(ω_(d−1)r^(d−1)). Equivalently, away from a source, the radial divergence is r^(1−d)∂_r(r^(d−1)F), which must vanish. For the projection, differentiate the displayed map X(u,v), take Euclidean inner products of its two coordinate derivatives and obtain a diagonal pullback metric with both diagonal entries 4/(1+r²)². Taking the square root of the determinant supplies the stated area factor. This second calculation concerns how one surface is charted, not how a physical flux changes across a family of spheres.

Worked example

Compare the large-distance factors in their own variables.

A three-dimensional radial flux decays as physical radius R^(−2). The stereographic area density decays as chart radius r^(−4). These statements do not contradict each other: R labels different spheres in a physical space, whereas r labels positions within a chart of one unit sphere. Identifying them would discard the very geometric distinction being calculated.

The original question now has a derivation of each factor and a concrete reason the coordinate formula does not imply the conservation law.

Scope of the result

A projected representation of a conserved field can of course be constructed. It must transform the field and the integration measure together; conservation is then transported, not created by projection.

Further development

Pull back a specified flux form into stereographic coordinates and verify the invariant integral explicitly, including the missing chart point.

M014 q-differentiation of the Gudermannian

2023-03-11 · historical evidence U · An exact secant integral and uniform bounds

There is a clean identity connecting q-differentiation and gd, but it is an instance of the fundamental theorem of calculus rather than evidence of a special learning mechanism. It allows the q-difference quotient to be bounded and its ordinary-derivative limit to be controlled.

Definitions and assumptions

For real q>0, q≠1 and x≠0, define D_qf(x)=(f(qx)−f(x))/((q−1)x).

Define D_qgd(0)=1 by continuous extension.

Let m=min(1,q) and M=max(1,q). On the interval between x and qx, the absolute coordinate lies between m|x| and M|x|.

Exact integral representation and bounds

D_qgd(x)=∫_0^1 sech(x(1+s(q−1)))ds. Consequently sech(M|x|)≤D_qgd(x)≤sech(m|x|). As q→1 these quotients converge to sech x uniformly on compact x-intervals.

Derivation or proof

Since gd′(t)=sech t, subtracting its values gives gd(qx)−gd(x)=∫_x^(qx)sech t dt. Substitute t=x+s(q−1)x; the oriented integral and denominator cancel even when q<1 or x<0, proving the formula. The function sech t is positive, even and decreasing as |t| increases, so its extrema on this same-sign interval give the bounds. For x in a compact interval and q in a sufficiently small neighborhood of one, all arguments remain in a common compact interval. Uniform continuity of sech there makes the integrand converge uniformly, hence the averaged quotients converge uniformly. At x=0 the integrand equals one, proving the claimed extension without division by zero.

Worked example

Compare q=1/2 and q=2 at x=2.

The first quotient averages sech between 1 and 2; the second averages it between 2 and 4. Both are positive, and the verifier checks their values against the exact endpoint bounds. These are secant averages over different intervals, not two names for the same local derivative.

The relation is exact and has a precise limit. It distinguishes an honest q-difference calculation from a purely verbal association.

Scope of the result

Replacing an ordinary gradient by D_q changes the optimization method and its chain rule. For products, D_q(fg)=g(x)D_qf+f(qx)D_qg; the ordinary Leibniz formula is recovered only in the limit. No improved learning performance follows from positivity or the bounds alone.

Further development

Analyze a specified q-gradient iteration, derive its actual descent conditions and compare it with ordinary gradient descent under matched evaluation costs.

References: R020, R002

M015 Quantitative Jacobian bounds for deep compositions

2023-03-11 · historical evidence U · Vanishing demonstrated and a residual-depth bound derived

A bounded activation does not protect its derivative from saturation. For gd the derivative is explicitly sech, so useful depth bounds can be derived without guessing from wave-propagation analogies. A residual scaling offers a separate, conditional way to control conditioning as depth grows.

Definitions and assumptions

For scalar layers h_l=gd(w_l h_(l−1)+b_l), write z_l=w_l h_(l−1)+b_l.

The input derivative is $J=\prod_{l=1}^{L} w_l \operatorname{sech} z_l$.

For vector layers the Jacobian is $D_l W_l$ with D_l diagonal and entries sech of the preactivations. For the residual construction take $T>0$ and $K\geq 0$. Singular-value bounds must use all relevant coordinates.

Saturation bound and a depth-scaled residual alternative

If $|w_l|\leq W$ and every scalar $|z_l|\geq a>0$, then $|J|\leq(W \operatorname{sech} a)^L$. For residual maps $x_{l+1}=x_l+(T/L)F_l(x_l)$ with $\|DF_l\|_2\leq K$ and $L>TK$, every total-Jacobian singular value lies between $(1-TK/L)^L$ and $(1+TK/L)^L$. For $L\geq 2TK$, the bounds e^{-2TK} and e^{TK} are uniform in depth.

Derivation or proof

The scalar chain rule gives the product, and monotonicity of sech on nonnegative arguments supplies the first estimate. In a residual layer, write its Jacobian $I+E$ with $\|E\|\leq TK/L$. For every vector v, the triangle and reverse triangle inequalities give $(1-\|E\|)\|v\|\leq\|(I+E)v\|\leq(1+\|E\|)\|v\|$. Multiply the singular-value bounds across layers. The upper uniform estimate follows from $\log(1+s)\leq s$. For $s\leq 1/2$, $\log(1-s)\geq -2s$ gives the lower estimate. These are worst-case deterministic bounds, conditional on a common derivative bound K; they make no distributional or performance claim.

Worked example

Use $h_{l+1}=g d(h_{l+2})$, $h_0=0$, with $L=20$.

Every preactivation is at least two. Therefore $|J|\leq \operatorname{sech}(2)^{20}\approx 3.09865\times 10^{-12}$. Direct accumulation of logarithmic derivatives gives approximately 1.82740×10^{-24}, far smaller still. The calculation avoids underflow and illustrates that the proved upper bound need not be sharp for a particular trajectory.

The original concern is now expressed as a measurable derivative contraction, alongside a different architecture with an explicit conditional depth guarantee.

Scope of the result

The residual lower bound does not imply successful training, identifiability or useful representations. The gd upper bound for a vector layer requires control of the operator norm of its whole diagonal derivative matrix, not merely one saturated coordinate.

Further development

Test whether a proposed layer family maintains the required K and singular-value conditions on its actual state domain; compare those certificates with measured gradients.

M016 Minimal universal recurrences on an observable quotient

2023-03-11 · historical evidence A · An exact annihilator theorem with a reduced example

Cayley-Hamilton always gives a recurrence for a finite linear state, but the observable behavior may require a strictly smaller polynomial. The correct object is the induced operator after all permanently invisible directions have been quotiented out.

Definitions and assumptions

Let V be a finite-dimensional vector space over a field, A:V→V and C:V→Y linear, with outputs y_n=CA^n x.

Let N=∩_(k=0)^(n−1)ker(CA^k), where n=dimV. Cayley-Hamilton makes this the intersection over all k≥0.

A universal output recurrence with polynomial p means CA^k p(A)x=0 for every state x and every k≥0.

The observable quotient’s minimal polynomial is the universal recurrence

N is A-invariant, so A induces A_bar on V/N. A polynomial p gives a universal output recurrence exactly when p(A_bar)=0. Thus the unique least-degree monic universal recurrence is the minimal polynomial of A_bar. Its degree is at most dim(V/N); for a scalar observable with an observable quotient it equals that quotient dimension.

Derivation or proof

If v∈N, all outputs CA^kv vanish, hence CA^k(Av)=CA^(k+1)v=0 and Av∈N. Now a recurrence holds exactly when p(A)V⊂N, which is precisely p(A_bar)=0 after applying the quotient map. This proves both directions and minimality. The dimension bound is the standard minimal-polynomial bound. For scalar output, the row span of C,CA,… on the quotient has full dimension by observability. A polynomial of smaller degree would make every later row a combination of fewer initial rows, contradicting full dimension. This last equality need not hold for multiple outputs: A=I with C=I observes an n-dimensional state but has universal polynomial z−1.

Worked example

Take A=diag(1,2,3) and C=(1,1,0).

The invisible space is span(e3). On the two-dimensional quotient the minimal polynomial is (z−1)(z−2), so y_(n+2)−3y_(n+1)+2y_n=0 for every initial state. The full characteristic polynomial has a redundant factor z−3. If the particular initial state is e1, its individual sequence satisfies the shorter recurrence y_(n+1)=y_n.

This distinguishes full-state, universal-observation and single-trajectory recurrences, rather than conflating their orders.

Scope of the result

A recurrence fitted to one finite prefix may be accidental. The quotient argument proves a recurrence for all times under the stated model; finite fitting alone does not.

Further development

Compute exact observability kernels and Hankel ranks, then certify the smallest realization compatible with the desired class of initial conditions.

References: R021

M017 The reachable and observable parts of a realization

2023-03-11 · historical evidence A · Minimal zero-initial-state quotient constructed

Reachability and observability answer different questions. Their intersection structure determines which internal modes affect driven input-output behavior from zero initial state. A mode can be unstable and entirely absent from the transfer function, making the initial-condition convention essential.

Definitions and assumptions

For a finite-dimensional real or complex system $\dot{x}=Ax+Bu$ and $y=Cx+Du$ with locally integrable inputs, define $R=\operatorname{span}\{A^k\,\mathrm{im}B : k\geq 0\}$, the reachable subspace.

Let $N=\bigcap_{k\geq 0}\ker(CA^k)$, the unobservable subspace.

Both R and N are A-invariant, and the transfer function is $H(s)=C(sI-A)^{-1}B+D$ where the resolvent exists.

The zero-initial input-output state lives on R/(R∩N)

> Restriction to R discards modes no input can excite from zero. Quotienting by R∩N then discards modes no output can detect. The resulting realization is reachable and observable and has the same zero-initial input-output map; its dimension is the standard minimal realization order.

Derivation or proof

Starting at zero, the variation-of-constants formula $x(t)=\int_0^t e^{A(t-s)}Bu(s)ds$ lies in R because that subspace is A-invariant and contains imB. Adding a state in R∩N never changes any subsequent output, including under common future inputs, so the quotient preserves the response. Conversely, all reachable quotient states are generated by the projected input, and an invisible quotient class would have a representative in N, hence in R∩N, so it is zero. These properties characterize the usual minimal finite-dimensional realization. Replacing the zero initial condition by arbitrary initial states changes the target: visible but unreachable modes can now contribute.

Worked example

Take $A=\operatorname{diag}(-1,-2,3)$, $B=(1,0,1)^T$ and $C=(1,1,0)$.

$R=\operatorname{span}(e1,e3)$, while $N=\operatorname{span}(e3)$. The minimal driven state therefore has dimension one and transfer function $1/(s+1)$. The internally unstable e3 mode is reachable but invisible. The e2 mode is visible but unreachable; it matters for initial condition e2, producing e^{-2t}, but not for zero-initial driven responses.

The example has reachable rank two and observable rank two, yet minimal driven order one. Merely taking either rank misses the intersection.

Scope of the result

A stable transfer function does not establish internal stability when an unstable canceled mode remains. No empirical sensor claim follows without a correct state and observation model.

Further development

Use exact basis changes to expose the four Kalman blocks and compare driven, arbitrary-initial-state and noisy-observation specifications.

References: R021

M018 Exact finite closure and a polynomial obstruction

2023-03-11 · historical evidence A · Invariant heat subspace and nonlinear no-go result proved

A finite matrix can represent an infinite-dimensional evolution exactly on a proved invariant subspace. A truncation outside such a subspace instead produces an approximation. The distinction can be settled in examples, and a useful degree argument rules out a broad polynomial closure proposal.

Definitions and assumptions

A subspace W of observables is Koopman-invariant for a map f when $p \in W$ implies $p \circ f \in W$.

For the periodic heat equation $u_t = u_{xx}$, trigonometric modes are eigenfunctions of the generator ∂_{xx}.

For the obstruction, work with real polynomials as functions on an interval, and let f have degree $d \geq 2$.

No finite polynomial Koopman space can contain the state coordinate

> There is no finite-dimensional subspace $W \subset \mathbb{R}[x]$ that contains x and is invariant under $p \mapsto p \circ f$ when $\deg f = d \geq 2$. In contrast, the heat equation exactly preserves every fixed finite span of Fourier modes.

Derivation or proof

If W contains x and is invariant, it contains $x \circ f = f$, then $f \circ f$, and inductively every iterate $f^{\circ n}$. Their degrees are d^n because the nonzero leading coefficient of a composition cannot cancel. Polynomials with distinct degrees are linearly independent: in any finite linear combination the largest degree has a coefficient contributed by exactly one summand, so that coefficient must vanish, and descending induction eliminates the rest. Thus W would contain infinitely many independent vectors, a contradiction. For heat flow, $\partial_{xx} \sin(kx) = -k^2 \sin(kx)$ and the same holds for $\cos(kx)$, while the constant mode has eigenvalue zero. This proves invariance of each chosen finite mode span and gives exact coefficient evolution $a_k(t) = e^{-k^2 t} a_k(0)$.

Worked example

Compare the logistic map and a nonlinear PDE stencil.

For $f(x) = 2x(1-x)$, exact iterates have degrees 1,2,4,8,16,32,64 starting with x. No finite polynomial feature list containing x closes exactly on an interval. For Burgers-type nonlinearity $u\, u_x$, the initial mode $\sin x$ immediately generates $\sin x \cos x = (1/2)\sin 2x$, so the single-mode span is not invariant even though it is invariant for heat flow.

Finite polynomial degree growth gives a proof of impossibility, while Fourier invariance gives a valid exact finite example. Neither conclusion rests on a numerical residual alone.

Scope of the result

The obstruction does not exclude nonpolynomial coordinate transformations, finite-state restrictions, approximate closures or special higher-dimensional nonlinear systems. On a finite orbit, polynomials are functions modulo additional relations and the interval argument no longer applies.

Further development

Before applying Cayley-Hamilton to a nonlinear model, certify an invariant feature space or quantify the projection error. Investigate special model classes separately rather than assuming universal finite closure.

References: R006, R007

B Harmonic analysis graph invariants and arithmetic geometry

M019 Hardy-BMO duality from atoms

2023-04-23 · historical evidence U · Classical estimate proved; duality theorem identified

The useful connection is a bounded pairing between a cancellation-sensitive signal space and a mean-oscillation space. This supplies an actual inequality; it does not make arbitrary behavior into a Hardy-space object.

Definitions and assumptions

A real H^1 atom a is supported in a finite interval I, satisfies $\int a=0$ and $||a||_\infty \le 1/|I|$. Higher dimensions use cubes or balls.

The BMO seminorm is $\sup_I |I|^{-1}\int_I |b-b_I|$, where b_I is the interval average. Constants have seminorm zero.

The atomic H^1 norm is the infimum of $\Sigma|\lambda_j|$ over convergent atomic decompositions $f=\Sigma\lambda_j a_j$. This is not the Sobolev H^1 norm.

Cancellation converts mean oscillation into a bounded functional

> For an atom a and $b\in BMO$, $|\int ab| \le ||b||_{BMO}$. Consequently $|\langle f,b\rangle| \le ||f||_{H^1,atomic}\,||b||_{BMO}$. The converse identification of every bounded H^1 functional with BMO modulo constants is the classical duality theorem, not a conclusion of the finite example alone.

Derivation or proof

Subtract the constant b_I inside the pairing: $\int ab=\int_I a(b-b_I)$, because $\int a=0$. The atom bound gives $|\int ab| \le |I|^{-1}\int_I |b-b_I| \le ||b||_{BMO}$. For a finite atomic sum, apply the triangle inequality and sum the coefficient magnitudes. Completion gives the bounded functional on the atomic space. The quotient by constants is forced by cancellation: adding c to b leaves every atom pairing unchanged. The difficult converse is a representation theorem and requires the established Hardy-space theory; no finite rank calculation can replace it.

Worked example

Take $I=[-1,1]$, $a(x)=-1/2$ on $(-1,0)$, $a(x)=1/2$ on $(0,1)$, and zero outside. Let $b(x)=\mathrm{sign}(x)$.

The atom has zero integral and supremum 1/2. For any interval containing positive fraction t and negative fraction $1-t$, the mean oscillation of sign is $4t(1-t) \le 1$. Symmetric intervals achieve 1. The pairing is $\int ab=1$, so the atomic estimate is sharp for this normalization.

A normalization, a cancellation condition and a sharp example replace the earlier loose invocation of two named function spaces.

Scope of the result

Pairings of general distributions require the continuous extension, not an unjustified pointwise integral. Graph analogues need their own measure, Laplacian and atomic definitions; the Euclidean theorem is not automatically inherited.

Further development

Define graph atoms with respect to an explicit reversible measure, then ask whether the chosen graph-BMO space is the dual of the chosen graph Hardy space. Proving just the atom bound is a useful first stage, but it is not full duality.

References: R022

M020 A complete heat-signature collision

2023-04-23 · historical evidence U · Classical pair with exact finite certificate

The 4×4 rook graph and the Shrikhande graph supply an exact obstruction to recovering a graph from its heat trace or diagonal heat signature. The full heat operator retains strictly more information.

Definitions and assumptions

Both graphs have vertex set $Z_4 \times Z_4$. Rook adjacency means sharing exactly one coordinate.

Shrikhande adjacency means the difference lies in $\{(\pm1,0),(0,\pm1),(1,1),(-1,-1)\}$ modulo four.

For adjacency A, L=6I−A and K_t=exp(−tL). The heat trace is Tr K_t; the vertex signature is (K_t)_{vv}.

Identical spectra and vertex signatures do not force isomorphism

> Both graphs have Laplacian eigenvalues 0 once, 4 six times and 8 nine times, and every vertex in both has signature (1+6e^(−4t)+9e^(−8t))/16. Nevertheless their clique numbers are respectively four and three.

Derivation or proof

The verifier constructs both integer adjacency matrices and checks $A^2=4I+2J$. The all-ones vector has adjacency eigenvalue six. On its orthogonal complement J vanishes, so adjacency eigenvalues are ±2. Trace A=0 and dimension sixteen force multiplicities six and nine. Both graphs are connected because their Cayley generators span Z_4^2, hence eigenvalue six is simple. Their vertex translations act transitively, so all diagonal entries of any matrix function of A agree and equal its trace divided by sixteen. The verifier exhaustively checks four-vertex subsets: rook four-cliques exist, whereas Shrikhande has none, and Shrikhande has triangles. Clique number is invariant under isomorphism, proving the graphs differ.

Worked example

Use the two explicit sixteen-vertex adjacency matrices produced by rook_adjacency() and shrikhande_adjacency() in verify.py.

Exact matrix multiplication verifies the common-neighbor identity. Exhaustive enumeration of C(16,4)=1820 subsets per graph certifies the four-clique distinction. This does not rely on floating eigenvalue comparisons or a graph-isomorphism heuristic.

Even the entire curve of diagonal heat signatures can lose structural information. This is a stronger failure than matching a handful of sampled times.

Scope of the result

The obstruction does not apply to the complete labeled K_t. For any fixed $t>0$, K_t is symmetric positive definite and its unique real symmetric logarithm gives $L=-\log(K_t)/t$. Equality up to a permutation therefore recovers graph isomorphism. Numerical recovery can still be ill-conditioned when small eigenvalues of K_t are noisy.

Further development

Study the minimum additional observations that separate this pair, such as selected off-diagonal heat entries or local clique counts. Specify whether the goal is pair separation, recovery up to isomorphism or uniformly stable reconstruction.

References: R011

M021 Exact modularity phases on a four-cycle

2023-04-24 · historical evidence U · Finite optimization solved exactly

A small graph already exhibits resolution-dependent optima and nontrivial ties. Reporting one partition without its resolution parameter would conceal both effects.

Definitions and assumptions

The graph is C_4 with four edges, degree two at every vertex and total edge weight $m=4$.

For a vertex partition P and $\gamma\geq 0$, $Q_\gamma(P)=\Sigma_{B\in P}[e(B)/4-\gamma(|B|/4)^2]$, where $e(B)$ counts internal undirected edges.

The feasible set contains all fifteen set partitions of four labeled vertices. Rotated partitions are distinct selections even when they have the same score.

The complete modularity phase diagram

> For $0\leq\gamma<1$ the all-in-one partition uniquely maximizes $Q_\gamma=1-\gamma$. For $1<\gamma<2$ the two partitions into adjacent pairs maximize $Q_\gamma=1/2-\gamma/2$. For $\gamma>2$ the singleton partition uniquely maximizes $Q_\gamma=-\gamma/4$. At $\gamma=1$ there are three maximizers; at $\gamma=2$ there are seven.

Derivation or proof

Group partitions by block pattern and internal-edge count. Besides the three candidate lines, an adjacent pair with two singletons has score $1/4-3\gamma/8$; a triple plus singleton has $1/2-5\gamma/8$; opposite pairs have $-\gamma/2$; a nonadjacent pair with two singletons has $-3\gamma/8$. Comparison of these finitely many affine functions gives the envelope above. The first two candidate lines meet at $\gamma=1$. The pair and singleton lines meet at $\gamma=2$, where the four adjacent-pair-plus-singletons partitions also tie. No discarded line is higher for $\gamma\geq 0$. The verifier independently enumerates the partitions and evaluates rational scores at interior and boundary values.

Worked example

At $\gamma=3/2$, compare the all-in-one, adjacent-pair and singleton candidates.

Their scores are $-1/2$, $-1/4$ and $-3/8$ respectively. Exactly two adjacent-pair partitions win. At $\gamma=2$ each has score $-1/2$, as do the four one-pair partitions and the all-singleton partition.

The preferred community structure changes under the chosen scale even though the graph has not changed. Boundary multiplicities are mathematically part of the answer.

Scope of the result

This solves one finite instance, not general modularity maximization. Numerical graph algorithms may miss ties, and graph perturbations require a separate sensitivity analysis.

Further development

For a weighted graph, compute the exact upper envelope of all partition score lines for manageable n. Identify parameter intervals with a unique optimizer and certify a score gap that bounds stability to weight perturbations.

References: R023

M022 Ensemble assumptions determine the spectral baseline

2023-04-24 · historical evidence U · Moment calculation proved

A random-matrix comparison must preserve dependence, centering and normalization. A simple fourth-moment calculation shows why a sparse adjacency matrix cannot be treated as a dense Wigner matrix without qualification.

Definitions and assumptions

Let A be a simple random graph on n vertices with independent Bernoulli(p) upper-triangular entries and zero diagonal.

Set $W_{ij}=(A_{ij}-p)/\sqrt{np(1-p)}$ for $i\neq j$ and $W_{ii}=0$. Then $E\,W_{ij}=0$ and $E\,W_{ij}^2=1/n$.

A sample covariance $S=X^TX/m$ is positive semidefinite and belongs to a different ensemble, even if X has independent entries.

Exact second and fourth normalized moments

$E[\mathrm{Tr}(W^2)/n]=(n-1)/n$. Moreover $E[\mathrm{Tr}(W^4)/n]=2(n-1)(n-2)/n^2+(n-1)[1-3p(1-p)]/[n^2p(1-p)]$. Thus $p=c/n$ with fixed $c>0$ yields fourth-moment limit $2+1/c$, whereas $np(1-p)\to\infty$ removes this extra term.

Derivation or proof

Expand the traces as sums over closed walks. For the second moment, each of $n(n-1)$ directed edges contributes $1/n$. For length four, centering kills walks containing an edge used only once. Two distinct edges traversed twice give two families of $n(n-1)(n-2)$ walks, each with expectation $1/n^2$. Alternating repeatedly on one edge gives $n(n-1)$ walks with $E\,W_{ij}^4=[1-3p(1-p)]/[n^2p(1-p)]$. Divide by n to obtain the formula. This derivation depends on upper-triangular independence; it fails for matrices whose entries share hidden common factors.

Worked example

Compare dense $p=1/2$ with $p=c/n$ and compare W with a covariance matrix.

The dense fourth-moment limit is two. The sparse fourth-moment limit retains $1/c$. A covariance matrix satisfies $x^TSx=||Xx||^2/m\geq 0$, while centered symmetric adjacency matrices can have negative eigenvalues.

One observed eigenvalue histogram cannot select its own null ensemble. The normalization and dependence structure must be specified before interpreting outliers.

Scope of the result

Two moments do not prove a complete limiting spectral law. The sparse calculation demonstrates a mismatch, not a universal law for every sparse network. No prediction about human behavior follows from these moment identities.

Further development

Estimate the dependence and degree heterogeneity that a null ensemble must preserve. Compare centered, variance-normalized and degree-corrected matrices, and use moment or resolvent bounds matched to each construction.

References: R024

M023 A reversible q-dependent chain with an exact gap

2023-04-25 · historical evidence A · Finite family solved analytically

Here q has one explicit meaning: the upward jump rate of a reflecting birth-death chain. This produces an actual probability model, stationary distribution and spectral answer.

Definitions and assumptions

The states are $0,\dots,N$. From $i<N$ the jump $i\to i+1$ has rate $q>0$; from $i>0$ the jump $i\to i-1$ has rate one.

The generator Q has diagonal entries equal to minus its outgoing rate and row sums zero. The process is irreducible on this finite state space.

Write $Z=\Sigma_{i=0}^N q^i$ and $\pi_i=q^i/Z$. At $q=1$ this is uniform.

Stationary distribution and complete nonzero spectrum

> Detailed balance holds, and the eigenvalues of $-Q$ are zero and $\lambda_k=q+1-2\sqrt{q}\cos(k\pi/(N+1))$, $k=1,\dots,N$. The spectral gap is $\lambda_1>0$.

Derivation or proof

For each adjacent pair, π_i q=π_{i+1}; summing these balances proves πQ=0. Conjugate −Q by diag(sqrt(π_i)) to obtain a symmetric tridiagonal matrix S with off-diagonal −s, s=sqrt(q), interior diagonal q+1 and endpoint diagonals q and one. The interior eigenvalue recurrence has solutions v_i=sin((i+1)θ)−s⁻¹sin(iθ) for λ=q+1−2s cosθ. The left endpoint condition holds directly. The right endpoint reduces to (2cosθ−s−s⁻¹)sin((N+1)θ)=0. Taking θ=kπ/(N+1), 1≤k≤N, gives N independent nonzero eigenmodes; the stationary mode sqrt(π) supplies zero. These N+1 eigenvectors exhaust the dimension.

Worked example

Take N=2 and q=4.

The stationary weights are (1,4,16)/21. The matrix −Q has characteristic roots 0,3,7. Exact determinant evaluation and balance checks in verify.py reproduce all three. At q=1 the gap is 2−2cos(π/(N+1)), asymptotic to π²/(N+1)².

The parameter has a measurable effect on equilibrium concentration and mixing, with no unspecified q-randomness.

Scope of the result

Reflecting boundaries are essential to this finite-chain answer. An infinite state space has additional recurrence and normalizability conditions: Σqⁱ is finite only when 0<q<1. The gap formula is not a formula for arbitrary q-deformed chains.

Further development

Use the exact reversible gap to bound L² convergence. For a point start x, a standard consequence is TV distance at time t at most (1/2)sqrt(1/π_x−1)e^(−λ_1 t). Study how boundary placement changes the useful mixing bound.

References: R025

M024 Continuous q-Hermite polynomials through a Jacobi operator

2023-04-25 · historical evidence A · Classical family with exact low-degree verification

A recurrence with positive coefficients constructs a spectral measure and explains orthogonality. This is a precise operator-polynomial connection rather than a generic appeal to q-special functions.

Definitions and assumptions

Fix 0<q<1. Let H₀=1, H₁=2x and H_{n+1}=2xH_n−(1−qⁿ)H_{n−1}.

On ℓ^2 of the nonnegative integers, define a bounded self-adjoint Jacobi operator J by off-diagonal entries a_n=sqrt(1−qⁿ)/2 for n≥1 and zero diagonal.

Let μ_q be the probability spectral measure of J at e₀. Since ||J||≤1, its support lies in [−1,1].

Orthogonality and exact norms

The recurrence gives H_n(J)e₀=Π_{j=1}^n sqrt(1−qʲ)e_n. Hence ∫H_mH_n dμ_q=δ_{mn}Π_{j=1}^n(1−qʲ). In particular μ_q has second moment (1−q)/4 and fourth moment (1−q)(2−q−q²)/16.

Derivation or proof

The spectral theorem identifies integration of a polynomial with its matrix element at e₀. Induct on n using 2Je_n=sqrt(1−q^{n+1})e_{n+1}+sqrt(1−qⁿ)e_{n−1}; the recurrence cancels the lower basis vector. Orthogonality then follows from orthogonality of e_n, and the norm is the squared coefficient product. For moments, count weighted return walks of length two and four from zero. At length two there is one path; at length four the paths 0→1→0→1→0 and 0→1→2→1→0 contribute a_1^4 and $a_1^2a_2^2$.

Worked example

Take q=1/2 and compute H₀ through H₅ using rational coefficients.

The probability measure has moments μ₂=1/8 and μ₄=5/128. Exact weighted-walk moments verify every pairing of H₀,…,H₅; squared norms begin 1,1/2,3/8,21/64. The finite computation does not numerically approximate an infinite oscillatory integral.

A recurrence, an operator and a measure now support the same calculation. The cited family name is backed by a fully stated normalization.

Scope of the result

The Jacobi construction proves the displayed orthogonality; it does not itself reproduce every q-difference identity or special-function representation. Changing q outside the stated interval may change positivity or moment determinacy.

Further development

Set x=sqrt(1−q)z/2 and divide H_n by (1−q)^(n/2). The recurrence becomes K_{n+1}=zK_n−[n]_qK_{n−1}, which tends coefficientwise to the probabilists’ Hermite recurrence as q→1. Analyze convergence of the rescaled measures as a separate limiting problem.

References: R026

M025 A graph game with explicit equilibria

2023-04-24 · historical evidence U · Finite equilibria and potential verified

A graph becomes a game only after strategies and payoffs are supplied. The four-cycle coordination example also separates a local equilibrium from a globally optimal configuration.

Definitions and assumptions

Each vertex of C₄ chooses s_i∈{0,1}. Its payoff is the number of neighbors choosing the same value, minus εs_i.

A pure Nash equilibrium permits no strictly profitable unilateral change. Indifference is allowed.

The potential is Φ(s)=Σ_{edges {i,j}}1[s_i=s_j]−εΣ_i s_i.

An exact potential and a perturbation-induced equilibrium change

Every unilateral payoff change equals the corresponding change in Φ. At ε=0 there are six pure equilibria: the two constant profiles and the four rotations of 0011. For 0<ε<2 only the two constant profiles remain; for ε>2 only all-zero remains.

Derivation or proof

Changing s_i changes exactly the edge agreement terms incident to i and its own action cost, proving the potential identity. Since the game is finite, any sequence of strict improving moves terminates. To classify C_4 equilibria, observe that at ε=0 an isolated action has two disagreeing neighbors and can improve, while blocks of two have one agreeing and one disagreeing neighbor and are weakly stable. These give exactly the listed profiles. If 0<ε<2, a boundary vertex choosing one has one neighbor of each type, so switching to zero saves ε without losing agreement. Thus mixed profiles cannot be equilibria. The constant-one profile stays stable until its cost exceeds the loss of two agreements. Constant-zero is stable for every positive ε.

Worked example

Enumerate all sixteen action profiles and each of their four possible unilateral flips.

The verifier finds six, two and one equilibria at ε=0,1/2,3 respectively and checks the potential identity for every profile and deviation at ε=1/2. At ε=0, 0011 is an equilibrium with two agreeing edges, while the global optimum has four.

Local stability, global optimization and a graph partition are three distinct notions. A small action cost can remove weak equilibria without changing the graph.

Scope of the result

The boundary ε=2 needs its own weak-inequality treatment; the displayed open-interval statement avoids hiding it. Mixed-strategy equilibria are not classified here, and a different payoff law can give a different game on the same graph.

Further development

Characterize strict equilibria on a general graph through neighbor majorities, then compare them with potential maximizers. For constrained interventions, optimize which action costs or edge weights eliminate a selected undesirable equilibrium.

References: R027

M026 Relative entropy along a finite Markov chain

2023-04-29 · historical evidence U · Contraction proved with a strict counterexample for Shannon entropy

Relative entropy has a robust monotonicity property relative to a stationary reference distribution. The unqualified claim that any diffusion must increase Shannon entropy is false.

Definitions and assumptions

Let P be a row-stochastic matrix on a finite state set and π a stationary probability with every $\pi_i>0$.

$D(p||\pi)=\Sigma_i\, p_i \log(p_i/\pi_i)$, with 0log0=0. The next distribution is pP.

Shannon entropy is $H(p)=-\Sigma_i\, p_i \log p_i$ and depends on the chosen state representation.

Data processing gives D(pP||π)≤D(p||π)

The inequality holds for every initial distribution p. If P is doubly stochastic, it implies H(pP)≥H(p) by taking π uniform. Without this extra structure, Shannon entropy can decrease.

Derivation or proof

Form joint arrays $a_{ij}=p_iP_{ij}$ and $b_{ij}=\pi_iP_{ij}$. Their relative entropy is $\Sigma_i p_i \log(p_i/\pi_i)$, because each row of P sums to one. Summing out i sends the arrays to pP and πP=π. Apply the log-sum inequality separately in each column, then add over j. Equality in a column requires the ratios p_i/π_i to be constant among its positive incoming transitions. In particular a strictly positive P gives strict contraction unless p=π. The uniform-reference identity $D(p||u)=\log n-H(p)$ proves the doubly stochastic corollary.

Worked example

Use p=(1/2,1/2), π=(9/10,1/10), and P with both rows equal to π.

One step gives pP=π. Shannon entropy decreases from log2≈0.693147 to 0.325083. Meanwhile relative entropy decreases from log(5/3)≈0.510826 to zero. The transition probabilities are exact rationals; only the displayed logarithms are numerical.

Convergence toward a concentrated equilibrium can reduce Shannon entropy without contradicting information contraction.

Scope of the result

This proves monotonicity, not a uniform convergence rate. Nonirreducible chains can preserve information in different closed classes. Continuous-state systems need densities, integrability and an appropriate reference measure.

Further development

For a strictly positive distribution evolving under the reversible q-chain of M023, differentiate D along the semigroup. Detailed balance produces $-(1/2)\Sigma_{ij}\pi_iQ_{ij}(r_i-r_j)(\log r_i-\log r_j)\le 0$, $r_i=p_i/\pi_i$. A log-Sobolev inequality would strengthen this to a quantitative entropy-decay rate.

References: R025

M027 Neat groups and a smooth modular-curve example

2023-05-11 · historical evidence U · Definition clarified; rank-one example derived

Neatness removes finite-order eigenvalue phenomena in an arithmetic group. It controls the open quotient, while boundary smoothness remains a separate geometric issue.

Definitions and assumptions

An element is neat when the multiplicative subgroup generated by its complex eigenvalues contains no nontrivial root of unity. A group is neat when each of its elements is.

For N≥3, let $\Gamma(N)=\{\gamma\in SL_2(Z):\gamma\equiv I \bmod N\}$, acting on the upper half-plane H by fractional linear transformations.

Write $Y(N)=\Gamma(N)\backslash\mathbb{H}$, using the image of $\Gamma(N)$ in $PSL_2(\mathbb{R})$ when discussing effective actions and indices.

Γ(3) has a free action on H and its quotient is a smooth noncompact curve

For $\gamma=I+3A\in\Gamma(3)$, determinant one implies $\operatorname{Tr}\gamma=2-9\det A$. Consequently the possible elliptic integer traces are excluded except the identity/unipotent trace two case. $\Gamma(3)$ is neat in its defining two-dimensional representation and acts freely on $\mathbb{H}$.

Derivation or proof

The identity $\det(I+3A)=1+3\operatorname{Tr}A+9\det A$ yields $\operatorname{Tr}A=-3\det A$. A real SL_2 matrix fixing a point of $\mathbb{H}$ is elliptic; an integer elliptic trace lies in $\{-1,0,1\}$, apart from the central finite-order endpoints. None is congruent to two modulo nine, and $-I$ has trace -2 and is absent. Nonidentity trace-two unipotents have no interior fixed point. Hyperbolic eigenvalues λ,λ^{-1} have $|\lambda|\neq1$, so their generated subgroup has no nontrivial torsion; unipotents contribute only eigenvalue one. Proper discontinuity and absence of stabilizers make the quotient smooth. This argument is specific and does not replace the general theorem that arithmetic groups contain neat finite-index subgroups.

Worked example

Compute $\Gamma(3)$'s effective index and cusp data.

$SL_2(F_3)$ has 24 elements, so the effective PSL_2 index is twelve. There are four cusps, each of width three. The torsion-free genus relation $2g-2+c=\mu/6$ then gives $g=0$ for the smooth compactification $X(3)$. The verifier directly enumerates determinant-one residue matrices.

The open curve is smooth and hyperbolic, yet its compactification is genus zero. These properties are compatible.

Scope of the result

Torsion-free and neat are distinct notions in general linear groups; their coincidence in this rank-one calculation must not be generalized indiscriminately. Smoothness of the open quotient alone does not determine the geometry of a chosen higher-dimensional boundary.

Further development

For a specified higher-rank group, compare level conditions that guarantee neatness with cone decompositions that guarantee regular toroidal charts. These are two separate certificates.

References: R028

M028 Minimal and toroidal compactifications in an explicit chart

2023-05-11 · historical evidence U · Curve case constructed; higher-dimensional distinction stated

In dimension one the basic boundary operation is tangible: a punctured disk acquires its missing point. Higher-dimensional toroidal compactification adds combinatorial choices that this curve example does not see.

Definitions and assumptions

A cusp of width N has local translation $\tau\mapsto\tau+N$. The invariant coordinate is $q_N=\exp(2\pi i\tau/N)$.

For $\mathrm{Im}\tau$ sufficiently large, the cusp quotient is a punctured disk $0<|q_N|<\varepsilon$. Adding $q_N=0$ gives a smooth disk.

For an arithmetic quotient of a Hermitian symmetric domain, the Baily-Borel construction supplies a canonical projective completion. Toroidal constructions use admissible rational cone decompositions near boundary components.

The rank-one cusp fills smoothly, but a fan can create or resolve toric singularities

The smooth compactification of a modular curve fills each cusp with one point; the normal curve completion is unique. In higher dimensions, a toroidal boundary chart is smooth only when its relevant cones are regular in the chosen lattice. Projectivity requires a compatible support-function condition rather than following from any arbitrary subdivision.

Derivation or proof

The q_N coordinate is unchanged by $\tau\mapsto\tau+N$ and maps a horizontal fundamental strip at large imaginary height bijectively to a punctured disk. The puncture can be filled by the standard complex coordinate. In one complex dimension normality already implies smoothness, which explains why the minimal/toroidal distinction is muted for curves. To see a genuinely higher-dimensional local issue, take a two-dimensional cone generated by $(1,0),(1,2)$. Its primitive generators have determinant two, so the affine toric chart is singular. Inserting the ray $(1,1)$ gives two determinant-one cones and hence smooth charts. This is a local toric illustration, not an assertion that this fan is the fan of a particular unstated arithmetic quotient.

Worked example

For $Y(3)$, fill its four cusps; separately perform the determinant-two cone subdivision.

The curve becomes $X(3)$, a smooth projective genus-zero curve. The fan computation changes a singular affine chart into a smooth toric modification. The two calculations demonstrate two distinct roles for boundary construction.

Canonical completion, smoothness and projectivity must be checked individually; they are not synonyms for making a space richer.

Scope of the result

The full Baily-Borel and toroidal existence theorems are cited classical results. The local calculations do not prove their general gluing, arithmetic descent or projectivity statements.

Further development

Specify a higher-dimensional locally symmetric variety and an admissible fan. Compute stabilizers, lattice regularity and support functions, then compare the resulting boundary strata and contraction to the minimal compactification.

References: R029

M029 General type and log-general type on modular curves

2023-05-11 · historical evidence U · Curve family calculated; higher-dimensional problem bounded

The canonical divisor of a compactification and the log-canonical divisor of its punctured interior answer different questions. Principal modular curves make the difference explicit without appealing to vague increases in structure.

Definitions and assumptions

For a smooth projective curve X of genus g, deg $K_X=2g-2$. It is of general type exactly when $g\geq 2$.

For a reduced divisor D of c punctures, $\deg(K_X+D)=2g-2+c$. The open curve is of log-general type exactly when this is positive.

For $N\geq 3$, the effective principal congruence group has index $\mu_N=|SL_2(Z/NZ)|/2$, all cusp widths N, and $c_N=\mu_N/N$. There are no elliptic points.

A smooth hyperbolic arithmetic interior need not have a general-type compactification

For X(3), $\mu=12$, $c=4$ and $g=0$. Thus $\deg K=-2$ but $\deg(K+D)=2$. For X(7), $\mu=168$, $c=24$ and $g=3$, so $\deg K=4$ and $\deg(K+D)=28$. Both interiors are log-general type, but only the latter compact curve is general type.

Derivation or proof

A torsion-free index-μ quotient of the upper half-plane has hyperbolic area $\mu\pi/3$, because $PSL_2(Z)$ has area $\pi/3$. Gauss-Bonnet on the surface with c cusps gives area $2\pi(2g-2+c)$, so $2g-2+c=\mu/6$. Since every cusp has width N, the cusp-width sum gives $c=\mu/N$. These equations calculate g. On a compact curve, a divisor is big exactly when its degree is positive: Riemann-Roch makes its large multiples grow linearly with positive leading coefficient, whereas nonpositive degree cannot have that growth. Applying this to K and K+D proves the distinction. The verifier directly counts determinant-one matrices modulo N for $N=3,\ldots,7$ and recovers genera 0,0,0,1,3.

Worked example

Use levels three and seven as contrasting cases and level six as a boundary case.

At $N=6$ the effective index is 72 and there are twelve cusps, giving genus one. Its canonical degree is zero while its log-canonical degree is twelve. It lies between the genus-zero and general-type compact examples.

A precise level-dependent calculation replaces the unsupported inference that smoothness or hyperbolicity already implies a big canonical divisor.

Scope of the result

The genus formula uses the classical torsion-free modular-curve geometry. It is not a proof of Tai-type higher-dimensional general-type theorems. Ramification, boundary discrepancy and compactification choice require explicit treatment outside this curve setting.

Further development

For a specified higher-dimensional congruence quotient, compute a canonical-divisor formula, identify boundary and ramification contributions and prove positivity at declared levels. Keep ordinary and logarithmic canonical divisors in separate columns.

References: R028, R030

M030 A projective embedding from a degree bound

2023-05-11 · historical evidence U · Curve criterion proved from Riemann-Roch

Projectivity can be demonstrated by producing enough sections of a line bundle to separate points and tangent directions. The curve case gives a concrete replacement for a list of named geometric structures.

Definitions and assumptions

Let X be a smooth projective curve over C of genus g and L a line bundle of degree d.

A line bundle is very ample when its complete linear system gives a closed embedding into projective space.

Riemann-Roch says $h^0(L)-h^0(K_X\otimes L^{-1})=d+1-g$. A line bundle of negative degree has no nonzero section.

Degree at least 2g+1 guarantees a projective embedding

> If $d\geq 2g+1$, then $h^0(L)=d+1-g$ and L separates points and tangent vectors. Hence its complete linear system embeds X into P^(d−g).

Derivation or proof

For any length-two subscheme Z of X, $\deg(K_X\otimes L^{-1}(Z))=2g-2-d+2\leq -1$. Serre duality therefore gives $H^1(L(-Z))=0$. The exact sequence $0\to L(-Z)\to L\to L|_Z\to 0$ shows that restriction of sections to Z is onto. Taking two distinct points gives separation of their values, and taking a double point gives tangent separation. The same argument for a length-one subscheme gives base-point freeness. These are the embedding criteria for the complete linear system. The section count follows from the vanishing of $H^1(L)$ and Riemann-Roch. This proof states exactly which positivity replaces an unsupported claim of projectivity.

Worked example

Take a genus-one curve and a line bundle of degree three.

The bound is met, $h^0(L)=3$, and the resulting embedding is into P^2 with degree three: a smooth plane cubic. For g=3 and d=7, there are five sections and an embedding into P^4. These are actual dimensions of spaces of sections, not a metaphorical richness measure.

A concrete line bundle can turn an abstract curve into an explicit projective object. Algebraic descriptions and projective embeddings are nevertheless distinct notions.

Scope of the result

The argument begins with a smooth projective curve and does not prove that every analytic space is algebraizable. An algebraic affine line already shows that being algebraic need not mean being projective. Higher-dimensional embedding criteria require their own positivity and vanishing results.

Further development

For a particular analytic compactification, identify a candidate line bundle, prove its positivity, obtain the required sections and show that their map is injective and immersive. State separately any algebraization theorem used to enter the algebraic category.

References: R031

M031 Adjunction fixes the quintic dimension

2023-05-13 · historical evidence A · Canonical bundle and characteristic classes computed

The phrase two-dimensional Calabi-Yau quintic was not a sufficiently defined geometric object. Adjunction determines which degree and ambient dimension actually produce trivial canonical bundle.

Definitions and assumptions

Let X_d be a smooth degree-d hypersurface in complex projective n-space. Its complex dimension is $n-1$.

Write $H=c_1(O_X(1))$; the canonical bundle of P^n is $O(-n-1)$.

The conormal sequence is $0\to O_X(-d)\to \Omega^1_{P^n}|_X\to \Omega^1_X\to 0$. Smoothness is essential to this vector-bundle form.

Adjunction and the smooth quintic calculation

> Taking determinants gives $K_X=O_X(d-n-1)$. A smooth quintic in P^4 has complex dimension three, real dimension six and trivial canonical bundle. A smooth plane quintic instead has genus six. A smooth quartic in P^3 is the familiar surface-degree case.

Derivation or proof

The determinant of the conormal sequence is $K_{P^n}|_X=O_X(-d)\otimes K_X$, yielding the displayed formula. For a plane curve, taking degrees gives $2g-2=d(d-3)$; $d=5$ yields $g=6$. For a smooth quintic threefold, the normal tangent sequence gives $c(TX)=(1+H)^5/(1+5H)$. Expand through degree three: $c_1=0$, $c_2=10H^2$ and $c_3=-40H^3$. Since $\int_X H^3=5$, its topological Euler characteristic is $\int c_3=-200$. Each step depends on the stated smooth hypersurface geometry rather than a guessed number of spatial dimensions.

Worked example

Compare degree-five hypersurfaces in P^2 and P^4.

The first is a one-complex-dimensional genus-six curve with $K=O(2)$. The second is a three-complex-dimensional hypersurface with $K=O$. The verifier independently expands the Chern polynomial and recovers coefficients $1,0,10,-40$ and Euler characteristic -200.

The ambient projective dimension and degree determine the canonical twist; the word quintic alone determines neither a surface nor a Calabi-Yau manifold.

Scope of the result

A trivial canonical bundle is not the entire definition of Calabi-Yau under every convention. Additional cohomological or holonomy conditions must be stated. The Chern-class calculation supplies no cryptographic hardness claim and no enumeration of all Calabi-Yau geometries.

Further development

For a smooth complete intersection of multidegree $(d_1,\ldots,d_r)$ in P^n, repeat adjunction to obtain $K=O(\Sigma d_i-n-1)$. Compute its dimension $n-r$ and characteristic classes before proposing algorithms on its coordinate ring or cohomology.

References: R032, R033

M032 Formal inversion has a public coefficient algorithm

2023-05-13 · historical evidence A · Algorithm and precision-doubling identity proved

Formal reciprocal coefficients are determined recursively. Inversion of a known truncated unit power series is therefore an available algorithm, not an unexplained source of computational hardness.

Definitions and assumptions

Work in $R[[q]]$ for a commutative ring R with identity. Let $F=1+\Sigma_{n\geq 1}a_nq^n$.

A unit series has an invertible constant coefficient. The normalization one avoids writing a_0^{-1} in every formula.

A congruence modulo q^m means equality of all coefficients below degree m; no analytic convergence is required.

A unique reciprocal and a doubling step

There is a unique $G=\Sigma_{n\geq 0}b_nq^n$ with $FG=1$, where $b_0=1$ and $b_n=-\Sigma_{i=1}^n a_i b_{n-i}$. If $FG\equiv 1 \bmod q^m$, then $G_{new}=G(2-FG)$ satisfies $FG_{new}\equiv 1 \bmod q^{2m}$.

Derivation or proof

Equating the degree-n coefficient of FG with zero isolates b_n because F has constant term one. This gives existence by recursive construction and uniqueness degree by degree. For doubling, write $E=1-FG$. Then $1-FG(2-FG)=(1-FG)^2=E^2$. If E has order at least m, its square has order at least $2m$. The update is valid in any characteristic because it is a polynomial identity; it does not require division by two. Naive coefficient multiplication takes quadratic ring-operation count in the truncation length, while fast multiplication and doubling improve that bound when the coefficient ring admits efficient arithmetic.

Worked example

Take $F=1-q-q^2$ over Z.

The recurrence is b_n=b_{n−1}+b_{n−2}, with $b_0=b_1=1$. The inverse begins $1+q+2q^2+3q^3+5q^4+8q^5+13q^6$. The verifier checks the convolution identity exactly through degree twelve.

A recipient knowing F can recover the same truncated reciprocal coefficients by a transparent deterministic calculation. A purported one-way map based solely on hiding this inversion would fail its first baseline test.

Scope of the result

This does not prove every cryptographic proposal involving formal series insecure. Some other specified hidden-data or inversion problem could be difficult. Bit complexity also depends on coefficient growth and cannot be read off solely from the number of ring operations.

Further development

For a concrete proposal, identify exactly what the adversary knows and which operation is claimed hard. Compare the intended task with coefficient recurrence, Newton inversion, factorization and reconstruction from observed coefficients before making any security assertion.

References: R013

M033 The A_2 Weyl denominator expanded exactly

2023-05-13 · historical evidence A · Classical identity verified in a complete low-rank case

A root system and group algebra turn the denominator slogan into a finite cancellation identity. The A_2 calculation exposes all six terms and the role of signs.

Definitions and assumptions

Choose simple roots α,β of A_2; positive roots are α,β,α+β and ρ=α+β.

The Weyl group is generated by s_α(a,b)=(−a+b,b) and s_β(a,b)=(a,a−b) on root coordinates.

Set x=e^(−α), y=e^(−β) in the group algebra. The denominator identity is Σ_w det(w)e^(wρ)=e^ρΠ_{γ>0}(1−e^(−γ)).

Both sides reduce to the same six monomials

> After division by e^ρ, the A_2 identity is $(1-x)(1-y)(1-xy)=1-x-y+x^2y+xy^2-x^2y^2$.

Derivation or proof

The orbit of ρ consists of (1,1),(0,1),(1,0),(−1,0),(0,−1),(−1,−1). Their reflection parities are respectively even, odd, odd, even, even and odd. Subtracting ρ in the exponents gives the displayed monomials and signs. Independently multiply the three factors: first 1−x−y+xy, then subtract xy times this expression. The two xy terms cancel, leaving precisely the orbit sum. This proves the full low-rank identity without relying on numerical evaluation. For a general root system, alternating behavior under simple reflections and the leading term characterize the denominator, but that argument requires a separately stated general theorem.

Worked example

Represent a bivariate polynomial by its integer coefficient dictionary.

The verifier multiplies all factors and obtains coefficients {(0,0):1,(1,0):−1,(0,1):−1,(2,1):1,(1,2):1,(2,2):−1}. It checks equality with the independently listed orbit-side dictionary.

The identity is a signed group-algebra statement. It is neither a generic randomizing device nor an unproved claim of algebraic complexity.

Scope of the result

The example does not prove the entire Weyl character formula or classify root systems. If exponentials are interpreted as analytic functions rather than formal symbols, their domain and specialization must be stated, especially at roots of unity or vanishing denominators.

Further development

Compute a character as a quotient of alternating sums in A_2 and show that apparent denominator singularities cancel to a genuine Laurent polynomial. Then test specialization and branching identities in exact arithmetic.

References: R034

M034 An affine A_1 identity with its normalization proved

2023-05-14 · historical evidence A · Classical product identity derived and coefficients checked

An explicit Jacobi triple product is a tractable affine-root-system identity. It is distinct from an unspecified identity involving Macdonald symmetric polynomials.

Definitions and assumptions

Write $(a;q)_\infty=\Pi_{m\geq 0}(1-aq^m)$. Analytically take $0<|q|<1$ and $z\neq 0$.

Formally work in q-series whose coefficients are finite Laurent polynomials in z. At every q-degree only finitely many bilateral summands contribute.

The identity is $\Sigma_{n\in Z}(-1)^n z^n q^{n(n-1)/2}=(z;q)_\infty(q/z;q)_\infty(q;q)_\infty$.

A functional equation fixes all Laurent coefficients up to one factor, and a partition identity fixes that factor

> The product satisfies $F(qz)=-z^{-1}F(z)$. Its Laurent coefficients therefore obey $c_{n+1}=-q^n c_n$. The constant coefficient is one, giving the entire bilateral series.

Derivation or proof

Shift the product indices: $(qz;q)_\infty=(z;q)_\infty/(1-z)$ and $(1/z;q)_\infty=(1-1/z)(q/z;q)_\infty$. Their ratio is $-1/z$, yielding the functional equation. To normalize, expand $(z;q)_\infty=\Sigma_{a\geq 0}(-1)^a q^{a(a-1)/2}z^a/(q;q)_a$; this expansion follows by its own product shift and constant term one. The analogous expansion for $(q/z;q)_\infty$ shows that the z^0 coefficient of F is $(q;q)_\infty\Sigma_{k\geq 0}q^{k^2}/(q;q)_k^2$. Every integer partition decomposes uniquely into its maximal $k\times k$ Durfee square, the cells to its right and the cells below. Those two remaining partitions have at most k parts, giving generating series $1/(q;q)_k^2$. Thus the sum is $1/(q;q)_\infty$ and the normalizing coefficient is exactly one.

Worked example

Truncate the product at q-degree twelve while retaining exact Laurent powers of z.

The verifier obtains exactly the bilateral terms with $n(n-1)/2 \le 12$, including positive and negative n. All other product coefficients cancel. The proof explains why agreement persists beyond the finite truncation.

The ring, normalization and combinatorial mechanism are explicit. This supplies a real model for revisiting a named affine identity.

Scope of the result

This is a classical identity, not a new Macdonald formula. It does not supply identities for every $P_\lambda(x;q,t)$, and analytic manipulations beyond the stated convergence region need justification.

Further development

Choose a higher-rank affine root system and specify its product and lattice sum. Use a low-rank proof, exact coefficient checks and a clear convergence/completion convention before claiming a general transport principle.

References: R035

M035 Young tableaux counted by recursion and hooks

2023-05-14 · historical evidence A · Small instance independently counted

The shape of a Young diagram is a combinatorial input, and its statistic must be declared. Counting standard tableaux and evaluating a two-parameter symmetric polynomial are different tasks.

Definitions and assumptions

A partition λ of n is a weakly decreasing list of row lengths summing to n.

A standard Young tableau fills its cells with $1,\dots,n$, increasing along every row and column.

The hook length of a cell is one plus the number of cells to its right and below. The classical hook-length theorem gives $f^\lambda = n!/\Pi_c\, h(c)$.

The corner-removal recurrence gives a separate exact count

> Let $f(\varnothing)=1$. Then $f^\lambda = \Sigma_{c \text{ removable corner}} f^{(\lambda \setminus c)}$. For $\lambda=(3,2)$, this count is five and matches the hook-length formula.

Derivation or proof

In any standard tableau, its largest entry n cannot have a cell to its right or below, so it occupies a removable corner. Removing it gives a unique standard tableau of the smaller shape. Conversely, insert n into any chosen removable corner of any valid smaller tableau. These inverse operations prove the recurrence without using the hook-length formula. For (3,2), the two smaller shapes are (2,2) and (3,1), with counts two and three; hence five. The hook lengths are 4,3,1,2,1 and their product is 24, so the independent closed formula gives $5!/24=5$. The general hook-length theorem is quoted classical background; the recurrence alone is a full independent proof of this finite count.

Worked example

Count the standard tableaux of shape (3,2) by both methods.

The verifier recursively removes corners with memoization, computes each hook directly from the diagram and checks that both answers are five. It does not infer the count from a floating approximation or from a representation-dimension claim.

The integer five has a definite interpretation: it counts exactly these tableaux and, by the relevant representation theorem, is the dimension of the corresponding irreducible S_5 module over characteristic zero.

Scope of the result

Semistandard tableaux, Schur-function evaluations, modular representations and Macdonald-polynomial specializations require different weights or hypotheses. The same diagram alone does not make their answers equal.

Further development

Compute a weighted generating polynomial on the same shape, declare the statistic being weighted and compare its specialization at one with the unweighted count. Then separate a proved representation-theoretic interpretation from a conjectural application.

References: R036

M036 A nonconstant Kazhdan-Lusztig calculation in S_4

2023-05-13 · historical evidence A · Classical recurrence implemented exactly

Coxeter data and normalization determine the polynomial. The smallest useful nonconstant example supplies an actual computation to replace claims that KL polynomials are universal complexity scores.

Definitions and assumptions

Let $W=S_4$ with simple generators s_i swapping adjacent positions and length $\ell(w)$ equal to the inversion count.

Use the Hecke convention $(T_s-q)(T_s+1)=0$ and the usual polynomials $P_{u,v}(q)$, with $P_{u,u}=1$, $P_{u,v}=0$ unless $u\le v$ in Bruhat order, and degree at most $(\ell(v)-\ell(u)-1)/2$ for $u<v$.

For permutations, $u\le v$ exactly when the sorted initial k entries of u are componentwise no larger than those of v for every k.

Two singular Schubert examples have P_{e,v}=1+q

The exact right-descent recurrence gives $P_{e,3412}(q)=1+q$ and $P_{e,4231}(q)=1+q$. In contrast $P_{u,4321}=1$ for every $u\in S_4$ in the same normalization.

Derivation or proof

Choose a right descent s of v and let c be one if us<u and zero otherwise. The recurrence is P_{u,v}=q^(1−c)P_{us,vs}+q^cP_{u,vs}−Σ_{u≤z<vs,zs<z} μ(z,vs)q^{(ℓ(v)−ℓ(z))/2}P_{u,z}. Here μ(z,vs) is the coefficient of q^{(ℓ(vs)−ℓ(z)−1)/2}, and is zero when that exponent is not an integer. Every polynomial on the right has a strictly shorter upper permutation, so the recurrence terminates. The verifier implements integer polynomial arithmetic, generates permutations, tests Bruhat order from the prefix criterion and evaluates the displayed examples. It also checks constant terms, degree bounds and coefficient nonnegativity for every comparable pair in S_4. These consistency checks support the implementation but do not replace the mathematical theorem defining the recurrence.

Worked example

Compute the interval from the identity to 3412, whose length is four.

The allowed degree is at most one and the resulting coefficient tuple is (1,1). The identity-to-4231 calculation agrees. The full S_4 pass tests 213 comparable pairs and the longest-element smooth case.

The output is a polynomial indexed by a specific ordered pair in a specified Coxeter group. Its geometric meaning concerns local intersection-cohomology data under the relevant theorem, not generic algorithmic difficulty.

Scope of the result

The finite computation proves no unprovided relation to error-correcting codes, traveling-salesperson optimization or cryptographic hardness. Different Hecke conventions can reparameterize the same canonical data.

Further development

State an explicit map from a proposed application into Bruhat intervals and prove that the statistic of interest is preserved. Without that map, keep the KL calculation as representation-theoretic mathematics in its own right.

References: R037

M037 An independently replayable BCH certificate

2023-05 · historical evidence A · Specified code completely enumerated

The parameters are not merely a label: a generator polynomial, field convention and finite enumeration identify the exact code and establish the correction guarantee.

Definitions and assumptions

Work over F_2 and take α as a root of x^4+x+1 in F_{16}. The verifier checks that α has multiplicative order fifteen.

Let $g(x)=(x^4+x+1)(x^4+x^3+x^2+x+1)=x^8+x^7+x^6+x^4+1$.

The cyclic code C consists of m(x)g(x) for deg m<7, represented as length-fifteen binary words. Hamming weight counts nonzero coordinates.

This standard narrow-sense BCH construction has parameters [15,7,5] and corrects any two errors

The generator divides $x^{15}-1$ and vanishes at $\alpha,\alpha^2,\alpha^3,\alpha^4$. There are exactly 128 codewords and the least nonzero weight is five. Consequently distinct radius-two Hamming balls are disjoint.

Derivation or proof

The root checks identify the designed-distance-five BCH construction. Polynomial division verifies cyclic divisibility. Multiplication by the nonzero monic g is injective on the 128 degree-below-seven messages, and the resulting degree is below fifteen; this establishes length and dimension. For a linear code, the distance between two codewords is the weight of their difference, another codeword, so minimum distance equals minimum nonzero weight. Exhaustive enumeration establishes that minimum as five. If a received word were within distance two of two codewords, the triangle inequality would put those codewords within distance four, a contradiction. Thus nearest-codeword decoding succeeds for every error pattern of weight at most two.

Worked example

Enumerate all messages and all error patterns of weights zero, one and two.

The exact weight enumerator is $1+18z^5+30z^6+15z^7+15z^8+30z^9+18z^{10}+z^{15}$. There are $1+15+C(15,2)=121$ allowed error patterns per codeword, and the verifier obtains $128\times121=15{,}488$ distinct received words in the union of the decoding balls.

The earlier one-error description understated the guarantee of this specified construction. The result is exhaustive for this finite code, including a reproducible generator and decoder-radius certificate.

Scope of the result

The code is not perfect: the 15,488 radius-two words do not exhaust all 32,768 binary words. The certificate says nothing about guaranteed correction of every three-error pattern. Other codes with the same length and dimension need not share the minimum distance.

Further development

Publish syndrome calculation and a bounded-distance decoder, then compare decoder output against exhaustive nearest-codeword results on all received words. Treat larger parameters with algebraic decoding proofs rather than assuming enumeration remains cheap.

References: R038

C Representations geometric phases and dynamical models

M038 A finite Peter-Weyl calculation with an exact truncation error

2023-05-16 · historical evidence U · Classical framework with worked cases proved here

The useful question is which observables are retained by a representation cutoff. An expansion becomes operational once the group, normalized Haar measure and error norm are fixed. Both the circle and a noncommutative compact group admit short examples with an exactly known discarded component.

Definitions and assumptions

On U(1), write $z=e^{i\theta}$ and use normalized measure $d\theta/(2\pi)$. The irreducible characters are z^n, $n\in\mathbb{Z}$.

On SU(2), conjugacy classes have eigenvalues $e^{i\theta},e^{-i\theta}$, $\theta\in[0,\pi]$. For class functions, normalized Haar integration is $(2/\pi)\int_0^{\pi} f(\theta)\sin^2\theta\, d\theta$.

A finite Peter-Weyl calculation with an exact truncation error

For $f(\theta)=3+2\cos\theta-\sin 2\theta$ on U(1), projection to frequencies $|n|\le 1$ has squared L^2 error 1/2. On SU(2), the square of the defining character decomposes as $\chi_{1/2}^2=\chi_0+\chi_1$, and retaining only χ_0 has squared error 1.

Derivation or proof

Orthogonality follows from integrating $e^{i(n-m)\theta}$: the integral is zero unless $n=m$. Here the Fourier coefficients are $c_0=3$, $c_1=c_{-1}=1$, $c_2=i/2$ and $c_{-2}=-i/2$. The discarded orthogonal terms therefore contribute $|i/2|^2+|-i/2|^2=1/2$. For SU(2), $\chi_{1/2}=2\cos\theta$ and $\chi_1=4\cos^2\theta-1$. Direct integration with the stated Haar density gives $\int\chi_1=0$ and $\int\chi_1^2=1$, proving both the decomposition and the error claim without appealing to numerical fitting.

Worked example

Take the SU(2) class observable $f(U)=(\operatorname{Tr} U)^2$. This is a function of the group element, not merely an arbitrary two-by-two mixing matrix.

The tensor product of two defining representations has characters $\chi_{1/2}^2$. Its invariant antisymmetric line gives $\chi_0=1$; its symmetric three-dimensional summand gives χ_1. Thus $f=1+\chi_1$, $\int f=1$, $\int|f|^2=2$. Projection to constants has residual χ_1 with norm one, while retaining the spin-one character makes the residual exactly zero.

A small representation cutoff can have a completely explicit observable loss. The calculation also shows why the group measure and the choice of function matter.

Scope of the result

This proves finite examples, not the full Peter-Weyl completeness theorem or a universal truncation rate. General noncentral functions require matrix coefficients rather than characters alone.

Further development

For a chosen smoothness norm on a compact group, derive a representation-tail estimate using eigenvalues of its Casimir operator; compare that estimate with the exact finite examples.

References: R039

M039 Rephasing invariance and a sharp three-flavor CP bound

2023-05-16 · historical evidence U · Classical formula with an explicit sharpness derivation

Complex entries are not themselves physical phase invariants: row and column conventions can change them. A four-entry product cancels those conventions, and in three dimensions its imaginary part admits a sharp elementary bound.

Definitions and assumptions

Let U be a three-by-three unitary mixing matrix. Row and column rephasings act by $U_{\alpha i} \mapsto e^{i r_\alpha} U_{\alpha i} e^{i c_i}$.

For distinct rows and columns, define $J_{\alpha\beta ij} = \mathrm{Im}(U_{\alpha i} U_{\beta j} U^*_{\alpha j} U^*_{\beta i})$. The sign depends on the ordered indices.

Rephasing invariance and a sharp three-flavor CP bound

> Each $J_{\alpha\beta ij}$ is invariant under these rephasings. In the usual three-angle one-Dirac-phase parameterization, $|J| \le 1/(6\sqrt{3})$, with equality attained by the normalized three-by-three discrete Fourier matrix.

Derivation or proof

The phases in the four factors cancel pairwise: $r_\alpha + r_\beta - r_\alpha - r_\beta$ and $c_i + c_j - c_j - c_i$ both vanish. In the standard parameterization, $J = c_{12} s_{12} c_{23} s_{23} c_{13}^2 s_{13} \sin\delta$. The first two sine-cosine pairs have absolute value at most 1/2. With $t = s_{13} \in [0,1]$, the remaining factor $t(1-t^2)$ has derivative $1-3t^2$ and maximum $2/(3\sqrt{3})$. Multiplication gives the bound. The Fourier example below attains it, so the bound is sharp rather than merely convenient.

Worked example

Set $\omega = \exp(2\pi i/3)$ and $U_{ab} = \omega^{ab}/\sqrt{3}$ for $a,b = 0,1,2$.

Geometric-series cancellation verifies $U^\dagger U = I$. The ordered quartet $U_{00} U_{11} U^*_{01} U^*_{10}$ equals $\omega/9$, whose imaginary part is $\sqrt{3}/18 = 1/(6\sqrt{3})$. Complex conjugation changes its sign. In vacuum, an oscillation amplitude can be written $A_{\alpha\beta} = \Sigma_i U_{\beta i} U^*_{\alpha i} e^{-i\varphi_i}$. Squaring shows directly that $P(U) - P(U^*) = 4\Sigma_{i>j} \mathrm{Im}(Z_{ij}) \sin(\varphi_i - \varphi_j)$, where $Z_{ij} = U_{\beta i} U^*_{\alpha i} U^*_{\beta j} U_{\alpha j}$.

A nonzero invariant is a mathematically meaningful obstruction to making the relevant matrix real by phase conventions; actual probability asymmetries also require nontrivial propagation phases.

Scope of the result

The calculation assumes the stated unitary three-flavor framework and says nothing about the experimentally fitted value of δ. Degenerate masses and inaccessible channels can suppress an observable asymmetry despite a formal matrix invariant. Column rephasings here concern oscillation equivalence; a Majorana mass term need not permit the same field rephasings in the full theory.

Further development

Use the invariant quartets to certify equality or inequivalence of numerically supplied mixing matrices under row and column rephasing, with rigorous tolerances.

References: R040

M040 A complete small persistence calculation

2023-05-22 · historical evidence A · Finite homology and barcode computed exactly

Topology becomes a calculation only after a space and a filtration are specified. Three vertices provide a minimal example in which components merge, a loop is born and a filling destroys it. Every rank can be checked over the two-element field.

Definitions and assumptions

Let K_0 contain vertices 0,1,2; add edge 01 at time 1, edge 12 at time 2, edge 20 at time 3 and face 012 at time 4.

Use homology over F_2. For a finite chain complex, $\beta_k = \dim C_k - \operatorname{rank}\partial_k - \operatorname{rank}\partial_{k+1}$.

A complete small persistence calculation

> Across times 0,1,2,3,4, the pairs (β_0,β_1) are (3,0),(2,0),(1,0),(1,1),(1,0). The persistent barcode is H_0:[0,1),[0,2),[0,∞), and H_1:[3,4).

Derivation or proof

The first two added edges each join different components and increase rank ∂_1 by one, from zero to two. The third edge has boundary equal to the sum of the first two boundaries, so rank ∂_1 stays two while $\dim C_1$ becomes three; one cycle is born. The face boundary is 01+12+20, which is precisely the unique nonzero cycle up to scalar. Its addition raises rank ∂_2 from zero to one and kills H_1. The component survival convention can select any representative among equally old vertices, but the interval multiset does not depend on that tie-breaking.

Worked example

Compare the boundary of a triangle with the filled triangle on the same three vertices.

Both spaces have $\beta_0=1$. The boundary has $C_2=0$ and $\beta_1=3-2=1$. The filled triangle has one face with nonzero boundary and $\beta_1=3-2-1=0$. At time 4, the loop has not shrunk by geometric deformation; it becomes a boundary because the available chain complex changed.

A Betti count is an invariant of the constructed space, while a barcode additionally records when a class appears and disappears in the chosen filtration.

Scope of the result

Different constructions from the same observations can yield different filtrations. Equal Betti numbers do not imply homeomorphism or homotopy equivalence. No universal data-to-topology interpretation is supplied by this example.

Further development

Perturb edge and face entry times and verify which persistence intervals remain long; then compare this explicit behavior with an appropriate stability theorem.

References: R041

M041 A two-level Berry holonomy with a gauge check

2023-05-22 · historical evidence A · Classical holonomy evaluated explicitly

The geometric object is an eigenline over a parameter loop. Choosing an eigenvector gives a connection formula, but the final phase must survive a change of that choice. A latitude on the Bloch sphere makes the distinction fully visible.

Definitions and assumptions

Take $H=n\cdot\sigma$ with $n=(\sin\theta\cos\varphi,\sin\theta\sin\varphi,\cos\theta)$, fixed $\theta\in(0,\pi)$, and φ from 0 to 2π.

Choose the normalized positive-energy eigenvector $v(\varphi)=(\cos(\theta/2),e^{i\varphi}\sin(\theta/2))$. Define $A\varphi=i\langle v,\partial_\varphi v\rangle$ and $\gamma=\oint A\varphi d\varphi$ modulo 2π.

A two-level Berry holonomy with a gauge check

> The positive eigenline has $\gamma=-\pi(1-\cos\theta)$. Under a periodic eigenvector gauge change, $\exp(i\gamma)$ is unchanged. At $\theta=\pi/3$ the holonomy is $-i$; at the equator it is -1.

Derivation or proof

Differentiation gives $\langle v,\partial_\varphi v\rangle=i\sin^2(\theta/2)$, hence $A\varphi=-\sin^2(\theta/2)$. Integrating yields $-2\pi\sin^2(\theta/2)=-\pi(1-\cos\theta)$. If $v'=e^{i\chi(\varphi)}v$, then $A'=A-\chi'$. A single-valued gauge around the loop has $\chi(2\pi)-\chi(0)$ an integer multiple of 2π, so $\exp(i\gamma')=\exp(i\gamma)$. This is a direct calculation of holonomy, independently of any physical interpretation of an adiabatic protocol.

Worked example

Sample the latitude $\theta=\pi/3$ at N equally spaced phases, with the final point identified with the first.

Form the normalized cyclic product $\Pi_k\langle v(\varphi_{k+1}),v(\varphi_k)\rangle/|\langle v(\varphi_{k+1}),v(\varphi_k)\rangle|$. Its phase tends to γ as N grows; reversing the overlap order conjugates the answer. Multiplying each sampled eigenvector by an arbitrary phase leaves the closed product unchanged because consecutive phase factors telescope. The verification program checks both convergence and exact cancellation of sampled gauge factors.

The loop phase is meaningful even though individual eigenvector phases and connection values are convention dependent.

Scope of the result

The Hamiltonian stays gapped on this unit sphere. A loop crossing a degeneracy does not admit this isolated-eigenline calculation, and adiabatic dynamical error requires a separate gap-and-speed estimate.

Further development

Replace the eigenline by a degenerate eigenspace and study the resulting matrix-valued parallel transport; retain projectors and overlap matrices rather than scalar phases.

References: R042

M042 Crossing symmetry and an exactly opened spectral gap

2023-05-22 · historical evidence A · Two-by-two spectral theorem proved directly

A level crossing and an avoided crossing can be separated by a single parameter that either preserves or couples two invariant sectors. This gives an exact test of what a proposed symmetry protects.

Definitions and assumptions

Let $H(t,\varepsilon)=[[t,\varepsilon],[\varepsilon,-t]]$ with real t and ε.

Write $r=\sqrt{(t^2+\varepsilon^2)}$. For $r>0$, set $P_\pm=(I\pm H/r)/2$.

Crossing symmetry and an exactly opened spectral gap

> The eigenvalues are $\pm r$, the gap is $2r$, and its minimum over t is $2|\varepsilon|$. At $\varepsilon=0$, the diagonal sectors are uncoupled and the levels meet at $t=0$; $\varepsilon\neq 0$ removes that crossing.

Derivation or proof

Direct multiplication gives $H^2=(t^2+\varepsilon^2)I$. Therefore its minimal polynomial divides λ^2-r^2 and, for $r>0$, $P_\pm^2=P_\pm$, $P_+P_-=0$, $P_++P_-=I$ and $HP_\pm=\pm rP_\pm$. The spectral gap formula follows. For $\varepsilon=0$, H commutes with σ_z, so each coordinate axis is an invariant symmetry sector. The $\varepsilon\sigma_x$ perturbation does not commute with σ_z and permits those sectors to mix. Thus protection is a statement about an operator relation, not a verbal appeal to symmetry.

Worked example

Set $\varepsilon=1/5$ and let t run through $-1,0,1$.

The gaps are $2\sqrt{26}/5$, $2/5$, $2\sqrt{26}/5$. The trace remains zero and the determinant is $-(t^2+1/25)$, so an eigenvalue cannot pass through zero at any real t. At $\varepsilon=0$, the determinant at $t=0$ vanishes and both eigenvalues coincide. The projector formulas explicitly cease to apply at that degenerate point because $r=0$.

A perturbation that changes a commutation relation can open a gap whose size is computed without approximation.

Scope of the result

This static spectral calculation does not determine transition probabilities during a time sweep. Those depend on a time law, an initial state and the dynamical equation; replacing the minimum gap by a guessed transition rate is unjustified.

Further development

For a larger Hermitian matrix with a known symmetry decomposition, classify which perturbation blocks couple crossing sectors and bound the resulting gaps.

References: R042

M043 A Moore spectrum as a precise chromatic example

2023-05-22 · historical evidence A · Elementary cofiber consequence with general theory bounded explicitly

Chromatic height refers to detection by homology theories at a fixed prime. It can already distinguish the sphere from a simple torsion object. This has no mathematical dependence on optical colors or on periodic measurements in an unrelated model.

Definitions and assumptions

Fix a prime p. Let M(p) be the cofiber of multiplication by p on the sphere spectrum: S→S→M(p).

Use K(0)=HQ. For n≥1, Morava K-theory has coefficient graded field K(n)*=F_p[v_n,v_n^{-1}], with |v_n|=2(p^{n}−1). A finite p-local spectrum has type t when K(n)* vanishes for n<t and is nonzero at n=t.

A Moore spectrum as a precise chromatic example

> The p-local sphere has type zero, while M(p) has type one. For each n≥1, the long exact sequence gives a short exact sequence 0→K(n)_q→K(n)_qM(p)→K(n)_(q−1)→0.

Derivation or proof

For rational homology, multiplication by p is invertible, so the cofiber has zero rational homology. For K(n), n≥1, multiplication by p is zero because the coefficients have characteristic p. The cofiber long exact sequence therefore has zero connecting multiplication maps on both sides, yielding the displayed short exact sequence. It in particular contains a nonzero injected copy of K(n)*. Thus the first nonvanishing height is one. As a graded vector space over the coefficient graded field, the sequence can be split, though there is no need to claim a canonical splitting.

Worked example

Compare S, the cofiber of the identity map on S, and M(p).

The identity cofiber is contractible and is detected by none of these theories. The sphere has nonzero rational homology and hence height zero. M(p) loses rational information but retains positive-height information. This is an actual consequence of the maps defining the objects, rather than a pattern assigned from a plot.

The example illustrates why a prime, a spectrum and a chosen detection theory must precede a claim about chromatic periodicity.

Scope of the result

No computation of the stable homotopy groups of spheres or proof of the periodicity theorem is supplied here. Existence and powers of self-maps at a given height are deeper statements and are not inferred from the coefficient period alone.

Further development

Construct a specified finite type-two example from a known self-map and verify its K(0) and K(1) acyclicity using the relevant cofiber sequences and hypotheses.

References: R043

M044 A conservative positive diffusion stencil with a sharp local condition

2023-05-30 · historical evidence A · Discrete conservation positivity and energy estimate proved

A numerical update should preserve the quantities its PDE is meant to preserve. On a periodic grid, the standard heat stencil admits exact mass, positivity and squared-norm statements before any simulation is run.

Definitions and assumptions

Let $N \geq 3$ periodic grid values be indexed modulo N. Set $F_r(x)_i = r\,x_{i-1} + (1-2r)x_i + r\,x_{i+1}$.

For heat diffusion, $r = D\Delta t/\Delta x^2$, $D \geq 0$. The grid mass is $\Sigma_i\, x_i$; the squared discrete norm is $\Sigma_i\, x_i^2$.

A conservative positive diffusion stencil with a sharp local condition

> For every r, the update preserves mass. It maps every nonnegative grid to a nonnegative grid exactly when $0 \leq r \leq 1/2$. In that range it also contracts the squared norm.

Derivation or proof

Summing the update cyclically makes each old value appear with coefficient $r+(1-2r)+r=1$, proving mass conservation. Nonnegative coefficients prove positivity. Conversely, a single nonzero grid basis vector detects a negative off-diagonal coefficient when $r<0$ or a negative central coefficient when $r>1/2$. For $0 \leq r \leq 1/2$, convexity of z^2 gives $F_r(x)_i^2 \leq r\,x_{i-1}^2 + (1-2r)x_i^2 + r\,x_{i+1}^2$. Summing proves contraction. This proof is finite and exact, requiring no floating-point time stepping.

Worked example

Take N=5, r=1/4 and initial grid (1,0,0,0,0).

One step gives (1/2,1/4,0,0,1/4), preserving mass one and reducing the squared norm from one to 3/8. At r=3/4 the same initial condition produces a central value $-1/2$, immediately disproving positivity. Fourier modes $e^{(2\pi i k/N)}$ have amplification factors $1-4r\sin^2(\pi k/N)$, providing a second independent stability calculation.

The update is a well-defined averaging operator in the positive range. A nonlinear reaction term must be analyzed separately rather than multiplied into the diffusion formula without checking these properties.

Scope of the result

For odd N the sharp spectral ℓ^2 stability threshold can differ from the positivity threshold, so those two notions should not be conflated. Boundary conditions, variable diffusivity and reactions change the proof obligations.

Further development

Write variable-diffusivity terms as edge flux differences so mass conservation still telescopes, then derive nodewise time-step restrictions for positivity.

M045 Reflection equivariance as an operator identity

2023-05-30 · historical evidence A · Exact symmetry criterion proved for a stencil family

Symbols for identity and reflection are meaningful only after their action on states has been defined. Commutation makes the claimed symmetry testable and exposes the effect of asymmetric coefficients or boundaries.

Definitions and assumptions

On a periodic N-grid, define $(Rx)i=x_{(-i \bmod N)}$, so $R^2=I$. Let S be the cyclic shift.

Consider $F=aS^{-1}+bI+cS$, and a pointwise reaction $G(x)i=g(x_i)$. Reflection equivariance means $F(Rx)=RF(x)$ for every x.

Reflection equivariance as an operator identity

> For $N\geq 3$, the three-point linear stencil commutes with reflection exactly when $a=c$. Any pointwise reaction using the same scalar function at every node commutes with reflection and every grid permutation.

Derivation or proof

The identity $RSR=S^{-1}$ follows by applying both sides to an indexed coordinate. Conjugation gives $RFR=aS+bI+cS^{-1}$. For $N\geq 3$ the two shifts are distinct and linearly independent from each other in this coefficient comparison, so $RFR=F$ exactly when $a=c$. For the nonlinear term, $G(Rx)i=g(x_{(-i)})=(RG(x))i$, proving equivariance directly. Consequently the composition of reflection-equivariant diffusion and reaction updates remains equivariant, whereas an asymmetric advection component generally does not.

Worked example

Use $N=5$ and compare $F=(1/4)S^{-1}+(1/2)I+(1/4)S$ with $A=(1/3)S^{-1}+(2/3)S$.

On the basis vector at node zero, the symmetric stencil produces equal mass at the two neighboring nodes, so reflection leaves the output unchanged. The asymmetric stencil produces unequal neighboring weights and its reflected output differs. This single vector is a complete counterexample to the universal commutation claim. A spatially varying reaction g_i can fail similarly even when each g_i is smooth.

Equivariance is a property of a typed update and group action. Writing an identity morphism or reflection symbol as an unexplained multiplicative factor does not supply it.

Scope of the result

The periodic reflection used here is not automatically a symmetry of a grid with unequal endpoint conditions or asymmetric geometry. A symmetry of equations also need not be a symmetry of a particular initial condition.

Further development

Compute the commutant of a chosen finite symmetry group and parameterize all linear updates that preserve it; then impose positivity and conservation as additional linear constraints.

M046 Affine coordinate changes and time evolution are different operations

2023-06-05 · historical evidence A · Composition and transformed velocity derived

An affine transformation changes a description of points; an evolution law describes a path. Their interaction is simple for constant frames and adds extra terms for moving frames. Mixing these roles can create a fictitious physical conclusion.

Definitions and assumptions

An affine map is $T(x)=Mx+b$. It is a coordinate isomorphism when M is invertible.

A path $x(t)$ has velocity $\dot{x}(t)$. A time-dependent frame is $y(t)=M(t)x(t)+b(t)$.

Affine coordinate changes and time evolution are different operations

> Affine composition satisfies $(M_2,b_2)\circ(M_1,b_1)=(M_2M_1,M_2b_1+b_2)$. Under a moving frame, $\dot{y}=M\dot{x}+Mx+\dot{b}$, and $\ddot{y}=M\ddot{x}+2M\dot{x}+Mx+\ddot{b}$.

Derivation or proof

Substitute x into the second affine map and collect terms to obtain the composition law. Differentiate the frame relation using the product rule once and twice. The M and $\dot{M}$ terms vanish only for a fixed linear frame; $\dot{b}$ and $\ddot{b}$ vanish only for a fixed translation. Thus even a stationary world point may have nonzero coordinate velocity in a rotating or translating frame. This is a change in the description of motion, not a newly derived external force.

Worked example

Let R be rotation by 90 degrees in the plane and T translation by $(1,0)$. Also take a free path $x(t)=x_0+vt$.

RT sends the origin to $(0,1)$, whereas TR sends it to $(1,0)$, exhibiting noncommutativity. In a constant affine frame, the free path becomes $y(t)=Mx_0+b+(Mv)t$, still affine in time. For a rotating frame $M(t)=R(\omega t)$, a stationary $x=(1,0)$ is described by $y=(\cos\omega t,\sin\omega t)$, with acceleration $-\omega^2 y$. The derivative terms account for that coordinate acceleration exactly.

The same geometric transformation may be used in a dynamical model, but its time dependence and reference-frame meaning must be stated.

Scope of the result

An arbitrary invertible linear map need not preserve Euclidean lengths or kinetic-energy expressions. Those require transforming the metric as well. Singular affine maps are projections rather than reversible coordinate changes.

Further development

Carry a covariance or noise metric through the same coordinate change and verify which estimation errors are invariant; this prevents apparent gains caused solely by rescaling.

M047 Mutual information detects dependence rather than a causal channel

2023-06-08 · historical evidence A · Finite identities and data processing proved

A dependence measure needs a joint distribution. Its nonzero value does not distinguish direct interaction from a shared source, while its change under a specified noisy channel can be calculated exactly.

Definitions and assumptions

For finite variables, $I(X;Y)=\Sigma_{x,y}\, p_{xy}\log_2(p_{xy}/(p_x p_y))$; terms with zero mass contribute zero.

Let $h_2(\varepsilon)=-\varepsilon\log_2\varepsilon-(1-\varepsilon)\log_2(1-\varepsilon)$. A binary symmetric channel flips an input bit independently with probability ε.

Mutual information detects dependence rather than a causal channel

> For a fair bit X and $Y=X\oplus E$ with E independent Bernoulli(ε), $I(X;Y)=1-h_2(\varepsilon)$. For a Markov chain $X\to Y\to Z$, $I(X;Z)\le I(X;Y)$. A common cause can give positive mutual information with no direct arrow between X and Y.

Derivation or proof

The channel output is fair, so $H(Y)=1$, while conditioning on X leaves exactly the flip uncertainty $H(Y|X)=h_2(\varepsilon)$. The entropy identity yields the formula. For data processing, expand the same quantity two ways: $I(X;Y,Z)=I(X;Y)+I(X;Z|Y)=I(X;Z)+I(X;Y|Z)$. The Markov property makes the first conditional term zero and nonnegativity of conditional mutual information makes the second nonnegative.

Worked example

Cascade two independent binary symmetric channels with flip probability 0.1, and compare with $X=Z$, $Y=Z$ for a fair common cause Z.

Two flips cancel, so the cascade error is $2\varepsilon(1-\varepsilon)=0.18$. Thus information drops from $1-h_2(0.1)\approx0.531004$ bits to $1-h_2(0.18)\approx0.319923$ bits. In the common-cause example X and Y coincide, giving $I=1$ bit, although no direct causal channel $X\to Y$ has been specified. The identical joint table can arise from different causal descriptions.

Information is a precisely calculable statistical quantity; assigning a mechanism requires additional assumptions or interventions.

Scope of the result

Plug-in estimates from finite samples are biased, and continuous differential entropies need separate treatment. The examples do not establish that a positive empirical estimate is significant or causally interpretable.

Further development

Construct a sampling protocol and a null distribution for a chosen estimator, then quantify how much dependence remains after conditioning on measured common causes.

References: R044

M048 One measurement can miss coherence that another detects

2023-06-08 · historical evidence A · Born-effect bounds and a two-observable example proved

The same numerical overlap can represent different things depending on normalization and the measurement rule. An explicit qubit calculation shows why observable choice is also an identifiability choice.

Definitions and assumptions

A density matrix satisfies $\rho \geq 0$ and $\mathrm{Tr}\rho = 1$. A binary measurement effect satisfies $0 \leq E \leq I$, with probability $p = \mathrm{Tr}(\rho E)$.

For a normalized pure state ψ and rank-one effect $|\varphi\rangle\langle\varphi|$ with normalized φ, the probability is $|\langle\varphi,\psi\rangle|^2$.

One measurement can miss coherence that another detects

> Every valid effect gives $0 \leq \mathrm{Tr}(\rho E) \leq 1$. The pure state $\psi = (\sqrt{3}/2, 1/2)$ and the mixed state $\mathrm{diag}(3/4, 1/4)$ have identical probabilities in the coordinate-basis measurement but are separated by the plus-state effect.

Derivation or proof

Diagonalize $\rho = \Sigma_k w_k |v_k\rangle\langle v_k|$ with $w_k \geq 0$ and $\Sigma w_k = 1$. Then $\mathrm{Tr}(\rho E) = \Sigma w_k \langle v_k, E v_k\rangle$ is a convex combination of numbers in $[0,1]$, proving the bound. The pure density matrix is $[[3/4, \sqrt{3}/4], [\sqrt{3}/4, 1/4]]$, whereas the mixed matrix removes the off-diagonal terms. A diagonal effect only pairs with diagonal entries, so it cannot distinguish these states.

Worked example

Take $E_0 = \mathrm{diag}(1,0)$ and $E_+ = (1/2)[[1,1],[1,1]]$.

Both states give $p(E_0) = 3/4$. The pure state gives $p(E_+) = |(\sqrt{3}/2 + 1/2)/\sqrt{2}|^2 = (2+\sqrt{3})/4 \approx 0.933013$, while the mixed state gives $1/2$. The probability gap $\sqrt{3}/4$ is exactly the contribution from coherence in this basis. If an unnormalized vector is used instead, multiplying it by two multiplies an unnormalized squared overlap by four and can exceed one.

A restricted observation panel defines equivalence classes of states. Adding a suitable observable can split those classes even when additional repeated measurements of the original observable cannot.

Scope of the result

A finite number of samples does not reveal these probabilities exactly. State discrimination, tomography and experimental error require separate statistical analysis; the example supplies the underlying mathematical separation only.

Further development

For a finite candidate set of density matrices, find the smallest allowed family of effects that separates them with a declared minimum probability gap.

M049 An observable timescale that saturates the energy-time bound

2023-06-10 · historical evidence A · Mandelstam-Tamm observable inequality derived

A time in an uncertainty relation must be defined operationally. For time-independent observables in a unitarily evolving state, the ratio between standard deviation and expectation-change rate yields a valid timescale. It is different from an arbitrary elapsed duration.

Definitions and assumptions

Let H and A be self-adjoint on a common suitable domain, with no explicit time dependence in A. Write $\Delta A^2=\langle A^2\rangle-\langle A\rangle^2$ and likewise ΔH.

Where $d\langle A\rangle/dt\neq 0$, define $\tau_A=\Delta A/|d\langle A\rangle/dt|$.

An observable timescale that saturates the energy-time bound

> Under the usual finite-variance and domain assumptions, $\Delta H\,\tau_A\geq\hbar/2$. A two-level precession example attains equality wherever the chosen observable has nonzero expectation derivative.

Derivation or proof

The Robertson inequality gives $\Delta H\Delta A\geq|\langle[H,A]\rangle|/2$. The Heisenberg expectation equation gives $d\langle A\rangle/dt=(i/\hbar)\langle[H,A]\rangle$. Substitute the latter into the former and divide by the nonzero derivative magnitude to obtain the result. No self-adjoint time operator was introduced, and nothing in this derivation permits replacing τ_A by an arbitrary measurement duration.

Worked example

Set $H=(\hbar\omega/2)\sigma z$, $\omega>0$, initial $\psi=(1,1)/\sqrt{2}$ and $A=\sigma x$.

Unitary evolution gives $\langle A\rangle=\cos\omega t$. Since $A^2=I$, $\Delta A=|\sin\omega t|$, while $|d\langle A\rangle/dt|=\omega|\sin\omega t|$ and $\Delta H=\hbar\omega/2$. Whenever $\sin\omega t\neq 0$, $\tau_A=1/\omega$ and $\Delta H\tau_A=\hbar/2$. At the isolated zeros the ratio is 0/0 and is not defined by the formula; its limiting value exists. The state first becomes orthogonal to its initial state at π/ω, a separate timescale with a different constant.

The inequality is exact once its time quantity and assumptions are specified. Saturation can be checked by elementary two-by-two dynamics.

Scope of the result

Unbounded operators require domain care. Open-system evolution, explicitly time-dependent observables or generators, and orthogonalization bounds need distinct statements rather than an unqualified $\Delta E\Delta t$ formula.

Further development

Derive a bound for the specific state-distinguishability target being used, and compare observable-change time, fidelity speed and orthogonalization time on the same model.

References: R045

M050 Modularity from lattice reindexing and a diagnostic zero

2023-06-12 · historical evidence A · Classical Eisenstein construction with explicit coefficient checks

A q-series is not modular merely because it is written with q. The transformation law can be proved for an actual lattice sum, while low-order coefficients and forced special values provide useful checks of the implementation.

Definitions and assumptions

For $\mathrm{Im}\tau>0$ define $G_4(\tau)=\Sigma(m,n)\neq(0,0)(m+n\tau)^{-4}$, an absolutely convergent lattice sum.

Normalize $E_4=G_4/(2\zeta(4))$. Its Fourier expansion is $E_4=1+240\Sigma n\geq 1\sigma_3(n)q^n$, $q=\exp(2\pi i\tau)$. Also $E_6=1-504\Sigma n\geq 1\sigma_5(n)q^n$.

Modularity from lattice reindexing and a diagnostic zero

$G_4((a\tau+b)/(c\tau+d))=(c\tau+d)^4G_4(\tau)$ for matrices in $SL_2(Z)$. E_4 begins $1+240q+2160q^2+6720q^3+17520q^4$. The weight-six transformation law forces $E_6(i)=0$.

Derivation or proof

In a term of the transformed lattice sum, $m+n(a\tau+b)/(c\tau+d)=[(md+nb)+(mc+na)\tau]/(c\tau+d)$. The integer map $(m,n)\mapsto(md+nb,mc+na)$ is a bijection because its determinant is $ad-bc=1$. Absolute convergence permits reindexing, proving weight four. The stated Fourier expansion supplies its holomorphic behavior at the cusp; coefficients follow by summing divisor cubes: $\sigma_3(1),\ldots,\sigma_3(4)=1,9,28,73$. For E_6, the substitution $\tau\mapsto-1/\tau$ fixes i and multiplies its value by $i^6=-1$, so its value must vanish.

Worked example

Evaluate a truncated E_6 series at $\tau=i$, where $q=e^{-2\pi}$ is small, and compare with the exact forced zero.

The first terms use divisor fifth-power sums. Partial sums approach zero rapidly, and an explicit geometric-polynomial tail bound can certify the residual. The verification program checks these coefficients and the numerical special-value residual. A truncated series agreeing at one point still does not prove its global modularity; the reindexing argument or a theorem supplies that part.

Transformation, holomorphy, cusp conditions and coefficients perform different jobs. The lattice construction connects them without treating notation as a theorem.

Scope of the result

Only the full modular group and these holomorphic even-weight examples are treated. Multiplier systems, congruence subgroups, weakly holomorphic forms and nonholomorphic completions require additional definitions.

Further development

For a chosen weight and subgroup, compare a computed q-series with a known basis using a proved coefficient bound such as a suitable Sturm bound, with all hypotheses stated.

References: R046

M051 A classical abelian Chern-Simons calculation with its global boundary

2023-06-12 · historical evidence A · Small-gauge invariance and a torus action evaluated

The Chern-Simons integrand needs a geometric setting before it becomes an action. A globally defined abelian connection one-form on a closed manifold gives a tractable case in which gauge variation and an actual integral can be derived directly.

Definitions and assumptions

Let M be a closed oriented three-manifold and A a globally defined real one-form representing a connection on a trivial U(1) bundle.

Choose $S_k(A)=(k/(4\pi))\int_M A\wedge dA$. A small gauge change is $A\mapsto A+d\chi$ with χ a globally defined real scalar.

A classical abelian Chern-Simons calculation with its global boundary

For this setting S_k is invariant under small gauge changes. Its first variation is $\delta S_k=(k/(2\pi))\int \delta A\wedge dA$. On the unit-period three-torus, $A=a\cos(2\pi z)dx+b\sin(2\pi z)dy$ has $S_k=-kab/2$ for orientation $dx\wedge dy\wedge dz$.

Derivation or proof

The gauge variation is $(k/(4\pi))\int d\chi\wedge dA=(k/(4\pi))\int d(\chi dA)$, which vanishes on a closed manifold by Stokes. For the first variation, expand the action to first order. Integration by parts makes the two terms $\delta A\wedge dA$ and $A\wedge d\delta A$ equal after integration, giving the factor two. For the torus example, differentiating A and expanding the wedge product yields $A\wedge dA=-2\pi ab(\cos^2(2\pi z)+\sin^2(2\pi z))dx\wedge dy\wedge dz$; its integral is $-2\pi ab$.

Worked example

Take $a=b=1$ and $k=2$ on the unit three-torus.

The classical action is -1. The form is globally periodic and smooth, but dA is nonzero, so it is not a stationary flat connection when $k\neq 0$. Varying a or b changes the action continuously. Thus even in this elementary example, quantization of a coupling level must not be confused with a claim that every classical action value is an integer.

The calculation supplies an actual well-defined action and variation in a restricted setting, while identifying exactly where stronger global statements would enter.

Scope of the result

Nontrivial bundles and large gauge transformations require a globally consistent definition beyond a single one-form integral. Level quantization depends on the group, normalization, and often spin data; it is not proved by the small-gauge calculation.

Further development

Construct the exponentiated action using differential cohomology or a four-dimensional extension for a stated class of bundles, and track the required integrality condition.

References: R047

M052 Explicit genus-one modular data with the anomaly retained

2023-06-12 · historical evidence A · Torus relations and fusion calculation verified; full functor not reconstructed

A modular functor contains gluing and labeling data for many surfaces; it cannot be specified by an unexplained arrow from a vector-space category to $SL_2(Z)$ representations. Nevertheless, one important piece can be calculated completely: the torus matrices of the two-label semion theory, corresponding to SU(2) level one.

Definitions and assumptions

Use labels 0 and 1 and torus space $V(T^2)=C^2$. Set $S=(1/\sqrt{2})[[1,1],[1,-1]]$ and $T_0=\mathrm{diag}(1,i)$.

A projective mapping-class action may acquire scalar phases. Let $T=\exp(-i\pi/12)T_0$ to incorporate the standard central-charge-one phase in this example.

Explicit genus-one modular data with the anomaly retained

$S^2=I$ and $(ST_0)^3=\exp(i\pi/4)I$; after the stated scalar adjustment, $(ST)^3=I$. The Verlinde expression using this S gives fusion $1\times1=0$. Conditional on the modular-functor gluing dimension formula, the unpunctured genus-g dimension is 2^g.

Derivation or proof

Multiply the displayed two-by-two matrices to check the relations. Replacing T_0 by its scalar multiple changes the cube by $e^{-i\pi/4}$ and cancels the projective scalar. For the fusion coefficient, N_ab^c=Σx S_ax S_bx conjugate(S_cx)/S_0x. Substitution gives N_11^0=1 and N_11^1=0. The genus dimension formula is Σx(S_0x)^(2−2g); both entries equal $1/\sqrt{2}$, yielding $2\cdot(1/\sqrt{2})^{(2-2g)}=2^g$. This last deduction explicitly uses the gluing theorem rather than deriving all its axioms from S and T.

Worked example

At genus one the dimension is two, at genus two it is four. Compare T_0 with the adjusted T in a relation check.

Ignoring the scalar phase produces a nontrivial residue $\exp(i\pi/4)$ in the cubic relation; it is a meaningful anomaly convention, not a numerical error. The verifier computes both relations and the full two-label fusion table to numerical precision with exact expected integers.

A small consistent torus example is available now, but consistency of torus matrices alone is not a construction of a modular functor on all labeled surfaces.

Scope of the result

No claim is made that arbitrary matrices satisfying these relations extend to a modular tensor category or a full gluing theory. General levels, punctures and anomaly extensions require additional data.

Further development

Write the genus-zero associativity and braiding data for the chosen two-label theory, verify their coherence relations and then reconstruct gluing maps explicitly.

References: R048

M053 Two-oscillator locking with stable and unstable branches

2023-06-12 · historical evidence A · Exact phase reduction solved

The missing time derivative in the old formula changes its mathematical type. Once the dynamics is restored, the smallest nontrivial network has a sharp synchronization threshold and an explicit stability classification.

Definitions and assumptions

Take $\theta_1=\omega_1+K\sin(\theta_2-\theta_1)$ and $\theta_2=\omega_2+K\sin(\theta_1-\theta_2)$, with $K>0$.

Let $\delta=\theta_1-\theta_2$ and $\varphi=(\theta_1+\theta_2)/2$. Phases are taken modulo 2π.

Two-oscillator locking with stable and unstable branches

The equations reduce to $\dot{\delta}=\Delta\omega-2K\sin\delta$ and $\dot{\varphi}=(\omega_1+\omega_2)/2$. If $|\Delta\omega|<2K$ there are two equilibrium phase differences: the branch with $\cos\delta>0$ is asymptotically stable, while the other is unstable. At equality they merge into a nonhyperbolic threshold.

Derivation or proof

Subtract and add the two original equations; the sine terms cancel in the mean and reinforce in the difference. A locked difference requires $\sin\delta=\Delta\omega/(2K)$, which has two solutions per period in the strict interior. The scalar linearization is $-2K\cos\delta$. Negative derivative gives local asymptotic stability and positive derivative gives instability. At $|\Delta\omega|=2K$ the derivative vanishes; a Taylor expansion gives a quadratic leading term and shows that the threshold equilibrium is not the same robust exponentially attracting situation as an interior stable branch.

Worked example

Let $K=1$ and $\Delta\omega=1$.

The stable difference is $\delta^*=\pi/6$ and the unstable one is $5\pi/6$, with linearized rates $-\sqrt{3}$ and $+\sqrt{3}$. The common locked frequency is $(\omega_1+\omega_2)/2$. If instead $\Delta\omega=3$, then $\dot{\delta}=3-2\sin\delta\geq 1$, so the unwrapped phase difference increases forever and no phase lock is possible. A common shift $\theta_i\mapsto\theta_i+\alpha$ changes neither difference dynamics nor locking.

Synchronization here means a constant relative phase, not necessarily identical phases or a zero absolute frequency.

Scope of the result

Larger heterogeneous networks need not have the same threshold. Finite-size locking, mean-field coherence transitions and noisy synchronization are different problems. At the boundary the absence of exponential stability must be recorded.

Further development

For a specified graph and frequency vector, solve the phase-balance equations modulo common shifts and certify the Jacobian on the quotient space.

References: R049

M054 Invariant gates and an exact identifiability obstruction in conductance ODEs

2023-06-12 · historical evidence A · ODE invariance and a passive reduction proved

The mathematical value of a conductance model comes from a typed ODE and an observation protocol. Two simple calculations already separate a well-defined state domain from what a voltage trace can identify.

Definitions and assumptions

A gate q satisfies $\dot{q}=\alpha(V)(1-q)-\beta(V)q$, with nonnegative locally Lipschitz rates.

With gates frozen, write $C\dot{V}=I-\Sigma_j g_j(V-E_j)$, $C>0$ and $g_j\geq 0$. Define $G=\Sigma g_j$ and $B=\Sigma g_j E_j$.

Invariant gates and an exact identifiability obstruction in conductance ODEs

The gate interval $[0,1]$ is forward invariant. For constant voltage, q approaches $\alpha/(\alpha+\beta)$ with time constant $1/(\alpha+\beta)$, when $\alpha+\beta>0$. The frozen-gate voltage depends on conductances only through G and B; three conductances are generally not separately identifiable from that passive protocol.

Derivation or proof

At $q=0$ the vector field is $\alpha\geq 0$, and at $q=1$ it is $-\beta\leq 0$, so uniqueness prevents crossing outward through either boundary. With constant rates, solve $\dot{q}=\alpha-(\alpha+\beta)q$ directly. For the voltage reduction, $C\dot{V}=I+B-GV$ gives $V_\infty=(I+B)/G$ and $V(t)=V_\infty+(V_0-V_\infty)e^{-Gt/c}$ for $G>0$. Any conductance perturbation δg satisfying $\Sigma\delta g_j=0$ and $\Sigma E_j\delta g_j=0$ leaves the entire voltage trace unchanged when it remains nonnegative.

Worked example

Use reversal values $E=(0,1,2)$, conductances $g=(2,2,2)$, and perturbation $\delta g=(1,-2,1)$.

For every sufficiently small λ, $g+\lambda\delta g$ stays positive. Both aggregate quantities remain $G=6$ and $B=6$, so identical C, I, V_0 produce exactly the same $V(t)$ despite different channel conductances. Independently, $\alpha=0.2$ and $\beta=0.3$ give $q_\infty=0.4$ and time constant 2 in the chosen time units. These are direct solutions, not fitted physiological claims.

State-domain validity and parameter identifiability are distinct mathematical requirements. A model may satisfy the first while failing the second under a limited observation protocol.

Scope of the result

The frozen-gate reduction is not a proof that the full nonlinear Hodgkin-Huxley model is unidentifiable under every experiment. Voltage dependence, varied inputs and additional observations can break this obstruction.

Further development

Specify a voltage-clamp or current-clamp input family, derive sensitivity equations and determine whether the added protocols remove the null directions demonstrated here.

References: R050

M055 Decision thresholds and genuine strategic equilibria

2023-07-18 · historical evidence U · Finite models solved explicitly

An uncertain state of nature does not automatically supply a second strategic player. The distinction can be made concrete by solving a one-agent decision and a separately defined coordination game.

Definitions and assumptions

In a decision model, nature selects state S with prior p and T with prior $1-p$. Action B has payoffs M and $-c$, while action N has zero in either state, with $M,c>0$.

In a two-player coordination game, each chooses B or N. Payoffs are (2,2) at (B,B), (1,1) at (N,N), and (0,0) at mismatched choices.

Decision thresholds and genuine strategic equilibria

In the decision problem B is optimal exactly when $p \geq c/(M+c)$, with indifference at equality. In the coordination game both (B,B) and (N,N) are Nash equilibria; there is also a mixed equilibrium where each chooses B with probability 1/3.

Derivation or proof

Expected utility of B is $pM-(1-p)c=p(M+c)-c$, while N gives zero, yielding the threshold. In the game, a player facing B obtains 2 by B and 0 by N; facing N, the player obtains 1 by N and 0 by B. Thus both coordinated profiles resist unilateral deviation. If the opponent chooses B with probability q, the expected payoff from B is $2q$ and from N is $1-q$. Equality gives $q=1/3$; this mutual indifference verifies the mixed equilibrium.

Worked example

Choose $M=9$ and $c=1$ in the decision model.

The action threshold is $p=0.1$. At $p=0.05$, B has expected payoff -0.5 and N is better; at $p=0.2$, B has expected payoff 1 and is better. No opponent changes strategy in this calculation. In the game, the mixed-equilibrium payoff is 2/3, lower than either coordinated pure payoff; these outcomes depend on the explicitly chosen bimatrix, not on the decision table.

The word equilibrium becomes meaningful only after strategic actors and their payoff dependence are specified.

Scope of the result

These are illustrative finite-utility mathematical models and do not validate any metaphysical premise or uniquely prescribe behavior. A zero-sum interpretation would require yet another payoff structure.

Further development

Add explicit private information or a perturbation of payoffs and determine whether the equilibrium set or the decision threshold changes continuously.

References: R051

M056 Infinite utilities through controlled finite limits

2023-07-18 · historical evidence U · Limit-order obstruction exhibited

Replacing very large payoffs by infinity can destroy the expectation being optimized. Finite truncations reveal when a conclusion is robust and when it depends on the order in which utility scales and probabilities approach their limits.

Definitions and assumptions

Take two states with probabilities p and $1-p$. For truncation $M>0$, action A has payoffs $(M,-M)$, action B has $(-M,M)$ and action C has $(0,0)$.

Expected utilities are $U_A=M(2p-1)$, $U_B=-U_A$ and $U_C=0$. Every finite model is well defined.

Infinite utilities through controlled finite limits

At fixed $p\neq 1/2$ the better of A and B is determined by the sign of $2p-1$. At $p=1/2$ all three actions tie for every M. But sequences of priors with the same limiting value 1/2 can produce opposite strict preferences that persist as $M\to\infty$.

Derivation or proof

Choose $p_M=1/2+a/M$ for fixed a and sufficiently large M so the prior is valid. Then $U_A=M(2a/M)=2a$ and $U_B=-2a$. If $a>0$, A strictly wins; if $a<0$, B strictly wins; if $a=0$, all tie. All these priors converge to 1/2. Hence the limiting prior alone does not determine the limiting decision problem. Directly assigning payoffs $+\infty$ and $-\infty$ with both states of positive probability would require subtracting two infinite contributions, leaving an undefined expectation rather than a usable optimization objective.

Worked example

For $M=100$ take priors 0.51,0.49 and 0.50.

The respective expected payoffs of A are $2,-2$ and 0, while B has the opposite values. Repeating with $M=1000$ and priors 0.501,0.499,0.500 gives exactly the same three outcomes. This is an exact algebraic example of a nonuniform limiting operation. The earlier finite wager threshold $c/(M+c)$ instead tends to zero at fixed c, but that conclusion depends on its asymmetric payoff structure.

A claim involving infinite utilities needs a declared finite approximation, probability assumptions and a limit argument; otherwise an asserted optimum may not exist as a defined mathematical object.

Scope of the result

This example shows sensitivity of one family of limits, not that every model with unbounded utility is inconsistent. Extended-valued expectations are legitimate when their positive and negative parts are not simultaneously infinite.

Further development

Specify a utility function and a class of probability measures under which expectations are well defined, then use uniform-integrability conditions to justify exchanging limits and optimization.

D Probability control relativity and foundations

M057 A Cauchy model has no ordinary mean or variance

2023-07-28 · historical evidence A · Classical result developed and worked

The useful revision is a complete separation between symmetry, existence of a mean, and concentration of an estimator. The Cauchy family gives an exact example in which averaging independent observations does not improve the distribution of the sample mean. This is a classical stability calculation, not a conclusion about any particular empirical dataset.

Definitions and assumptions

A Cauchy law with location μ and scale $\gamma>0$ has density $\gamma/[\pi((x-\mu)^2+\gamma^2)]$. Its characteristic function is $\varphi(t)=\exp(i\mu t-\gamma|t|)$. Independence multiplies characteristic functions.

Exact failure of sample-mean concentration

> If $X_1,\dots,X_n$ are independent Cauchy(μ,γ), then their arithmetic mean is again Cauchy(μ,γ), for every positive integer n. Neither $E[X]$ nor $\mathrm{Var}(X)$ exists as an ordinary finite moment.

Derivation or proof

The positive and negative parts of the centered first-moment integral each diverge logarithmically because $x/(1+x^2)$ has a logarithmic antiderivative. Symmetric truncation cancels these divergences, but that principal value is not a Lebesgue expectation. For the mean, independence gives $E\exp(it\Sigma X_j/n)=\varphi(t/n)^n=\exp(i\mu t-\gamma|t|)$. Uniqueness of characteristic functions identifies the law. Thus no finite-variance central-limit estimate can be invoked. The same tail calculation gives a divergent second moment.

Worked example

Take standard Cauchy observations and ask whether the mean is within one unit of zero.

For every n, $P(|\bar{X}_n|\le 1)=F(1)-F(-1)=1/2$, using $F(x)=1/2+\arctan(x)/\pi$. The probability remains one half even when n increases. In contrast, the population median is μ and independent sample quantiles can consistently estimate it because F is continuous and strictly increasing there.

A stable distribution can make an apparently reasonable estimator permanently noisy. Quantile estimation and distribution fitting must be evaluated under their actual assumptions.

Scope of the result

This does not establish that observations are Cauchy, that every heavy-tailed law lacks a mean, or that all robust estimators have the same behavior. Dependence would also change the characteristic-function multiplication.

Further development

For a specified tail exponent, determine which absolute moments exist, then compare mean, median and trimmed estimators using exact or justified asymptotic distributions.

References: R052, R053

M058 Feedback instability is a mechanism to specify

2023-07-28 · historical evidence A · Classical result developed and worked

Feedback questions become testable once an update matrix and a measured norm are fixed. Eigenvalues determine asymptotic decay in finite dimension, but they do not bound all finite-time amplification. A two-dimensional Jordan block makes the distinction completely explicit.

Definitions and assumptions

A discrete linear system is $x_{n+1}=Ax_n$. Spectral radius $\rho(A)$ is the largest eigenvalue modulus. The Euclidean amplification at step n is the induced norm $||A^n||_2$. A real matrix is normal when $AA^T=A^TA$.

Exact transient amplification in an asymptotically stable system

For $A=aI+bN$ with $N=[[0,1],[0,0]]$, $N^2=0$ and $0<a<1$, $A^n=a^nI+n\,a^{n-1}bN$. Although $\rho(A)=a<1$ and $A^n\to0$, intermediate norms can be arbitrarily large as $|b|$ increases.

Derivation or proof

The binomial theorem applies because I and N commute and every power N^k with $k\geq2$ vanishes. Only the terms k=0 and k=1 survive. The scalar sequences a^n and $n\,a^{n-1}$ both tend to zero, proving asymptotic stability. Acting on e_2 yields $A^ne_2=(n\,a^{n-1}b,a^n)$, so its Euclidean norm supplies a lower bound on $||A^n||_2$. At a fixed step n this lower bound grows linearly in $|b|$ without changing either eigenvalue.

Worked example

Choose $A=[[1/2,4],[0,1/2]]$ and initial state e_2.

At one step the output is (4,1/2), with norm $\sqrt{65}/2>4$. At two steps the output is (4,1/4), although the eigenvalues remain 1/2. The exact formula predicts eventual decay after this transient. The corresponding continuous example A_c=[[-1,4],[0,−1]] has exp(tA_c)=e^(−t)[[1,4t],[0,1]].

A stability statement and a peak-amplification statement answer different questions. A mechanism for a large transient can be proved without asserting any probability distribution for external shocks.

Scope of the result

The calculation assumes a fixed finite matrix. Nonlinear saturation, delayed feedback, time-varying matrices and stochastic inputs require additional models. A large matrix norm does not by itself imply a particular event occurs.

Further development

Optimize $\sup_n||A^n||$ over a parameter family, then compare Euclidean bounds with a quadratic Lyapunov norm that contracts at every step.

References: R001, R054

M059 Frobenius energy across disjoint blocks

2023-08-29 · historical evidence A · Classical result developed and worked

A multiscale norm construction needs a bookkeeping theorem. The Frobenius norm behaves like a Euclidean norm on the list of matrix entries, so squared energies add over disjoint blocks. Overlapping descriptions need multiplicity corrections instead of an unqualified addition rule.

Definitions and assumptions

For a real or complex matrix A, $||A||_F^2=\Sigma_{ij}|a_{ij}|^2=\mathrm{Tr}(A^* A)$. A block here means a subset of entry indices; blocks may have irregular shapes. A partition uses every entry exactly once.

Partition additivity and the overlap correction

If entry blocks $B_1,\ldots,B_m$ partition the indices, then $||A||_F^2=\Sigma_b||A|B_b||_F^2$. If they cover entries with multiplicities c_{ij}, the uncorrected sum is $\Sigma_{ij}c_{ij}|a_{ij}|^2$.

Derivation or proof

Expand every block norm as a sum of squared moduli and interchange two finite sums. The coefficient of a given entry is exactly the number of blocks containing it. For a partition that number is one. When every covered entry has positive multiplicity, assigning weight $1/c_{ij}$ to that entry within each occurrence restores the original energy. The argument needs no independence or probabilistic assumptions. An equivalent Hilbert-space statement uses orthogonal coordinate projections P_b and the identity $\Sigma P_b=I$.

Worked example

Let $A=[[1,2],[3,4]]$. Partition it by rows, then deliberately count its first row twice.

The complete squared Frobenius norm is $1+4+9+16=30$. Row energies are 5 and 25, adding to 30. Repeating the first row gives 35, exposing five units of double counting. Dividing the two first-row occurrences by multiplicity two restores 30. By contrast, the unsquared norms $\sqrt{5}+5$ do not equal $\sqrt{30}$.

Disjoint block decomposition preserves total squared energy. It does not automatically make values comparable across blocks of different cardinalities.

Scope of the result

When comparing differently sized systems, one must choose total energy, mean squared entry, an operator norm or a dimensionless normalization. Those choices encode different questions. Orthogonal-coordinate arguments also do not apply unchanged to arbitrary overlapping learned features.

Further development

For nonorthogonal feature maps F_b, study frame bounds $a||x||^2\le\Sigma||F_bx||^2\le b||x||^2$ and determine whether a tight-frame normalization can preserve the intended energy.

References: R001

M060 Population weights versus amplitude weights

2023-08-29 · historical evidence A · Classical result developed and worked

The distinction between probability weights and amplitude weights can be settled before any elaborate multiscale model is built. The former defines an expected square; the latter first attenuates the signal and then squares it. They agree only under special circumstances.

Definitions and assumptions

Let $p_i \geq 0$ with $\Sigma p_i = 1$ and let a_i be real or complex values. Expected square is $E|A|^2 = \Sigma p_i |a_i|^2$. Entrywise amplitude weighting by p gives $||p \odot a||_2^2 = \Sigma p_i^2 |a_i|^2$.

Probabilistic weighting is implemented by square-root amplitudes

The probability-weighted squared magnitude equals $||\mathrm{diag}(\sqrt{p_i})a||_2^2$. Weighting the amplitudes by p_i instead yields a different quadratic form. Since $0 \leq p_i \leq 1$, the second is no larger than the first.

Derivation or proof

Expand the Euclidean norm after applying the diagonal matrix. Each squared coefficient is $(\sqrt{p_i})^2 = p_i$. The ordinary amplitude weight p_i is squared by the norm, producing p_i^2. The inequality follows term by term from $p_i^2 \leq p_i$. Equality requires $p_i(1-p_i)|a_i|^2 = 0$ for every i: nonzero signals can occur only at weights zero or one, apart from degenerate cancellations that are impossible in this nonnegative sum.

Worked example

Use values $a = (2,4)$ and probabilities $p = (1/4, 3/4)$.

Expected square is $(1/4)4 + (3/4)16 = 13$. Squared probability-amplitude weighting gives $(1/16)4 + (9/16)16 = 37/4 = 9.25$. The weighted mean is $7/2$, so the variance is $13 - (7/2)^2 = 3/4$. None of these three quantities is a substitute for the others.

A correct implementation starts by deciding whether the data describe sampling frequencies, deliberate attenuation or an averaging measure. The square-root representation then supplies a precise matrix formula.

Scope of the result

If some weights vanish, the resulting quadratic form is only a seminorm on the full coordinate space. Importance weights need not sum to one, while signed weights need not produce a positive quadratic form at all.

Further development

For a general positive measure μ, formulate the same distinction as an $L^2(\mu)$ norm versus a weighted amplitude map, and establish invariance under a change of sampling representation.

References: R001

M061 Covariance inequalities with finite moments

2023-08-29 · historical evidence A · Classical result developed and worked

Covariance optimization is legitimate only after existence of the required moments is established. Once variables are square integrable, covariance becomes a positive semidefinite Gram matrix in an L^2 space, and the familiar bounds follow from one short argument.

Definitions and assumptions

For square-integrable X and Y, Cov(X,Y)=E[(X−EX)overline(Y−EY)]. For a real random vector Z, its covariance matrix Σ has entries Cov(Z_i,Z_j). Positive semidefinite means $w^{T}\Sigma w \geq 0$ for every w.

Covariance as a Gram matrix and its equality condition

$|\mathrm{Cov}(X,Y)|^2 \leq \mathrm{Var}(X)\mathrm{Var}(Y)$. For a finite square-integrable real vector, $\mathrm{Var}(w^{T}Z)=w^{T}\Sigma w$. If both variances are positive, equality in the scalar bound occurs precisely when the centered variables are proportional almost surely.

Derivation or proof

Centered variables are elements of the Hilbert space L^2. Cauchy-Schwarz gives the bound. Alternatively, for real variables, nonnegativity of $E[(X-EX-t(Y-EY))^2]$ for every real t forces the discriminant of this quadratic in t to be nonpositive. Expanding the centered weighted sum proves the matrix identity and hence positive semidefiniteness. Equality in Cauchy-Schwarz identifies linear dependence in L^2, where two representatives differing only on a null set are equivalent.

Worked example

Let X be equally likely to be −1 or 1, and let Y=2X.

Both means vanish. Variances are 1 and 4, covariance is 2, and Σ=[[1,2],[2,4]] has determinant zero. The combination 2X−Y vanishes almost surely, so its variance is zero. This explains the singularity structurally rather than treating it as a numerical failure.

A singular covariance matrix records an exact L^2 dependency. It can be essential information for a constrained quadratic calculation.

Scope of the result

The Cauchy example M057 is outside these hypotheses; finite-sample covariance numbers do not create a population covariance that fails to exist. For complex variables, conjugation must be retained. Estimated matrices may require a separate analysis of sampling error.

Further development

Characterize which dependencies are preserved under an observation map and how eigenvalue perturbation bounds distinguish a true null direction from one introduced by finite precision.

References: R001

M062 A quadratic program with singular cases included

2023-08-29 · historical evidence A · Classical result developed and worked

The positive-definite inverse formula is only one branch of the quadratic-program problem. The singular case admits a clean complete classification, and inequality constraints can change the optimizer even when the matrix is nonsingular. These are mathematical statements about quadratic forms, not practical allocation recommendations.

Definitions and assumptions

Let Σ be a symmetric positive semidefinite real matrix and $c \neq 0$. Minimize $w^{\mathsf{T}}\Sigma w$ subject to $c^{\mathsf{T}}w=1$. Write Σ^{+} for its Moore-Penrose inverse and $\ker\Sigma$ for its nullspace.

Complete equality-constrained PSD classification

If some $z \in \ker\Sigma$ has $c^{\mathsf{T}}z \neq 0$, the minimum is zero and all minimizers are the feasible null vectors. Otherwise $c \in \operatorname{range}\Sigma$, $q=c^{\mathsf{T}}\Sigma^{+}c>0$, the minimum is $1/q$, and all minimizers are $\Sigma^{+}c/q+z$ with $z \in \ker\Sigma$.

Derivation or proof

In the first case, $z/(c^{\mathsf{T}}z)$ is feasible and has zero objective; positive semidefiniteness rules out a lower value. In the second case, c is orthogonal to the nullspace and hence lies in the range. Set $w_0=\Sigma^{+}c/q$. It is feasible and $\Sigma w_0=c/q$. For any feasible $w=w_0+h$, $c^{\mathsf{T}}h=0$, so the cross term $2h^{\mathsf{T}}\Sigma w_0$ vanishes and the objective is $1/q+h^{\mathsf{T}}\Sigma h$. Equality holds exactly when $h \in \ker\Sigma$. This also proves uniqueness precisely when the feasible tangent space contains no nonzero null vector.

Worked example

Compare a diagonal matrix, a rank-one matrix, and a nonnegative constraint.

For $\Sigma=\operatorname{diag}(1,4)$, $c=(1,1)$, $w^{*}=(4/5,1/5)$ and the minimum is $4/5$. For $\Sigma=11^{\mathsf{T}}$, every feasible w has objective one. For $\Sigma=[[1,2],[2,9]]$, the unrestricted optimizer is $(7/6,-1/6)$ with value $5/6$; imposing $w \geq 0$ gives $(1,0)$ with value one. Its KKT multiplier for the second coordinate is positive.

Nullspaces, budgets and inequality constraints must be handled together; merely replacing an inverse by a pseudoinverse does not cover every singular case.

Scope of the result

If Σ is indefinite the convex proof fails and the objective may be unbounded below along feasible directions. Additional bounds or penalties define different programs.

Further development

For rational inputs, produce exact KKT and nullspace certificates, then study the parameter loci where the active set or nullspace changes.

References: R012

M063 Open-system dynamics must preserve physical states

2023-08-16 · historical evidence A · Classical result developed and worked

An open-system equation must preserve the convex set of density operators, including when an auxiliary system is attached. An exactly solvable qubit channel provides a stronger check than inspection of trace alone. Its Kraus form supplies complete positivity directly.

Definitions and assumptions

A density operator is a positive semidefinite matrix ρ with trace one. A map is completely positive when its tensor product with the identity map on any finite auxiliary system preserves positivity. Set $\hbar=1$.

Exact amplitude-damping semigroup

For H=0 and L=√γ |0⟩⟨1| with γ≥0, the Lindblad equation sends ρ=[[a,b],[b*,d]] to [[a+(1−q)d,√q b],[√q b*,qd]], where q=e^(−γt), t≥0. This defines a trace-preserving completely positive semigroup.

Derivation or proof

Direct matrix multiplication in LρL†−{L†L,ρ}/2 gives $\dot{d}=-\gamma d$, $\dot{b}=-\gamma b/2$ and $\dot{a}=\gamma d$, which solve to the displayed formula. Define $K_0=\mathrm{diag}(1,\sqrt{q})$ and $K_1=\sqrt{(1-q)}|0\rangle\langle 1|$. Their adjoint squares sum to I, and ΣK_jρK_j† reproduces the solution. Tensoring each K_j with an auxiliary identity shows complete positivity. Since q(t+s)=q(t)q(s), composition satisfies the semigroup law.

Worked example

Start from |+⟩⟨+|, so a=d=b=1/2.

At q=1/4 the output is [[7/8,1/4],[1/4,1/8]], with trace one and determinant 3/64>0. More generally detρ(t)=q detρ(0)+q(1−q)d^2≥0, giving an additional two-by-two positivity check. As t tends to infinity the state approaches |0⟩⟨0|.

The exact solution verifies what the mathematical generator does. It does not establish that this generator describes any proposed physical mechanism.

Scope of the result

A reduced microscopic model can require weak-coupling, coarse-graining or Markov assumptions before it has this form. Trace preservation alone is weaker than positivity, and positivity alone is weaker than complete positivity. Negative-time inversion is not a physical damping channel.

Further development

Add a specified Hamiltonian or dephasing operator and test whether an exact solution, a spectral decomposition of the generator, or a controlled splitting error is available.

References: R055, R056

M064 Population growth and the actual logarithmic time

2023-08-10 · historical evidence A · Classical result developed and worked

The original numerical claim can be resolved entirely within its assumed exponential model. The corrected answer is a model time, not a prediction of a real population. The factor of one half changes the initial condition and must not be silently discarded.

Definitions and assumptions

For discrete-time growth, $P(t)=P_0(1+r)^t$ with $r>-1$; allowing real t is an interpolation of that formula. A continuous exponential with the same unit-step multiplier has rate $\lambda=\log(1+r)$, not r exactly.

Exact target time and rate sensitivity

> For $P(t)=154000(1.0005)^{t/2}$, the starting value is 77000. The time to $P^*=10^{12}$ is $t^*=\log(10^{12}/77000)/\log(1.0005)=32767.109878086\ldots$ in the specified time units. If only integer times are allowed, the first crossing is 32768.

Derivation or proof

Take logarithms of the positive equation $P^*=77000(1.0005)^t$. Strict monotonicity of the exponential gives a unique real solution, and integer rounding follows from the same monotonicity. For a general multiplier $1+r>1$, differentiation yields $\partial t^*/\partial r=-\log(P^*/P_0)/[(1+r)\log(1+r)^2]$. This makes the long extrapolation highly sensitive to the assumed rate. Replacing $\log(1+r)$ by r is only a small-r approximation whose cumulative error should be reported.

Worked example

Compare the stated discrete interpolation with a continuous model using $\lambda=0.0005$.

The latter target time is $\log(10^{12}/77000)/0.0005\approx32758.92083$, about 8.18905 units earlier. Substituting $t^*=32767.109878086$ back into the original formula recovers 10^{12} to the working floating-point precision. The previously reported value near 11788 does not satisfy the equation.

Correct arithmetic settles a question about a formula; it does not supply evidence that the formula remains valid over tens of thousands of units.

Scope of the result

The model assumes a fixed rate and no carrying capacity, migration, stochasticity or resource constraint. Any logistic replacement needs a stated carrying capacity larger than the target, otherwise the target is never reached in finite time.

Further development

Specify a bounded-growth ODE, derive its target-time formula and classify unreachable targets before computing a numerical crossing.

M065 A unified equation needs shared variables and invariances

2023-08-23 · historical evidence A · Classical result developed and worked

A useful unification identifies a common mathematical structure and proves a transportable statement. Gradient flows provide one such structure, but the metric and mobility are part of the model. Sharing an energy symbol does not make two systems dynamically equivalent.

Definitions and assumptions

Let $E:R^n \to R$ be differentiable. A mobility $K(x)$ is a symmetric positive semidefinite matrix. The dynamics $\dot{x} = -K(x)\nabla E(x)$ uses the chosen Euclidean coordinates; positive-definite K defines a corresponding metric interpretation.

Energy dissipation and coordinate transport

> Every sufficiently differentiable solution satisfies $dE/dt = -\nabla E^T K \nabla E \le 0$. Under an invertible linear change $y = Sx$, the transported energy $\tilde{E}(y) = E(S^{-1}y)$ and mobility $\tilde{K}(y) = SK(S^{-1}y)S^T$ preserve this dissipation identity.

Derivation or proof

The chain rule gives $\dot{E} = \nabla E^T \dot{x}$, and positive semidefiniteness gives the sign. For the coordinate change, $\nabla_y \tilde{E} = S^{-T} \nabla_x E$ and $\dot{y} = S\dot{x} = -SKS^T \nabla_y \tilde{E}$. Substitution recovers the same scalar energy derivative. This calculation explains why transforming only the state while keeping an arbitrary identity mobility generally changes the dynamics. The result requires neither thermodynamic terminology nor a claim that all dissipative systems are gradient flows.

Worked example

Take $E(x) = x^T A x/2$ with symmetric A positive semidefinite, and $K = I$.

Then $\dot{x} = -Ax$ and $\dot{E} = -x^T A^2 x$. A finite weighted-graph heat equation has this form with A equal to the graph Laplacian, while the scalar equation $\dot{x} = -x$ is its one-dimensional instance. A constant component in the Laplacian nullspace is preserved rather than dissipated, which shows the difference between nonincrease and convergence to a unique state.

The shared theorem is energy monotonicity under a specified positive mobility. A proof of equivalence requires an actual state-space map such as S and the correctly transported energy and mobility.

Scope of the result

Nonsymmetric transport, Hamiltonian components, constraints and nonlocal metrics require extensions. Energy decrease alone does not prove boundedness, uniqueness, convergence rate or identification from observations.

Further development

Separate a generator into dissipative and skew parts, prove the resulting balance law and ask which observations distinguish the mobility from the energy.

References: R012

M066 Bessel versus Parseval

2023-09-01 · historical evidence A · Classical result developed and worked

The missing distinction in the original inequality list is completeness. Bessel’s inequality is an exact statement about projection error; Parseval’s equality appears when the chosen orthonormal family spans the relevant Hilbert space. A residual formula makes the practical meaning immediate.

Definitions and assumptions

Use an inner product linear in its first argument. For an orthonormal family $e_1,\dots,e_m$, define $P_m x=\Sigma\langle x,e_j\rangle e_j$. Its span is a closed finite-dimensional subspace of the Hilbert space H.

Projection residual and completeness criterion

For every x, $||x||^2=\Sigma_{j=1}^m|\langle x,e_j\rangle|^2+||x-P_m x||^2$. For a countable orthonormal family, the infinite coefficient sum is at most $||x||^2$. Equality for every x holds exactly when its closed linear span is H.

Derivation or proof

The residual is orthogonal to every e_j because taking its inner product with e_j subtracts the same coefficient. Pythagoras therefore splits the norm into projection and residual, and orthonormality gives the coefficient sum. Finite partial sums are increasing and bounded, so their limit exists. The partial projections form a Cauchy sequence, with squared differences equal to sums of omitted squared coefficients. Their limit projects onto the closed span. Equality means the residual projection onto the orthogonal complement vanishes. If equality holds for all x, that complement is zero and the span is dense.

Worked example

Take $x=(2,3,6)$ in R^3 and initially observe only e_1,e_2.

The observed squared coefficient sum is $4+9=13$, total squared norm is 49, and residual squared norm is 36. Adding e_3 removes the residual exactly. Omitting e_3 never becomes harmless merely because the first two coefficients were calculated to high precision.

This is both an approximation certificate and a statement of what the measurement family cannot see.

Scope of the result

An infinite orthonormal family need not be complete. For nonorthogonal features, coefficients from raw inner products are not automatically orthogonal projections, and frame or Gram-matrix analysis is needed.

Further development

Develop a frame analogue with lower and upper frame bounds, or identify the cheapest observation family whose residual vanishes on a specified finite-dimensional model class.

References: R057

M067 Hölder and Young with equality cases

2023-09-01 · historical evidence A · Classical result developed and worked

Conjugate exponents and equality cases turn a generic inequality into a sharp tool. The proof below also states what must be checked before applying it: measurability, the relevant integrability and the distinction between the absolute-value bound and the signed integral.

Definitions and assumptions

Let $1<p<\infty$ and $q=p/(p-1)$, so $1/p+1/q=1$. Norms are $||f||_p=(\int|f|^p)^{(1/p)}$ and the analogous q-norm, on one fixed measure space.

Young and Hölder with sharp equality

For $a,b\geq 0$, $ab\leq a^p/p+b^q/q$, with equality exactly when $a^p=b^q$. Consequently $\int|fg|\leq||f||_p||g||_q$. For nonzero finite norms, equality holds exactly when $|f|^p/||f||_p^p=|g|^q/||g||_q^q$ almost everywhere.

Derivation or proof

Fix a and minimize $h(b)=b^q/q-ab+a^p/p$ over $b\geq 0$. Its derivative is $b^{(q-1)}-a$, with its unique minimum at $b=a^{(p-1)}$, and the minimum equals zero. Normalize nonzero f,g to unit p- and q-norms, apply the scalar inequality pointwise and integrate. Equality in the integrated nonnegative difference requires pointwise equality almost everywhere, which yields the normalized-power condition. To obtain equality for $|\int fg|$ as well, the product must also have an almost-everywhere constant phase or sign.

Worked example

Choose $p=3$, $q=3/2$, $a=2$ and $b=4$.

Here $a^p=b^q=8$, so Young is exact: $ab=8=8/3+8/(3/2)$. For a finite counting measure, proportional normalized powers supply equally explicit Hölder equality examples. Choosing disjointly supported nonzero functions makes the product integral zero while both norms are positive, showing a strict case.

The equality condition identifies when a proposed estimate is sharp and which directions are responsible for slack.

Scope of the result

The endpoints $p=1,q=\infty$ require essential-supremum language and a different equality statement. If either required norm is infinite, the finite useful estimate is unavailable even when a formal product expression can be written.

Further development

Use the equality characterization to optimize a linear functional on an L^p unit ball and compare existence and uniqueness in endpoint versus strictly convex cases.

References: R057

M068 Matrix norms and the quantity being bounded

2023-09-01 · historical evidence A · Classical result developed and worked

Choosing a norm means choosing how input and output error are measured. A norm catalogue becomes useful only when attached to its induced error inequality. A two-dimensional example shows that several plausible norms report different numerical amplification.

Definitions and assumptions

The induced p-norm is $||A||_p=\sup_{x\neq0}||Ax||_p/||x||_p$. The induced 1-norm is the largest absolute column sum, the induced infinity-norm is the largest absolute row sum, and the Frobenius norm is a Euclidean norm on entries.

Sharp induced bounds and singular-value comparisons

For every vector x, $||Ax||_p\leq||A||_p||x||_p$. In Euclidean norm, $||A||_2$ is the largest singular value, while $||A||_F^2$ is the sum of squared singular values; hence $||A||_2\leq||A||_F\leq\sqrt{r}||A||_2$ for rank $r>0$.

Derivation or proof

The first inequality is the definition of the supremum. For p=1, triangle inequalities bound each output coordinate and collecting each $|x_j|$ gives the maximum column sum; choosing a coordinate vector at a maximizing column attains it. The infinity case uses a maximizing row and matching signs. Orthogonal or unitary invariance reduces the Euclidean claims to a diagonal singular-value matrix, where the largest entry bounds their squared sum above and below. The rank-zero case has both norms zero.

Worked example

Let $A=[[1,2],[0,1]]$.

Its induced 1- and infinity-norms are both three. Its Frobenius norm is $\sqrt{6}$. The eigenvalues of $A^TA=[[1,2],[2,5]]$ are $3\pm2\sqrt{2}$, so $||A||_2=\sqrt{(3+2\sqrt{2})}=1+\sqrt{2}$. These are distinct measures of the same operator. A claim of amplification three is sharp for a 1-norm or infinity-norm question, but overestimates the sharp Euclidean factor.

Reporting the input and output norms is part of stating the bound, not a cosmetic implementation detail.

Scope of the result

Condition numbers also depend on the norm and require invertibility or a specified pseudoinverse/restricted subspace. An entrywise norm cannot replace an induced one without tracking the comparison constants.

Further development

For a chosen observation map, compute the restricted operator norm on the admissible state subspace and quantify how much a generic full-space bound overestimates the actual errors.

References: R001

M069 Lyapunov conditions in the right direction

2023-09-03 · historical evidence A · Classical result developed and worked

A Lyapunov certificate is a sufficient argument tied to a chosen function. Failure of one function is not a system-instability certificate. The exact example below resolves that logical error and produces a contracting quadratic geometry for the same stable system.

Definitions and assumptions

For $\dot{x}=Ax$ and symmetric P, define $V(x)=x^{T}Px$. Positive definiteness $P>0$ makes V comparable with $||x||^{2}$. Along a trajectory, $\dot{V}=x^{T}(A^{T}P+PA)x$. A Hurwitz matrix has all eigenvalues in the open left half-plane.

Quadratic certificate and a counterexample to the failed converse

If $P>0$ and $A^{T}P+PA=-Q$ with $Q>0$, then $V(t)\leq\exp(-\lambda_{min}(Q)t/\lambda_{max}(P))V(0)$, proving exponential stability. A stable A can nevertheless make $d||x||^{2}/dt$ positive at some state.

Derivation or proof

The derivative identity follows by the product rule. Since $x^{T}Qx\geq\lambda_{min}(Q)||x||^{2}$ and $V\leq\lambda_{max}(P)||x||^{2}$, $\dot{V}\leq-[\lambda_{min}(Q)/\lambda_{max}(P)]V$. Integration gives exponential decay of V; the lower eigenvalue bound on P translates this into a Euclidean state bound. The proof makes no assertion that every positive function has a negative derivative. For Hurwitz A, the integral $P=\int_{0}^{\infty}\exp(tA^{T})Q\exp(tA)dt$ provides the converse existence result, with convergence following from exponential matrix decay.

Worked example

Take $A=[[-1,4],[0,-1]]$, whose eigenvalues are both -1.

At $x=(1,1)$, $Ax=(3,-1)$ and $d(x^{T}x)/dt=2x^{T}Ax=4>0$. Yet $P=[[1/2,1],[1,9/2]]$ has positive leading minor and determinant $5/4$, and direct multiplication gives $A^{T}P+PA=-I$. Thus the Euclidean norm can temporarily grow while the certified P-norm decreases.

The apparent contradiction disappears once the certificate is interpreted in the right direction and its metric is stated.

Scope of the result

A certificate for one linearization is not automatically a global nonlinear proof. Semidefinite Q needs an invariance argument for asymptotic convergence, and strict inequalities matter near marginal modes.

Further development

Find a common quadratic Lyapunov function for a specified family of matrices, or exhibit why switching among individually stable matrices defeats a proposed common bound.

References: R054

M070 Units of the Einstein equation

2023-09-09 · historical evidence U · Classical result developed and worked

Dimensional statements about tensor components must name the coordinate or frame convention. Using a local orthonormal frame and $x^0=ct$ makes the audit simple, while also explaining why coordinate components in other charts can carry different units.

Definitions and assumptions

Use metric signature $(-,+,+,+)$, spacetime coordinates of length dimension, and a dimensionless orthonormal metric. Write G_N for Newton's constant, c for speed, ε for energy density and $G_{\mu\nu}$ for the Einstein tensor; the two symbols G have unrelated types.

A consistent dimensional audit of Einstein's equation

In this convention $G_{\mu\nu}$ and $\Lambda g_{\mu\nu}$ have dimensions L^{-2}; $T_{\mu\nu}$ has dimensions $ML^{-1}T^{-2}$. Since $[G_N]=L^3M^{-1}T^{-2}$ and $[c^4]=L^4T^{-4}$, $(G_N/c^4)T_{\mu\nu}$ has dimension L^{-2}. The factor 8π is dimensionless and is not selected by dimensional analysis.

Derivation or proof

A connection coefficient contains one derivative of a dimensionless metric, giving L^{-1}, and curvature has a derivative of the connection or a product of two connections, giving L^{-2}. Energy per volume is $(ML^2T^{-2})/L^3=ML^{-1}T^{-2}$. Multiplying exponents in $G_N\,c^{-4}T$ gives zero mass exponent, length exponent -2 and zero time exponent. The cosmological constant must have the same curvature dimension when it multiplies the dimensionless metric. All statements are frame and coordinate dependent at the component level, while the tensor equation is geometric.

Worked example

Consider a perfect-fluid stress tensor in its local rest frame.

$T_{00}=\varepsilon$ and $T_{ii}=p$ in the orthonormal convention, so pressure and energy density have matching units. If instead time is represented by t rather than ct, changing the coordinate basis introduces c factors into the time components; one must transform the tensor rather than recycle the orthonormal component units.

The repaired equation is $G_{\mu\nu}+\Lambda g_{\mu\nu}=(8\pi G_N/c^4)T_{\mu\nu}$. It passes a dimensional audit without confusing Newton's constant, curvature or matter units.

Scope of the result

The audit cannot fix signs, the numerical coupling or a curvature convention. Those require definitions and a limiting calculation. Natural units $c=1$ suppress dimensions in notation but do not remove the need to restore them consistently.

Further development

Carry the same equation through an explicit coordinate rescaling and verify the transformed component units and physical observables.

References: R058

M071 Why the coefficient is 8π

2023-09-09 · historical evidence U · Classical result developed and worked

The coefficient 8π is determined by matching a conventioned relativistic field equation to Newtonian Poisson gravity. The trace reversal supplies a crucial factor one half. This is a derivation of the classical normalization, not a new gravitational law.

Definitions and assumptions

Set $\Lambda=0$ for the local weak-field limit. Use signature $(-,+,+,+)$, $x^0=ct$ and $R^\rho_{\sigma\mu\nu}=\partial_\mu\Gamma^\rho_{\nu\sigma}-\partial_\nu\Gamma^\rho_{\mu\sigma}+\Gamma^\rho_{\mu\alpha}\Gamma^\alpha_{\nu\sigma}-\Gamma^\rho_{\nu\alpha}\Gamma^\alpha_{\mu\sigma}$, with $R_{\sigma\nu}=R^\rho_{\sigma\rho\nu}$. Let $G_{\mu\nu}=\kappa T_{\mu\nu}$ initially have an unknown κ.

Newtonian matching yields $\kappa=8\pi G_N/c^4$

For a static weak potential with $g_{00}=-(1+2\Phi/c^2)$, slow matter and negligible pressure, $R_{00}\approx\nabla^2\Phi/c^2$ and the trace-reversed source is $T_{00}-g_{00}T/2\approx\rho c^2/2$. Matching $\nabla^2\Phi=4\pi G_N\rho$ therefore gives $\kappa=8\pi G_N/c^4$.

Derivation or proof

Contract the field equation in four dimensions: $-R=\kappa T$. Substituting back gives $R_{\mu\nu}=\kappa(T_{\mu\nu}-g_{\mu\nu}T/2)$. For dust at rest, $T_{00}\approx\rho c^2$, spatial diagonal stresses are negligible, and $T\approx-\rho c^2$; with $g_{00}\approx-1$ the displayed source becomes $\rho c^2-\rho c^2/2$. The weak, static connection is $\Gamma^i_{00}\approx\partial_i\Phi/c^2$. In R_{00}, time derivatives vanish and connection products are higher order, leaving $\partial_i\Gamma^i_{00}\approx\nabla^2\Phi/c^2$. Thus $\nabla^2\Phi=(\kappa c^4/2)\rho$. Equating its coefficient with Poisson's coefficient fixes κ.

Worked example

Track the factors before inserting the answer.

If one incorrectly equates R_{00} with $\kappa\rho c^2$ without trace reversal, the inferred coupling would differ by a factor of two. If one uses time rather than ct, the connection and stress components must both be rescaled. Either mistake can produce superficially plausible but inconsistent equations.

The numerical factor and powers of c are fixed together once geometric, matter and Newtonian conventions are stated.

Scope of the result

This leading-order derivation assumes weak fields, slow motion, small stresses and a regime in which Λ is negligible. Reversing the Riemann-tensor sign requires changing the field-equation sign convention consistently. The result does not derive Einstein's equation uniquely from Newtonian gravity.

Further development

Continue to the first post-Newtonian corrections and identify which additional observations distinguish relativistic field equations sharing the same Newtonian limit.

References: R058

M072 Sixteen entries but ten independent components

2023-09-09 · historical evidence U · Classical result developed and worked

There are several counts in general relativity, and conflating them produces incorrect conclusions. Matrix entries, independent tensor components, differential identities, initial-data constraints and physical propagating modes belong to different stages of the analysis.

Definitions and assumptions

Use indices $\mu,\nu\in\{0,1,2,3\}$. A symmetric covariant tensor satisfies $S_{\mu\nu}=S_{\nu\mu}$. The Einstein tensor obeys four contracted Bianchi identities. In a 3+1 formulation, the spatial metric and its conjugate momentum have six components each.

Component count and local vacuum degree-of-freedom count

A symmetric four-by-four tensor has ten independent components: four diagonal and six off-diagonal. The four contracted Bianchi identities are differential identities, not four duplicate entries. For regular vacuum general relativity, four first-class constraints acting on twelve spatial canonical variables leave four physical phase-space variables, or two configuration polarizations per spatial point.

Derivation or proof

For an n-by-n symmetric matrix there are n diagonal entries and $n(n-1)/2$ unordered off-diagonal pairs, giving $n(n+1)/2$. At $n=4$ this is ten, although the full display contains sixteen slots. The Bianchi relations differentiate combinations of field-equation components; they do not identify any selected four slots algebraically. In the canonical count, a first-class constraint imposes one phase-space condition and generates one gauge direction. Four such constraints therefore remove eight of the twelve spatial canonical variables. Dividing the remaining four phase-space variables into coordinate-momentum pairs gives two local propagating degrees of freedom.

Worked example

Write the first row as $S_{00},S_{01},S_{02},S_{03}$ and the remaining rows with $S_{10}=S_{01}$ and the corresponding symmetries.

There is no S_{44} in the chosen zero-through-three convention. A one-through-four convention may use it, but mixing the two is invalid. In a linearized plane-wave gauge, the remaining transverse-traceless amplitudes can be represented by h_+ and $h_\times$, illustrating rather than replacing the constrained Hamiltonian count.

The repaired statement retains all ten independent equations while explaining their differential dependence and gauge structure.

Scope of the result

Boundary conditions, global moduli, matter fields and singular constraint surfaces can affect global degrees of freedom. Subtracting four identities from ten components is not on its own a derivation of two propagating modes.

Further development

Perform an explicit linearized gauge reduction for a plane wave and compare it with the canonical count in a stated background and boundary setting.

References: R059

M073 Covariant conservation is a structural condition

2023-09-09 · historical evidence U · Classical result developed and worked

Covariant conservation is the compatibility condition supplied by the geometry of the field equation. It does not say that coordinate components are constant, or that a global conserved total energy exists in every spacetime. A homogeneous expanding metric gives an exact example of the distinction.

Definitions and assumptions

For the Levi-Civita connection, $\nabla g=0$ and the contracted Bianchi identity is $\nabla^{\mu}G_{\mu\nu}=0$. Let κ and Λ be constants. A perfect fluid has rest energy density ε, pressure p and four-velocity normalized by $u\cdot u=-1$ in a $c=1$ convention.

Source conservation and the FLRW continuity equation

The field equation $G_{\mu\nu}+\Lambda g_{\mu\nu}=\kappa T_{\mu\nu}$ with $\kappa\neq 0$ implies $\nabla^{\mu}T_{\mu\nu}=0$. In a homogeneous FLRW metric, this includes $\dot{\varepsilon}+3H(\varepsilon+p)=0$, where $H=\dot{a}/a$. For constant equation-of-state parameter $p=w\varepsilon$, $\varepsilon a^{3(1+w)}$ is constant.

Derivation or proof

Apply the covariant divergence to the field equation. The geometric identity and metric compatibility eliminate the left side, and constant κ gives the conclusion. In comoving FLRW coordinates, expanding $\nabla_{\mu}T^{\mu 0}=0$ combines the ordinary time derivative of ε with the connection terms associated with the changing spatial volume a^3; isotropic pressure contributes three identical work terms. Substitution $p=w\varepsilon$ yields $d\log\varepsilon/dt=-3(1+w)d\log a/dt$, whose integral gives the power law wherever ε and a are positive.

Worked example

Compare dust and radiation under a scale factor that doubles.

For dust $w=0$, energy density falls by $2^3=8$. For radiation $w=1/3$, it falls by $2^4=16$. Neither changing component violates covariant conservation. The additional radiation factor expresses the pressure-work or redshift contribution in this model.

A conservation law can involve geometric transport and changing volume. Ordinary component constancy is not the right test.

Scope of the result

A global conserved charge generally needs an appropriate symmetry or boundary construction. If κ or Λ varies, additional derivative terms enter and the simple inference must be revised. Matter models must satisfy their own evolution equations consistently.

Further development

Given a Killing field ξ, derive conservation of J^μ=T^{μν}ξ_ν for symmetric covariantly conserved T, and identify the boundary conditions needed to turn it into a constant integrated charge.

References: R058

M074 Grothendieck universes and their logical cost

2023-09-30 · historical evidence A · Classical result developed and worked

Universe language is a relative-size discipline backed by a set-theoretic assumption. It is not a mechanism for eliminating every paradox without cost. The useful mathematical task is to state the closure axioms and show exactly which categorical constructions they permit.

Definitions and assumptions

A Grothendieck universe U is a transitive set closed under pairing, power sets and unions indexed by a set belonging to U, with the family’s members in U. To support ordinary infinite mathematics one asks ω∈U. A U-small set is an element of U, which differs from merely being a subset of U.

The inaccessible-cardinal construction of a universe

If κ is a strongly inaccessible cardinal, then V_κ is a Grothendieck universe containing ω. Consequently U-small sets, maps, finite products and function sets can be handled inside one fixed relative-size level. The existence of such an uncountable inaccessible cardinal is an additional assumption, not a theorem of bare ZFC.

Derivation or proof

Transitivity of the cumulative hierarchy is immediate. If x,y∈V_κ, their ranks are below κ, so pairing and power sets still have rank below κ because κ is a limit ordinal. For an indexed family (x_i)_{i∈I} with I∈V_κ, strong inaccessibility ensures every set in V_κ has cardinality below κ. Regularity then bounds the supremum of the fewer-than-κ ranks of x_i below κ, so their union remains in V_κ. For function sets X→Y, their graphs are subsets of X×Y and are therefore contained in a suitable iterated power-set construction inside U.

Worked example

Take two U-small sets X and Y.

Ordered pairs can be constructed by the Kuratowski definition, yielding X×Y∈U. Every function graph lies in P(X×Y), and the collection of all such graphs is a subset of that power set, hence U-small under the closure axioms. By contrast, the entire collection of all U-small sets is U itself; asserting U∈U would contradict foundation. It belongs to a larger universe only when another size assumption is made.

The construction makes size bookkeeping precise while retaining a visible logical boundary.

Scope of the result

This treatment proves the conditional construction, not consistency or existence of inaccessible cardinals. Degenerate finite universes and conventions excluding or including ω must be distinguished. Different foundational treatments can avoid universe axioms for specific constructions.

Further development

Rebuild one category-of-structures construction using a fixed cardinal bound instead, then compare its smallness guarantees with the universe-based version.

References: R060, R061

M075 Stirling partitions do not supply torsional energy

2023-09-30 · historical evidence A · Classical result developed and worked

The original torsion comparison combined two unrelated notions of structure: set partitions and elastic deformation. Both can be developed, but a counting theorem does not determine an energy. An explicit discrete energy and continuum limit provide the missing bridge when an actual rod model is intended.

Definitions and assumptions

S(N,k) counts partitions of N labeled objects into k nonempty unlabeled subsets. A contiguous partition of an ordered chain instead cuts k−1 of its N−1 gaps. For a rod of length L, θ(x) is an angular displacement and GJ>0 is a torsional stiffness parameter.

Distinct partition counts and a consistent torsional energy

Arbitrary partitions satisfy S(N,k)=kS(N−1,k)+S(N−1,k−1); contiguous k-block partitions number C(N−1,k−1). For grid spacing h=L/N, E_N=(GJ/2)Σ_{i=0}^{N−1}[(θ_{i+1}−θ_i)/h]²h equals the continuum Dirichlet energy of the piecewise-linear interpolant.

Derivation or proof

For Stirling’s recurrence, the last label either joins one of k existing blocks or forms a singleton. For contiguous partitions, choosing the cut gaps is a bijection. For energy, the interpolant derivative is constant on each grid interval and equals the difference quotient; integrating its squared derivative gives the formula exactly. Thus strong H^1 convergence of interpolants implies energy convergence. A bounded-energy sequence with a fixed endpoint has an H^1 bound; weak lower semicontinuity gives a lower energy bound at weak limits, while interpolating smooth functions supplies elementary recovery approximations. No set-partition count is needed for this argument.

Worked example

First take N=4,k=2, then a constant total twist Θ.

The partition counts are seven and three respectively. For θ_i=Θi/N, every difference quotient is Θ/L and E_N=GJΘ²/(2L) for every N. For L=GJ=1 and θ(x)=x², endpoint interpolation gives E_N=2/3−1/(6N²), converging to (1/2)∫₀¹4x²dx=2/3.

The continuum energy follows from a stated stiffness law and interpolation, not from taking a limit of the Stirling numbers.

Scope of the result

The one-dimensional quadratic energy assumes a chosen linear elastic model. Material nonlinearities, warping, varying stiffness and topology of configurations require additional specification. Algebraic torsion in a group is a different concept again.

Further development

If partitions represent actual latent material domains, define a weighted sum over contiguous partitions with a stated domain energy and examine its continuum or large-system limit.

M076 Vector motion and rotation conventions

2023-09-23 · historical evidence A · Classical result developed and worked

Movement commands can be formulated without ambiguous appeals to trigonometry. Linear algebra combines the local commands; a rotation converts the resulting vector into another frame. Keeping magnitude conventions explicit resolves diagonal-speed discrepancies.

Definitions and assumptions

In a right-handed plane use column vectors and counterclockwise rotation $R(\theta)=[[\cos\theta,-\sin\theta],[\sin\theta,\cos\theta]]$. Let a local command $u=(u_1,u_2)$ specify components in orthonormal local axes. World velocity is $v=su$ rotated by R, namely $v=sR(\theta)u$.

Rotation preserves speed but raw diagonal input changes it

> Because $R^TR=I$, $||v||_2=s||u||_2$. A single unit-axis command has speed s, while raw input $u=(1,1)$ has speed $s\sqrt{2}$. Normalizing nonzero input by its Euclidean norm produces constant speed s regardless of its direction.

Derivation or proof

Multiply the displayed matrix by its transpose and use $\cos^2\theta+\sin^2\theta=1$ and cancellation of the cross terms. Taking squared norms then gives the identity. The difference between axis and diagonal commands follows from $||(1,0)||=1$ and $||(1,1)||=\sqrt{2}$. If the desired control preserves analog intensity below unit radius, the mapping $u\mapsto u/\max(1,||u||)$ leaves small commands unchanged and caps only those outside the unit disk. This is a specified control law, not a consequence forced by coordinate geometry.

Worked example

Use the exact rotation with $\cos\theta=3/5$ and $\sin\theta=4/5$.

The unit forward command maps to $(3/5,4/5)$, of squared norm one. The unnormalized diagonal maps to $(-1/5,7/5)$, of squared norm two. Its normalized version divides both components by $\sqrt{2}$, returning squared norm one. These checks use exact rational arithmetic before the final normalization.

Coordinate conversion and speed policy can be independently tested. This makes it possible to change a frame without silently changing the physical or numerical interpretation of commands.

Scope of the result

Nonorthogonal coordinate bases, perspective projection, acceleration limits and a non-Euclidean metric alter the formula or the appropriate norm. Time-dependent frames introduce additional derivative terms when differentiating positions.

Further development

Specify an anisotropic metric or acceleration constraint, then derive the admissible velocity set and a frame-consistent control projection onto it.

M077 A proposed analogy needs a dimensionless correspondence

2023-09-23 · historical evidence A · Classical result developed and worked

A spectral analogy becomes mathematical only when state spaces, operators, units and observables are related by an explicit map. A small periodic hopping model supplies a genuine Bloch calculation, while a similarity-transform example separates shared eigenvalues from shared dynamics.

Definitions and assumptions

Consider a two-site unit cell on an infinite one-dimensional lattice with real hoppings t_1,t_2. Its Hermitian Bloch matrix is H(k)=[[0,t_1+t_2e^(−ik)],[t_1+t_2e^(ik),0]], where k is a dimensionless crystal phase. Hopping parameters and eigenvalues have a common energy unit.

An explicit band gap and a warning about spectral equivalence

The two bands are E_±(k)=±√(t_1^2+t_2^2+2$t_1t_2$cosk). Their minimum separation is 2||t_1|−|t_2||. Equality of eigenvalue sets in another model does not establish equivalence of observation maps or Euclidean amplification.

Derivation or proof

The characteristic polynomial is E^2−|t_1+t_2e^(−ik)|². Expanding the squared modulus gives the dispersion relation. Its minimum is (|t_1|−|t_2|)² because cosk ranges over [−1,1]. If $B=SAS^{-1}$, A and B share characteristic polynomials, but exp(tB)=S exp(tA)S^{-1} and Euclidean norm bounds acquire the condition number of S. Only a unitary S preserves every Euclidean norm and the inner-product structure automatically. Observables must also transform, for example $C_B=C_A S^{-1}$, to preserve measured outputs.

Worked example

Take $t_1=2,t_2=1$ and compare an unrelated rescaled two-state model.

The bands at k=0 are ±3 and at k=π are ±1, so the gap is two. To transfer this statement, another model must supply a compatible periodic operator, an energy-scale conversion and a map respecting the relevant measurements. A match between an energy number and some unrelated flux number has no meaning until units and maps are supplied.

This example yields an actual operator theorem and a precise test for proposed cross-domain correspondences.

Scope of the result

The toy model does not establish that another physical system has Bloch periodicity, the same Hilbert space or the same operator domain. Isospectral operators can remain distinguishable by eigenvectors, locality and response functions.

Further development

Classify intertwiners BS=SA that also preserve a chosen observation map, and determine which spectral or transport identities survive after taking dimensionless variables.

References: R062

E Analysis geometry and computational certificates

M078 Residual energy and exact Gaussian calibration

2023-10-04 · historical evidence A · Developed classical result with worked example

The original squared-deviation formula becomes useful once its units, reference model and covariance are specified. A squared residual is a geometric discrepancy. An entropy is a functional of a probability distribution. Neither name determines the other.

Definitions and assumptions

Let $r=X-\mu\in R^d$. The raw residual energy is $E=r^Tr$. If Σ is a specified positive-definite covariance, the whitened discrepancy is $Q=r^T\Sigma^{-1}r$.

Here μ and Σ are known constants under the null model, rather than estimates obtained from the same observations.

Whitened Gaussian discrepancy

If X has multivariate normal distribution $N(\mu,\Sigma)$ with Σ positive definite, then Q has the χ^2 distribution with d degrees of freedom. Consequently $E[Q]=d$ and $Var(Q)=2d$. The unweighted E generally depends on the coordinate units.

Derivation or proof

Factor $\Sigma=LL^T$ with L invertible and put $Z=L^{-1}r$. The change-of-variables formula gives independent standard normal coordinates $Z_1,\ldots,Z_d$. Then $Q=Z^TZ=\Sigma Z_i^2$. Its moment-generating function is the product $(1-2t)^{(-d/2)}$, obtained by completing the square in each one-dimensional Gaussian integral. Differentiation at zero gives the stated moments. Changing the scale of one coordinate changes E but is canceled by the corresponding covariance transformation in Q.

Worked example

Take $r=(1,2)$ and $\Sigma=diag(1,4)$.

The raw residual is $E=1+4=5$. Whitening gives $Z=(1,1)$, so $Q=2$. If both coordinates are expressed in ten times larger numerical units, r becomes (10,20) and Σ becomes diag(100,400); E becomes 500, whereas Q remains 2.

The standardized discrepancy is dimensionless and has an exact reference distribution under the declared Gaussian null. It has not thereby become Shannon entropy.

Scope of the result

Estimating means, regression coefficients or covariance from the same sample changes the null distribution. In ordinary known-variance linear regression, orthogonal projection gives $n-p$ residual degrees of freedom; estimated covariance can instead lead to F or other statistics. Nonnormal residuals need separate calibration.

Further development

Study how Q changes under covariance uncertainty by bounding it between eigenvalue-controlled quadratic forms. This produces an honest robustness interval rather than assigning an entropy interpretation to a residual.

M079 The chain rule as a statement about local linear maps

2023-10-05 · historical evidence A · Developed classical result with worked example

The derivative is the linear part of a local approximation. The chain rule composes these linear parts and tracks their domains; a secant slope or a scalar symbol does not replace a multivariable derivative.

Definitions and assumptions

For differentiable $g:R^m \to R^n$ and $f:R^n \to R^k$, $Dg(x):R^m \to R^n$ and $Df(g(x)):R^n \to R^k$ are linear maps.

Differentiability means $g(x+h)=g(x)+Dg(x)h+o(||h||)$, with the remainder small relative to $||h||$.

Typed chain rule

$D(f \circ g)(x)=Df(g(x))Dg(x)$. If a scalar F has Hessian operator norm at most M along the segment from x to x+h, its linearization remainder is at most $M||h||^2/2$.

Derivation or proof

Insert the differentiability expansion of g into the expansion of f at g(x). Since $g(x+h)-g(x)=O(||h||)$, the outer remainder is $o(||h||)$. The coefficient of h is exactly the displayed composition. For the second statement, integrate the directional derivative along the segment twice; the second integral is $\int_0^1 (1-t)h^T H_F(x+th)h \, dt$, whose absolute value is bounded by $M||h||^2/2$.

Worked example

Let $g(u,v)=(u^2+v,e^u)$ and $f(a,b)=ab$, so $F(u,v)=(u^2+v)e^u$.

$Dg=[[2u,1],[e^u,0]]$, while $Df(g)=[e^u,u^2+v]$. Multiplication gives $DF=[e^u(2u+u^2+v),e^u]$. At (1,2), DF=(5e,e). Direct differentiation of F gives the same row vector and therefore independently checks the composition order.

A perturbation (h_1,h_2) changes F to first order by $5eh_1+eh_2$. The two inputs have different sensitivities; one scalar slope would omit that distinction.

Scope of the result

The Hessian remainder bound requires a neighborhood with a uniform bound. A pointwise derivative alone supplies no global finite-step error estimate. Nondifferentiable coordinate changes need different calculus.

Further development

For a long computation, propagate both Jacobians and explicit remainder bounds. This distinguishes a mathematically controlled sensitivity estimate from an uncalibrated finite-difference plot.

M080 Graph heat flow and the stability of a discrete gradient step

2023-10-05 · historical evidence A · Developed classical result with worked example

Heat evolution and gradient descent share a variational structure after choosing a metric and boundary conditions. Their discrete stability restrictions come from the spectrum of the actual discretized operator.

Definitions and assumptions

Let L be a symmetric positive-semidefinite graph Laplacian and $E(x)=x^{T}Lx/2$.

Continuous heat evolution is $\dot{x}=-Lx$. A forward Euler step is $x^{+}=(I-\eta L)x$. The constant vector is in the kernel when L is an ordinary graph Laplacian.

Spectral energy decrease

Continuous evolution satisfies $dE/dt=-||Lx||^{2}$. A discrete step is nonexpansive in Euclidean norm for every x exactly when $0\le\eta\le 2/\lambda_{max}$, assuming $\lambda_{max}>0$; the same interval makes E nonincreasing. Strict decay on every positive eigenspace holds for the explicit open interval $0<\eta<2/\lambda_{max}$; $\eta=0$ produces no decay.

Derivation or proof

Choose an orthonormal eigenbasis $Lq_{i}=\lambda_{i}q_{i}$ and write $x=\Sigma c_{i}q_{i}$. Continuous coefficients are $c_{i}(t)=e^{(-\lambda_{i}t)}c_{i}(0)$. For the step, $c_{i}^{+}=(1-\eta\lambda_{i})c_{i}$. Thus $||x^{+}||^{2}-||x||^{2}=\Sigma[(1-\eta\lambda_{i})^{2}-1]c_{i}^{2}$. This is nonpositive for every coefficient vector precisely when every factor has magnitude at most one. Multiplying each term by $\lambda_{i}/2$ proves the energy version. Also $1^{T}x^{+}=1^{T}x$, so mass is preserved independently of decay.

Worked example

Use the four-cycle Laplacian with eigenvalues 0,2,2,4.

For $\eta=0.4$, the update factors are $1,0.2,0.2,-0.6$, hence every nonconstant mode contracts. For $\eta=0.6$, the largest-frequency factor is -1.4 and that mode grows. The equilibrium constant mode remains unchanged in both cases.

A large time step can turn a dissipative differential equation into an unstable numerical scheme. Continuous dissipation alone cannot certify the discretization.

Scope of the result

Energy monotonicity does not imply positivity of each coordinate. Positivity of forward Euler is governed by entrywise update weights and can require a smaller bound. Nonsymmetric operators require a different argument.

Further development

Compare forward, backward and exponential steps for a common graph, recording preservation of mass, positivity and decay separately.

M081 Exact exponent arithmetic before approximation

2023-10-07 · historical evidence A · Developed classical result with worked example

Very large magnitudes are best handled by separating an exact exponent from a modest mantissa. That separation avoids overflow and makes errors in scale immediately visible.

Definitions and assumptions

Scientific notation writes a positive number as $a\times10^{k}$ with $1\le a<10$.

Relative error for an approximation $\tilde{y}$ to $y\neq0$ is $|\tilde{y}-y|/|y|$. A decimal logarithm adds $\log_{10}a$ to the exponent.

Square-root normalization

For an integer m and a>0, √(a×10^(2m))=√a×10ᵐ. For the recovered expression, √(5×10^54004)=√5×10^27002 exactly.

Derivation or proof

Both factors on the right are positive, and squaring gives 5×10^(2×27002), which is the original radicand. Uniqueness of the positive square root completes the proof. If an exponent is odd, move one factor of 10 into the mantissa before taking the square root; there is no need to evaluate the enormous number itself.

Worked example

Approximate the exact answer by 10^27002, or instead retain eleven significant mantissa digits.

The exact logarithm is $27002+(\log_{10}5)/2$. Omitting √5 gives relative error $1-1/\sqrt{5}\approx0.5527864045$, over 55 percent. Keeping 2.2360679775 as mantissa gives a small rounding error whose size can be bounded by squaring rational decimal endpoints around √5. The old comparison with 25,000×10^50,000 differs by tens of thousands of decimal orders.

An order-of-magnitude summary and a numerical approximation answer different questions. Stating the exact exponent costs essentially no extra computation.

Scope of the result

Exponent manipulations assume the usual positive real square root. Complex powers require a branch convention, and expressions with uncertain exponents need interval propagation rather than a single rounded exponent.

Further development

Use rational interval endpoints for the mantissa and retain an integer exponent in all later products, quotients and roots. This gives a small certificate that can be replayed without arbitrary-precision libraries.

M082 Competing response laws and identifiable parameters

2023-10-07 · historical evidence A · Developed classical result with worked example

A named empirical law is a family of fitted functions with a domain and an error model. A rigorous mathematical comparison first asks whether the parameters and competing shapes can be distinguished at all.

Definitions and assumptions

For $x>0$, a power response is $y=ax^b$ with $a>0$. Its logarithm is $\log y=\alpha+b \log x$, $\alpha=\log a$.

An additive Gaussian model for y and an additive Gaussian model for $\log y$ imply different likelihoods. Hick-type logarithmic response models use a different independent variable and cannot be substituted without defining the experiment.

Two-point identifiability in the noiseless power family

For two distinct positive inputs $x_1 \neq x_2$ and positive outputs, there is a unique power-law interpolant: $b=\log(y_2/y_1)/\log(x_2/x_1)$ and $a=y_1/x_1^b$. This identifies a family member, not the truth of the family.

Derivation or proof

Taking logarithms makes the problem a two-equation linear system with design rows $(1,\log x_1)$ and $(1,\log x_2)$. Its determinant $\log(x_2/x_1)$ is nonzero, so the solution is unique. A third input is therefore already an opportunity to reject the noiseless family rather than simply refit its two parameters. Under noise, a likelihood and uncertainty model are needed; logging observations changes the effective error structure.

Worked example

Consider synthetic observations at $x=1,2,4$ with $y=3,12,48$.

The first two pairs give $b=2$ and $a=3$. The third output predicted by the same parameters is $3\times4^2=48$. If the third output were 45, the exact interpolation claim would fail. A measurement error model would decide whether that discrepancy is tolerable; the name of the law cannot decide it.

The example supplies a clear algebraic diagnostic without asserting any real psychophysical mechanism.

Scope of the result

Three agreeing points do not establish extrapolation, uniqueness among all functional families, or a causal explanation. Fitting and evaluating on the same observations measures interpolation.

Further development

Compare out-of-sample log likelihoods or explicitly bounded approximation errors for power, logarithmic and saturating families, with the same observation protocol and parameter accounting.

M083 Tensor order and matrix rank are different invariants

2023-10-07 · historical evidence T · Developed classical result with worked example

A second-order tensor may be represented by a matrix, but its order is not the number of nonzero eigenvalues. A dyadic decomposition adds a third meaning of rank that needs a stated field and symmetry convention.

Definitions and assumptions

For a real matrix A, rank(A) is the dimension of its image. A rank-one dyad uv^T maps x to $u(v^Tx)$.

For a real symmetric A, the spectral theorem gives an orthonormal eigenbasis and $A=\Sigma\lambda_i v_i v_i^T$. Negative coefficients are permitted in this signed decomposition.

Minimal dyadic count for a symmetric matrix

The minimum number of unrestricted real rank-one matrix summands is rank(A). A symmetric matrix has a signed symmetric dyadic representation with exactly that many nonzero terms. A decomposition into positive-weight vv^T terms exists only for positive-semidefinite A.

Derivation or proof

The image of a sum of r rank-one maps is contained in the span of their r output vectors, so its rank is at most r. Therefore at least rank(A) summands are necessary. The spectral theorem supplies exactly rank(A) nonzero signed symmetric dyads. Finally every sum $\Sigma c_i v_i v_i^T$ with $c_i \geq 0$ has nonnegative quadratic form, so an indefinite matrix cannot possess such a decomposition.

Worked example

Let $A=\mathrm{diag}(-2,0,3)$.

Its tensor order is two, its matrix rank is two, and its signed dyadic decomposition is $-2e_1e_1^T+3e_3e_3^T$. At e_1 the quadratic form equals -2, excluding any sum of positive dyads. Replacing the zero entry by $\varepsilon \neq 0$ changes rank to three even when $||A_\varepsilon - A||_2 = |\varepsilon|$ is arbitrarily small.

Exact rank is discontinuous at a rank-deficient matrix; numerical effective rank requires a declared threshold and noise scale.

Scope of the result

These conclusions concern matrices. Higher-order tensor rank, symmetric rank and border rank are different problems, and a matrix spectral decomposition does not solve them.

Further development

Study a noisy matrix through singular-value intervals and separate exact algebraic rank from a tolerance-dependent approximation rank.

M084 Directional cancellation as a limited shearlet illustration

2023-10-07 · historical evidence T · Developed classical result with worked example

A useful entry point to shearlets is to show why a directional high-pass filter can avoid wasting coefficients along a straight edge. This finite calculation illustrates a mechanism, not a theorem of universal representation superiority.

Definitions and assumptions

An anisotropic dilation can be A_a=diag(a,√a), and a shear S_s=[[1,s],[0,1]]. A normalized analyzing function has form a^(−3/4)ψ(A_a⁻¹S_s⁻¹(x−t)).

The factor a^(−3/4) preserves the L² norm because det A_a=a^(3/2). A full shearlet frame additionally requires a designed generator, sampling and frame bounds.

Exact cancellation along a discrete straight edge

For the finite binary image F(i,j)=1[j≥i], the tangent difference F(i+1,j+1)−F(i,j) vanishes wherever both samples exist. The horizontal difference F(i+1,j)−F(i,j) is nonzero on the diagonal.

Derivation or proof

Adding one to both i and j preserves the inequality j≥i, so the tangent statement follows pointwise. Holding j fixed while increasing i by one changes the truth value exactly when j=i. Thus a coordinate direction aligned with the edge recognizes a coherent geometric feature that an unaligned difference repeatedly crosses. The normalization formula follows separately from a change of variables in the L² integral.

Worked example

Use a 16×16 sample grid and compare valid interior differences.

All 225 tangent differences are zero. Among the 240 valid horizontal differences, 15 are nonzero. Reversing or rotating the edge changes which directional operator cancels, so the favorable orientation is part of the setup. A separable wavelet can itself be extremely sparse for an axis-aligned edge.

The calculation validates an orientation mechanism and its restrictions. It does not compare complete reconstruction errors, coefficient budgets or full shearlet and wavelet systems.

Scope of the result

Nearly optimal approximation theorems concern specified cartoon-like regularity classes and particular frames. They are not established by this image experiment and do not extend to every signal.

Further development

Implement matched complete transforms and plot best-N-term reconstruction errors under a declared edge model, coefficient budget and boundary convention.

References: R063

M085 A fully normalized spin one-half Wigner representation

2023-10-07 · historical evidence T · Developed classical result with worked example

The spin phase space is a sphere, and its kernel and measure must be chosen together. For a qubit the complete reconstruction can be derived from elementary spherical moments, including the appearance of negative quasiprobabilities.

Definitions and assumptions

Write $A=a_0 I+a\cdot\sigma$, with Pauli matrices $\sigma=(\sigma_x,\sigma_y,\sigma_z)$. For Hermitian A, the four coefficients are real.

Choose $\Delta(n)=(I+\sqrt{3}\, n\cdot\sigma)/2$ for unit $n\in S^2$ and measure $d\mu=d\Omega/(2\pi)$, whose total mass is two. Define $W_A(n)=\mathrm{Tr}(A\Delta(n))$.

Self-dual reconstruction and trace pairing

$W_A(n)=a_0+\sqrt{3}a\cdot n$, $A=\int W_A(n)\Delta(n)d\mu$ and $\mathrm{Tr}(AB)=\int W_A W_B d\mu$. For a density matrix $\rho=(I+r\cdot\sigma)/2$, $\int W_\rho d\mu=1$ even though W_ρ need not be nonnegative.

Derivation or proof

Use $\mathrm{Tr}\sigma_i=0$ and $\mathrm{Tr}(\sigma_i\sigma_j)=2\delta_{ij}$. The spherical moments are $\int 1 d\mu=2$, $\int n_i d\mu=0$ and $\int n_i n_j d\mu=(2/3)\delta_{ij}$. Substitution gives the symbol immediately. Expanding $W_A\Delta$ and integrating leaves $a_0 I+a\cdot\sigma$. Expanding $W_A W_B$ gives $2a_0b_0+2a\cdot b$, exactly $\mathrm{Tr}(AB)$. These calculations also fix the measure normalization; using $d\Omega/(4\pi)$ without changing other factors would break reconstruction.

Worked example

Take the pure spin-up state $r=(0,0,1)$.

Its symbol is $W(n)=(1+\sqrt{3}n_z)/2$, with minimum $(1-\sqrt{3})/2<0$ at the south pole. The six directions $\pm e_x,\pm e_y,\pm e_z$ with weights $1/3$ each reproduce all required spherical moments exactly, so the reconstruction and trace identity can be checked by a six-point sum.

Negativity is compatible with reconstruction because $\Delta(n)$ is not a positive measurement effect. Treating W as an ordinary probability density would lose this distinction.

Scope of the result

This derivation concerns spin one-half and a specific self-dual convention. Higher-spin kernels have more multipoles, and changing conventions changes constants and symbol products.

Further development

Derive the induced star product from matrix multiplication and study its finite spherical-harmonic content before comparing it with a flat Moyal product.

References: R064, R065

M086 Exact polynomial Moyal products

2023-10-07 · historical evidence T · Developed classical result with worked example

On polynomial symbols the Moyal exponential terminates, so one can verify noncommutativity and associativity with exact arithmetic rather than relying on an informal infinite series.

Definitions and assumptions

For canonical coordinates (q,p), let $B=\overleftarrow{\partial}_q\overrightarrow{\partial}_p-\overleftarrow{\partial}_p\overrightarrow{\partial}_q$ and define $f\star g=\Sigma_{n\geq 0}(i\hbar/2)^n B^n(f,g)/n!$.

For polynomials only finitely many differentiated terms survive. The corresponding formal-series definition can be used beyond polynomials without claiming analytic convergence.

Canonical commutator and classical limit

$q\star p-p\star q=i\hbar$. More generally the antisymmetrized product divided by $i\hbar$ equals the Poisson bracket plus terms of formal order $\hbar^2$. The product is associative in the polynomial or formal setting.

Derivation or proof

At first order $q\star p=qp+i\hbar/2$ and $p\star q=pq-i\hbar/2$; higher derivatives vanish. Interchanging f and g changes the sign of B, so only odd n contribute to the commutator. After division by $i\hbar$, the n=1 term is the Poisson bracket and the next possible term has order $\hbar^2$. For associativity, introduce three tensor factors. The constant-coefficient bidifferential operators B_{12}, B_{13} and B_{23} commute. Either parenthesization is ordinary multiplication after $\exp[(i\hbar/2)(B_{12}+B_{13}+B_{23})]$, establishing equality.

Worked example

Set $\hbar=1$ and take $f=q^2$, $g=p^2$.

The finite expansion gives $q^2\star p^2=q^2p^2+2iqp-1/2$, while $p^2\star q^2=q^2p^2-2iqp-1/2$. Their commutator is $4iqp$, agreeing with i times $\{q^2,p^2\}=4qp$. The verifier also compares both parenthesizations for a family of low-degree monomials with rational complex coefficients.

The example checks a finite, exact algebra. It requires no discretized phase-space grid and therefore has no grid-resolution error.

Scope of the result

Formal associativity does not prove convergence of the Moyal series for arbitrary smooth functions. Trace-integral identities require integrability and boundary conditions; polynomial symbols on the full plane are usually not integrable.

Further development

Specify a Schwartz-class or other analytic symbol space and prove continuity or trace properties there, keeping that analysis separate from the formal calculation.

References: R066

M087 A spectral definition of fractional quantum evolution

2023-10-07 · historical evidence T · Developed classical result with worked example

A fractional differential symbol becomes a definite evolution only after specifying its self-adjoint operator, domain and boundary convention. A periodic box gives an explicit example with no ambiguous boundary prescription.

Definitions and assumptions

On the circle of length 2π, use normalized Fourier modes $e_n(x)=e^{(inx)}/\sqrt{(2\pi)}$. For $\alpha>0$, define $(-\Delta)^{(\alpha/2)}e_n=|n|^{\alpha}e_n$.

Take $\hbar=1$ and $H=D_\alpha(-\Delta)^{(\alpha/2)}$ with $D_\alpha>0$. Its domain consists of coefficients satisfying $\Sigma|n|^{(2\alpha)}|c_n|^2<\infty$.

Unitary fractional evolution on the circle

The solution of $i\psi_t=H\psi$ is $\psi(t)=\Sigma c_n e^{(-itD_\alpha|n|^{\alpha})}e_n$. It preserves the L^2 norm and the spectral energy wherever that expectation is finite. The case $\alpha=2$ recovers the usual quadratic dispersion with $D_2=1/(2m)$ in $\hbar=1$ units.

Derivation or proof

The Fourier transform is unitary from L^2 of the circle to $\ell^2(Z)$. Under it H is multiplication by the real nonnegative sequence $D_\alpha|n|^{\alpha}$ on its maximal domain, hence self-adjoint. Exponentiation multiplies each coefficient by a complex number of modulus one. Parseval therefore gives $||\psi(t)||^2=\Sigma|c_n|^2$. Multiplying by the same unchanged spectral weights proves energy conservation for states in the form domain.

Worked example

Choose $D_\alpha=1$, $\alpha=3/2$ and $\psi(0)=(e_0+e_1+e_2)/\sqrt{3}$.

The three energies are $0,1,2\sqrt{2}$. The coefficients at time t are $(1,e^{(-it)},e^{(-i2\sqrt{2}t)})/\sqrt{3}$, so their squared magnitudes still sum to one. At $\alpha=2$ the energies become 0,1,4; this is a change in dispersion, not a change of normalization.

The finite-mode example can be replayed exactly at the spectral level and numerically in physical space.

Scope of the result

On an interval, spectral, restricted and regional fractional Laplacians are generally different operators. This periodic calculation does not choose among them or validate a physical interpretation for every α.

Further development

Compare two explicitly defined bounded-domain operators using the same initial data and observable, and distinguish operator-domain effects from fractional-order effects.

References: R067

M088 Lagrange identity and its equality geometry

2023-10-07 · historical evidence T · Developed classical result with worked example

The finite-dimensional identity is an exact decomposition of the Cauchy-Schwarz deficit. An operator boundary identity sharing the same historical name is a separate theorem with separate domains.

Definitions and assumptions

For $a,b\in R^n$, define the antisymmetric coordinate minors $m_{ij}=a_i b_j-a_j b_i$.

The exterior product $a\wedge b$ has squared norm $\Sigma_{i<j}m_{ij}^2$ in the orthonormal exterior basis.

Lagrange identity

$||a||^2||b||^2-(a\cdot b)^2=\Sigma_{i<j}(a_i b_j-a_j b_i)^2$. Equality in Cauchy-Schwarz holds exactly when all these minors vanish, equivalently when the two vectors are linearly dependent.

Derivation or proof

Expand the right-hand side and rewrite the sum over $i<j$ as one half of the sum over all ordered pairs. The squared terms contribute $\Sigma_i a_i^2\Sigma_j b_j^2$. The cross terms contribute $-\Sigma_{ij} a_i a_j b_i b_j=-(a\cdot b)^2$. If $a\neq 0$ and every minor vanishes, select an index i with $a_i\neq 0$. The identities $a_i b_j=a_j b_i$ give $b=(b_i/a_i)a$. If $a=0$, dependence is immediate.

Worked example

Let $a=(1,2,3)$, $b=(2,-1,4)$.

Their squared norms are 14 and 21, and their dot product is 12, giving $294-144=150$. The three minors are $-5,-2,11$, whose squares sum to $25+4+121=150$. In three dimensions this is also $||a\times b||^2$.

The identity quantifies the area spanned by two vectors; it does not itself introduce a field equation or an elasticity model.

Scope of the result

For complex vectors the Hermitian inner product and absolute squares must be used. Green or Lagrange identities for differential operators introduce integration by parts, boundary traces and domain assumptions not present here.

Further development

For a stated PDE operator, derive its own boundary pairing and use the finite-dimensional inequality only at the particular algebraic step where it is applicable.

M089 Winding number and Gauss-Bonnet with boundaries explicit

2023-10-12 · historical evidence A · Developed classical result with worked example

Topological charges are attached to specified spaces and deformation classes. Winding number and Euler characteristic are both invariant quantities, but they are not interchangeable measurements of a generic defect.

Definitions and assumptions

For a closed C^1 loop $z:[0,2\pi]\to C\setminus\{0\}$, the winding number is $w=(1/(2\pi i))\int z'(t)/z(t)dt$.

For a closed oriented smooth surface, Gauss-Bonnet states $\int K\, dA=2\pi\chi$. A surface with boundary needs a boundary-geodesic-curvature term and possible corner contributions.

Integer winding and homotopy stability

> The displayed winding integral is an integer and remains unchanged during a continuous homotopy of loops that avoids zero. Passing through zero removes the hypothesis that protects the invariant.

Derivation or proof

Lift the argument along the parameter interval: $z(t)=r(t)e^{i\theta(t)}$ with $r>0$ and continuous θ. Then $z'/z=r'/r+i\theta'$ wherever derivatives exist. The radial integral vanishes because r has equal endpoints. Since the loop closes, $\theta(2\pi)-\theta(0)$ is an integer multiple of 2π. Under a zero-avoiding homotopy, the resulting winding is continuous in the homotopy parameter and integer-valued, hence constant. Smooth approximation extends the argument to continuous loops.

Worked example

Use $z(t)=e^{3it}$, and separately use a round sphere of radius R.

For the loop, $z'/z=3i$ and the winding is three. For the sphere, $K=R^{-2}$ and area $=4\pi R^2$, so the curvature integral is 4π and $\chi=2$. A flat disk has zero interior curvature but $\chi=1$; its circular boundary contributes 2π, demonstrating why the closed-surface formula cannot simply be reused.

Each calculation answers a specific topological question and exhibits its own necessary boundary condition.

Scope of the result

Winding depends on a chosen omitted point or target circle. Gauss-Bonnet requires sufficient regularity and the correct boundary terms. Neither invariant completely classifies arbitrary spaces or dynamical defects.

Further development

Construct a one-parameter family that crosses a zero and calculate the jump in winding; separately examine a surface family where a topology change destroys smooth-manifold hypotheses.

M090 An explicit marked-torus deformation

2023-10-12 · historical evidence A · Developed classical result with worked example

A torus supplies a complete elementary setting in which deformation parameters, marking, Beltrami coefficient and dilatation can all be written down. This replaces the old undefined sum of named geometric quantities.

Definitions and assumptions

For τ in the upper half-plane, the marked complex torus is $C/(Z+\tau Z)$. The marking identifies the two chosen lattice generators with a fixed reference basis.

A quasiconformal map has $f_{\bar{z}}=\mu f_z$ with $|\mu|<1$ almost everywhere. For a real-linear orientation-preserving map its maximal dilatation is $K=(1+|\mu|)/(1-|\mu|)$.

Rectangular torus stretch

The map $F(x+iy)=x+iby$ from the square marked torus to the rectangular marked torus of parameter ib has $\mu=(1-b)/(1+b)$ and $K=\max(b,1/b)$, for $b>0$. It realizes the classical torus Teichmüller distance $|\log b|/2$.

Derivation or proof

Since $x=(z+\bar{z})/2$ and $iy=(z-\bar{z})/2$, $F=((1+b)/2)z+((1-b)/2)\bar{z}$. Thus $F_z=(1+b)/2$, $F_{\bar{z}}=(1-b)/2$ and the coefficient follows. The real singular values are 1 and b, giving the same K as their ratio. For optimality among maps in the marking class, the standard extremal-length distortion bound applied to horizontal and vertical curve families gives K at least $\max(b,1/b)$; this affine map attains it. The extremal-length bound is the classical geometric input, not a consequence merely of the algebraic derivative calculation.

Worked example

Set $b=2$.

The map doubles the vertical period, has $\mu=-1/3$ and $K=2$. Its Teichmüller distance is $(\log 2)/2$. Replacing b by $1/2$ reverses the stretch and has the same distance. Changing the marking by an integer modular transformation is a separate quotient operation.

Every number in this example has a defined geometric role, unlike an arbitrary linear combination of unrelated invariants.

Scope of the result

The affine torus formula does not classify extremal maps of higher-genus surfaces. Teichmüller space and the moduli-space quotient should not be identified.

Further development

For a specified higher-genus surface, choose a quadratic differential and formulate the associated deformation with its normalization and marking retained.

References: R068

M091 Certified first logistic-map period doublings

2023-10-16 · historical evidence A · Developed classical result with worked example

The first two bifurcation parameters can be obtained algebraically, so they provide exact calibration points before any numerical Feigenbaum-ratio experiment.

Definitions and assumptions

Let f_r(x)=rx(1−x), with r in [0,4] and x in [0,1]. A period-k orbit is stable when the absolute value of the derivative of f_r^k along it is less than one.

A flip bifurcation is detected when this multiplier passes through −1, subject to the usual nondegeneracy conditions.

First two flip thresholds

The nonzero fixed point $x^*=1-1/r$ loses linear stability at $r_1=3$. The genuine two-cycle, existing for $r>3$, has multiplier $m_2=-r^2+2r+4$ and is stable for $3<r<1+\sqrt{6}$. At $r_2=1+\sqrt{6}$ the multiplier crosses −1 transversely; the classical flip theorem applies and produces the standard supercritical period-four branch.

Derivation or proof

At the fixed point, $f'_r(x^*)=r-2r(1-1/r)=2-r$, so its stability interval is $1<r<3$. Dividing $f_r^2(x)-x$ by the fixed-point factor leaves the two-cycle quadratic $r^2x^2-r(r+1)x+(r+1)$. Its roots are $x_\pm=(r+1\pm\sqrt{((r-3)(r+1))})/(2r)$, giving multiplier $-r^2+2r+4$ by their sum and product. The equation $m_2=-1$ selects $r_2=1+\sqrt{6}$, with derivative $-2\sqrt{6}\neq 0$. To check the flip nondegeneracy rather than infer birth from the multiplier alone, let $g=f_r^2$ near one cycle point. The logistic Schwarzian is $Sf=-6/(1-2x)^2$, and $Sg=(Sf\circ f)(f')^2+Sf<0$ on the threshold cycle, where both first derivatives are nonzero. At $g'=-1$, the cubic normal-form coefficient is $g'''/6+(g'')^2/4=-Sg/6>0$. Transversality and this positive coefficient are the hypotheses of the classical supercritical flip theorem. The algebra verifies those hypotheses; the local bifurcation theorem is classical input.

Worked example

Certify the second threshold using rational decimal endpoints.

Squaring 2.449489742783178 and 2.449489742783179 as exact fractions brackets six. Therefore $3.449489742783178<r_2<3.449489742783179$. At r=3.2 the multiplier is 0.16; at r=3.5 it is −1.25, independently locating stability and instability.

The first low-order thresholds are now exact benchmarks, not points inferred from a bifurcation picture.

Scope of the result

Two thresholds do not estimate the asymptotic Feigenbaum constant reliably. Universality is a theorem with hypotheses, not an inference from any finite list of ratios. Higher-period orbit equations require root isolation and exclusion of lower periods.

Further development

Isolate subsequent flip parameters with interval arithmetic, verify minimal period, and report rigorous intervals for successive scaling ratios rather than bare decimal limits.

References: R069

M092 Parameter loci, orbit closures and dissipative flows

2023-10-16 · historical evidence A · Developed classical result with worked example

Pictures of the Mandelbrot set, a logistic bifurcation diagram and a Lorenz trajectory refer to different mathematical spaces. Defining the plotted object prevents geometric resemblance from being mistaken for a common theorem.

Definitions and assumptions

For $z_0=0$ and $z_{n+1}=z_n^2+c$, the Mandelbrot set is the set of parameters c whose critical orbit remains bounded.

The Lorenz vector field is $\dot{x}=\sigma(y-x)$, $\dot{y}=x(\rho-z)-y$, $\dot{z}=xy-\beta z$. Its state space is R^3; σ,ρ,β are separate parameters.

Escape and volume contraction are different tests

> For the quadratic family, an orbit point with $|z|>\max(2,|c|)$ grows without bound. For positive Lorenz parameters the divergence is the constant $-\sigma-1-\beta<0$, so volume transported by a smooth local flow contracts exponentially. Volume contraction alone does not prove chaos.

Derivation or proof

The triangle inequality gives $|z^2+c|\geq|z|^2-|c|$. If $|z|>\max(2,|c|)$, then $|z|^2-|c|>|z|^2-|z|>|z|$. The same inequality persists and gives eventual unbounded growth. For the flow, the diagonal derivatives are $-\sigma,-1,-\beta$. Liouville's formula for the derivative flow gives $\det D\Phi_t=\exp(-(\sigma+1+\beta)t)$ while the solution exists. This controls transported volume, not orbit separation in every direction.

Worked example

Compare $c=0$, $c=-1$ and $c=1$, and compare the Lorenz divergence at $\sigma=10,\beta=8/3$.

The critical orbit for $c=0$ stays zero. For $c=-1$ it alternates $0,-1$. For $c=1$ it begins $0,1,2,5$ and then escapes by the criterion. The Lorenz volume factor is $\exp(-41t/3)$, regardless of ρ; this cannot alone distinguish periodic from chaotic parameter regimes.

A bounded critical orbit and a contracting state-space volume are statements about different objects and require different evidence.

Scope of the result

A finite run without detected escape does not certify Mandelbrot membership. Negative divergence does not imply a strange attractor, positive Lyapunov exponent or a complete global existence theorem.

Further development

Use rigorous escape certificates or validated trapping regions for individual questions, and label parameter space separately from state space in every visualization.

M093 Convex conjugates with their actual domains

2023-10-21 · historical evidence A · Developed classical result with worked example

A Legendre-Fenchel transform is a supremum over a declared domain. Solving a derivative equation is sufficient only after boundary and finiteness cases are checked.

Definitions and assumptions

For an extended-real function $f:R\to R\cup\{+\infty\}$, $f^*(p)=\sup_x(px-f(x))$. A domain restriction is encoded by $f=+\infty$ outside the domain.

Fenchel's inequality says $f(x)+f^*(p)\geq px$. Equality holds exactly when p is a subgradient of a closed proper convex f at x.

Quadratic and exponential conjugates

For $k>0$, $f(x)=kx^2$ on R gives $f^*(p)=p^2/(4k)$. Restricting to $x\geq 0$ gives $f^*(p)=\max(p,0)^2/(4k)$. For $f(x)=e^x$, $f^*(p)=p\log p-p$ if $p>0$, $f^*(0)=0$ and $f^*(p)=+\infty$ if $p<0$.

Derivation or proof

Complete the square: $px-kx^2=p^2/(4k)-k(x-p/(2k))^2$. On the half-line, the interior optimizer is feasible only if $p\geq 0$; otherwise the maximum is at $x=0$. For the exponential, differentiating gives $e^x=p$, so $p>0$ has the unique maximizer $x=\log p$. If $p=0$, the supremum of $-e^x$ is zero but is approached only as $x\to-\infty$. If $p<0$, sending $x\to-\infty$ makes $px-e^x$ diverge to $+\infty$. These cases prove the complete extended-real formula.

Worked example

Take $k=2$ and $p=-4$.

On the full line the maximizer is $x=-1$ and the conjugate value is 2. On $x\geq 0$, the maximizer is $x=0$ and the value is zero. The same stationary equation cannot be used unchanged in both domains. For e^x at $p=0$, no finite maximizer exists despite a finite conjugate value.

The example distinguishes finite supremum, attainment and an interior critical point.

Scope of the result

Biconjugacy $f^{**}=f$ requires f to be proper, lower semicontinuous and convex; otherwise f^{**} gives the appropriate closed convex envelope rather than recovering arbitrary input data.

Further development

Develop a multivariable constrained example using normal cones and KKT conditions, recording both primal feasibility and dual equality as the certificate.

M094 The CHSH bound and a finite-sample test

2023-10-21 · historical evidence A · Developed classical result with worked example

The mathematical content of a Bell test begins with a probability model and a clear null hypothesis. A violation claim needs a sampling argument in addition to a formal inequality.

Definitions and assumptions

A deterministic local assignment specifies $A_0, A_1, B_0, B_1 \in \{-1, 1\}$. Shared randomness makes a convex mixture of such assignments when settings are independent of that randomness.

The CHSH expression is $S = E[A_0B_0] + E[A_0B_1] + E[A_1B_0] - E[A_1B_1]$. In its uniformly sampled game version, a trial wins when A_xB_y=(−1)^(xy).

Classical bound and an iid tail certificate

> Every local model with measurement independence obeys $|S| \le 2$, equivalently game success probability at most 3/4. For independent identically distributed trials under this null, $P(\hat{w} \ge 3/4 + \varepsilon) \le \exp(-2n\varepsilon^2)$.

Derivation or proof

For a deterministic assignment, $S = A_0(B_0 + B_1) + A_1(B_0 - B_1)$. Exactly one bracket is zero and the other is ± 2, so $S = \pm 2$. Convex averaging preserves the bound. Uniform setting probabilities give $w = 1/2 + S/8$. For the finite-sample statement, wins are independent variables in [0,1] with mean at most 3/4, and Hoeffding’s inequality supplies the displayed one-sided tail. Exhausting the sixteen deterministic assignments independently verifies the algebraic bound.

Worked example

Suppose $n = 1000$ independent uniformly sampled trials have observed win fraction 0.80.

Here $\varepsilon = 0.05$, so the upper bound on the null tail is $\exp(-5) \approx 0.00673795$. This calculation uses the declared iid sampling and measurement-independence assumptions; it is not a p-value for an unspecified physical dataset.

The finite count, the null model and the test statistic are all explicit, making the strength and limitations of the evidence assessable.

Scope of the result

Adaptive devices, memory, nonuniform settings, missing detections or postselection require revised statistical reasoning and may alter the applicable null. Metaphysical conclusions do not follow from the probability algebra alone.

Further development

Replace the iid assumption by a clearly specified sequential local model and use a valid supermartingale test without discarding trials.

M095 An exact free resolution and a separate geometric blowup

2023-10-26 · historical evidence A · Developed classical result with worked example

A free resolution records relations among polynomial generators. A resolution of singularities changes a geometric space by a birational morphism. The two constructions can inform one another but their outputs and correctness criteria are different.

Definitions and assumptions

Let $R=k[x,y]$ and $I=(x^2,xy,y^2)$. Define $d_0:R^3\to I$ by $d_0(a,b,c)=ax^2+bxy+cy^2$.

Let $d_1:R^2\to R^3$ have columns $s_1=(-y,x,0)$ and $s_2=(0,-y,x)$. With grading, the complex is $0\to R(-3)^2\to R(-2)^3\to I\to 0$.

A minimal graded free resolution of I

> The displayed complex is exact, and its entries lie in the homogeneous maximal ideal, so it is minimal as a graded free resolution.

Derivation or proof

First $d_0s_1=d_0s_2=0$. Suppose $ax^2+bxy+cy^2=0$. Reducing modulo x gives $c\equiv 0$ modulo x, so $c=xc_1$. Dividing by x gives $ax+by+c_1y^2=0$. Reducing modulo y gives $a\equiv 0$ modulo y; write $a=-ya_1$. Then $b=xa_1-yc_1$. Thus $(a,b,c)=a_1s_1+c_1s_2$, proving equality of image and kernel. If $u\,s_1+v\,s_2=0$, the first coordinate gives $yu=0$ and the last gives $xv=0$, so $u=v=0$ because R is a domain. This proves injectivity. The standard graded minimality criterion applies since no differential entry is a unit.

Worked example

Compare the plane cusp $y^2=x^3$ over a field of characteristic zero.

In the blowup chart $y=xv$, the total transform is $x^2(v^2-x)=0$. The strict transform $v^2-x=0$ is smooth because its derivative with respect to x is -1. Its parametrization $x=v^2,y=v^3$ gives the familiar normalization map. It remains tangent to the exceptional divisor $x=0$; a normal-crossings embedded resolution requires further work.

The syzygy calculation proves exactness of a module complex. The blowup calculation proves smoothness of a strict transform in the relevant chart. Neither should be described as the other.

Scope of the result

The one-chart cusp example is not a general resolution algorithm or a completed embedded normal-crossings calculation. Schreyer methods do not automatically supply the geometric blowup data.

Further development

Construct a separate certificate for each stage of an embedded resolution, including exceptional multiplicities and intersection behavior.

References: R070

M096 A complete two-dimensional toric subdivision

2023-10-26 · historical evidence A · Developed classical result with worked example

A rational cone contains enough combinatorial data to detect and resolve a simple affine toric singularity. The lattice, primitive generators and subdivision must be specified rather than inferred from a picture.

Definitions and assumptions

Use $N=Z^2$ and the cone σ generated by $v_0=(1,0),v_2=(1,2)$. A full-dimensional two-dimensional cone is regular when its primitive ray generators form a lattice basis.

The affine toric variety is Spec $C[\sigma^\vee\cap M]$, where M is the dual lattice. Subdividing a cone without changing its support defines a proper birational toric morphism.

Regular subdivision of an index-two cone

The original cone has determinant two and is singular. Adding the primitive ray $v_1=(1,1)$ divides it into two determinant-one cones, giving a smooth toric surface mapping properly and birationally to the original affine surface.

Derivation or proof

The determinant $\det(v_0,v_2)=2$ is the index of their generated sublattice. The adjacent determinants $\det(v_0,v_1)=\det(v_1,v_2)=1$, so both charts have ray bases of N and are smooth. The fan has the same support and agrees with the original torus, establishing properness and birationality by the standard toric fan criterion. The dual semigroup is generated by $(0,1),(1,0),(2,-1)$, with the relation $(0,1)+(2,-1)=2(1,0)$. Hence the original coordinate ring is $C[u,v,w]/(uw-v^2)$, whose gradient vanishes at the origin.

Worked example

Calculate the determinant and ring descriptions directly.

The defining polynomial $F=uw-v^2$ has gradient $(w,-2v,u)$, so the only complex singular point is the origin. Each subdivided chart is a polynomial two-variable chart because its ray basis is unimodular. The exceptional curve records the new ray and is contracted by the toric morphism.

This supplies a full combinatorial smoothness certificate for this fan and a concrete algebraic model of the singular target.

Scope of the result

A determinant test at one cone does not establish smoothness of every cone in a larger fan. Properness depends on support and fan compatibility, not only local determinants.

Further development

Implement a two-dimensional subdivision certificate that checks primitive rays, angular ordering, unchanged support and determinant one for every resulting cone.

M097 Hodge numbers constrained by geometry

2023-10-26 · historical evidence A · Developed classical result with worked example

A Hodge diamond is meaningful only after choosing a cohomology theory and geometric hypotheses. Smooth projective complex examples provide exact checks of both the numbers and their symmetries.

Definitions and assumptions

For a smooth projective complex variety X, h^(p,q)=dim H^q(X,Ω_X^p) and b_k=Σ_{p+q=k}h^(p,q).

For a smooth connected projective complex curve of genus g, the only nonzero Hodge numbers are h^(0,0)=h^(1,1)=1 and h^(1,0)=h^(0,1)=g.

Curve and product examples

A genus-g curve has Betti numbers (1,2g,1) and Euler characteristic 2−2g. For $P^1\times P^1$ the nonzero Hodge numbers are h^(0,0)=h^(2,2)=1 and h^(1,1)=2, giving Euler characteristic four.

Derivation or proof

For curves, holomorphic one-forms have dimension g, and complex conjugation gives the antiholomorphic dimension g. Connectedness and orientation give the degree-zero and degree-two classes. Summing along diagonals proves the Betti numbers. For P^1, the Hodge polynomial is 1+uv. The Künneth formula for the product gives $(1+uv)^2=1+2uv+u^2v^2$, from which all displayed Hodge numbers follow. Their alternating total agrees with the product of the two sphere Euler characteristics.

Worked example

Use an elliptic curve and a smooth plane quintic.

At g=1, the elliptic curve has Betti numbers (1,2,1) and χ=0. A smooth plane curve of degree d has genus (d−1)(d−2)/2 by adjunction; at d=5 this gives g=6, hence Betti numbers (1,12,1) and χ=−10. This also independently diagnoses the earlier confusion between a plane quintic curve and a quintic Calabi-Yau threefold.

The numbers are linked by proven geometric structure, rather than predicted merely from a name or a numerical pattern.

Scope of the result

Singular, nonproper or non-Kähler examples require modified statements; ordinary pure Hodge symmetries cannot be assumed unchanged. Numerical estimates are not exact Hodge-number certificates.

Further development

For a specified smooth hypersurface, compute its relevant graded Jacobian pieces and reconcile the resulting Hodge numbers with independent Euler and symmetry checks.

M098 A rational parametrization and a nonexistence certificate

2023-10-26 · historical evidence A · Developed classical result with worked example

Finding many points and proving a point set complete are different achievements. Elementary conics already show how parametrization, integer constraints and local obstructions lead to distinct certificates.

Definitions and assumptions

For an equation X defined over Q, X(Q) is the rational point set, X(Z) the integral point set and X(F_p) a reduction over a finite field when defined.

Clearing denominators in a rational point yields a primitive integer tuple; primitivity must be retained in divisibility arguments.

Two conic certificates

> Every rational point of $x^2+y^2=1$ except $(-1,0)$ is $x=(1-t^2)/(1+t^2)$, $y=2t/(1+t^2)$ for $t\in Q$. The conic $x^2+y^2=3$ has no rational point.

Derivation or proof

For the circle, draw the line $y=t(x+1)$ through $(-1,0)$. Substitution factors out the known intersection and yields the displayed second point. Conversely, for a rational circle point with $x\neq -1$, $t=y/(x+1)$ is rational and reconstructs the point, proving completeness. For $x^2+y^2=3$, a rational point would yield primitive integers a,b,c with $a^2+b^2=3c^2$. Modulo three, squares are zero or one; their sum can be zero only when a and b are divisible by three. Substituting $a=3a_1, b=3b_1$ gives $c^2=3(a_1^2+b_1^2)$, so c is also divisible by three, contradicting primitivity.

Worked example

Set $t=2$ for the parametrized circle.

This gives $x=-3/5, y=4/5$ and verifies the equation exactly. By contrast no finite search is needed for the second conic: the divisibility contradiction rules out every rational denominator at once. Its lack of integer points also follows from squares modulo four, but that easier integer argument alone would not automatically prove the rational statement.

A complete parametrization and a nonexistence proof are both finite certificates, though of different kinds.

Scope of the result

Local solvability is not a general guarantee of global rational solvability for arbitrary varieties. A bounded search with no point is not itself a nonexistence theorem.

Further development

Choose a higher-degree curve and state exactly which global theorem or height bound would turn a point search into a complete result.

M099 A typed glossary that prevents invalid compositions

2023-10-09 · historical evidence A · Developed classical result with worked example

A useful glossary is attached to one mathematical argument. Recording domains, codomains, fields and adjoints prevents expressions that look symbolic but cannot be composed or interpreted consistently.

Definitions and assumptions

Use $F:R^2\to R$, $g:R^3\to R^2$, $DF(g(x)):R^2\to R$ and $Dg(x):R^3\to R^2$. Their composition $DF(g(x))Dg(x):R^3\to R$ is well typed.

Over C, A^T denotes transpose and $A\dagger$ conjugate transpose. They serve different bilinear and Hermitian pairings.

Type correctness and positivity are substantive conditions

> The matrix product rule requires the inner dimensions to match. A complex squared norm is $z\dagger z$, not z^Tz. The latter can vanish for a nonzero vector, so substituting transpose for adjoint can invalidate a positivity argument.

Derivation or proof

An m×n matrix represents a map from n-dimensional coordinate vectors to m-dimensional vectors. Composition $A\circ B$ is represented by AB when B has matching output dimension n; coordinate expansion proves the familiar summation over the shared index. For positivity, $z\dagger z=\Sigma|z_i|^2$ is real and nonnegative, and equals zero only at $z=0$. In contrast z^Tz is a complex bilinear expression whose terms can cancel. These are algebraic differences, not stylistic notation choices.

Worked example

Take $z=(1,i)$ and a derivative composition with $DF=[2,3]$, $Dg=[[1,0,4],[0,1,5]]$.

Here $z^Tz=1+i^2=0$ although z is nonzero, whereas $z\dagger z=2$. The derivative product is [2,3,23], a 1×3 row representing a map $R^3\to R$. Reversing the factors is not even dimensionally defined. The checker verifies the legal product and rejects incompatible dimensions.

A short typed glossary can therefore expose a false proof before any sophisticated theorem is invoked.

Scope of the result

Matching dimensions is necessary but not sufficient: boundedness, differentiability, measurability, convergence and geometric compatibility may also be required. A formal type checker cannot silently supply them.

Further development

Attach assumptions and units to the same signatures, then propagate them through a complete worked calculation so every invoked theorem has a visible hypothesis ledger.

F Singularities integrable systems and quantum field mathematics

M100 Bernstein-Sato certificates and a minimality proof

2023-11-01 · historical evidence A · Classical result developed and example checked

The original broad application question becomes an exact operator identity with a lower-bound argument. A certificate proves that a polynomial works; minimality proves that it is the b-function. Both steps can be completed for monomials, and a linear coordinate change supplies a nonmonomial example.

Definitions and assumptions

Work over C in the Weyl algebra generated by x and ∂ with $\partial x - x\partial = 1$. The parameter s commutes with both generators.

A Bernstein-Sato relation is $P(s)f^{s+1} = b(s)f^s$ in the formal localization module. The global b-function is the monic polynomial of least degree satisfying such a relation.

Exact monomial b-function

For $m \geq 1$, $b_{x^m}(s) = \prod_{j=1}^{m}(s + j/m)$, with certificate $P(s) = m^{-m}\partial^m$.

Derivation or proof

Differentiate $x^{m(s+1)}$ exactly m times. The coefficient is $\prod_{k=0}^{m-1}(m(s+1)-k) = m^m \prod_{j=1}^{m}(s+j/m)$, proving the certificate. To prove minimality, put an arbitrary polynomial-coefficient operator in normal order $\Sigma\, a_{ij}(s)x^i\partial^j$. Its action contributes the Laurent monomial $x^{ms+m+i-j}$. The coefficient of x^{ms} can only receive terms with $j = m+i$. Every such falling-factorial coefficient contains all m factors from the displayed certificate. The other Laurent powers are independent over C(s), so any possible b(s) is divisible by that monic product. Thus the candidate is minimal, not merely an annihilating multiple.

Worked example

Take $f = x^2y^3$ and then the nonmonomial $f = x^2 + y^2$.

For x^2y^3, $P = (1/108)\partial_x^2\partial_Y^3$ gives $b(s) = (s+1/2)(s+1/3)(s+2/3)(s+1)^2$. The same exponent-comparison argument in both variables proves minimality. Over C set $u = x + iy$ and $v = x - iy$. Then $x^2 + y^2 = uv$ and $\partial_u\partial_v = (1/4)(\partial_x^2 + \partial_Y^2)$, so $(1/4)\Delta f^{s+1} = (s+1)^2 f^s$.

A polynomial automorphism preserves the defining differential-operator problem, so the nonmonomial quadratic has $b(s) = (s+1)^2$. Repeated roots carry actual multiplicity; they must not be discarded.

Scope of the result

These are global polynomial examples in characteristic zero. General singularities may require local rather than global b-functions, elimination in a Weyl algebra, or additional singularity theory. The derivation does not establish that every proposed physical or statistical application admits a useful b-function.

Further development

A worthwhile next calculation is to compute a local b-function for a chosen non-normal-crossing plane singularity, retaining the operator certificate and a separate minimality argument. An unverified candidate formula should be labeled as such.

References: R071, R072

M101 Conic polarity as a typed point-to-line map

2023-11-21 · historical evidence A · Classical result developed and example checked

The corrected construction identifies the quadratic form first, then derives polarity, the dual conic and coordinate covariance. It removes the undefined polar-line formula and the invented idea that one point dualizes to a conic.

Definitions and assumptions

Let Q be an invertible symmetric 3×3 matrix over a field of characteristic different from two. Points are nonzero column vectors modulo scale; lines are nonzero covectors modulo scale.

The nonsingular conic is $C=\{x : x^{T}Qx=0\}$. The polar line of p has covector $\ell=Qp$, because its equation is $p^{T}Qx=0$.

Polarity and the dual quadratic form

> The polarity $p \mapsto Qp$ is invertible, incidence is reciprocal, and the tangent-line locus in the dual projective plane is $\ell^{T}Q^{-1}\ell=0$.

Derivation or proof

At a conic point a, differentiating $x^{T}Qx$ gives tangent covector $2Qa$, which defines the same projective line as Qa. Conversely any ℓ has pole $p=Q^{-1}\ell$. That line is tangent precisely when its pole lies on C, which is exactly $\ell^{T}Q^{-1}\ell=0$. Reciprocity follows from $p^{T}Qq=q^{T}Qp$. Under $x'=Mx$, $Q'=M^{-T}QM^{-1}$ and $\ell'=M^{-T}\ell$; hence $Q'p'=M^{-T}Qp$. The construction therefore respects projective coordinate changes rather than depending on one affine drawing.

Worked example

Use the real unit-circle conic $x^{2}+y^{2}-z^{2}=0$ and $p=(2,0,1)$.

Here $Q=\mathrm{diag}(1,1,-1)$, so the polar is $2x-z=0$. Its intersections with the conic are $a_{\pm}=(1/2,\pm\sqrt{3}/2,1)$. The tangents at these points have covectors $(1/2,\pm\sqrt{3}/2,-1)$, and each contains p because their pairing with p is zero. The polar itself is not tangent: its dual quadratic value is $2^{2}-1^{2}=3$.

The polar joins the two contact points of tangents drawn from the exterior point. Applying Q^{-1} to its covector returns p. This verifies the point-line correspondence and the dual-conic test separately.

Scope of the result

Invertibility of Q and the characteristic assumption matter. A degenerate quadratic form does not produce a global polarity. Real exterior/interior terminology has no direct analogue over every field; projective incidence survives, while real tangent counts may change.

Further development

For higher-dimensional quadrics the same matrix argument defines polar hyperplanes. A useful further problem is to classify how the construction degenerates as det Q tends to zero and which incidence statements remain valid.

M102 Cross-ratio with a fixed ordering and homogeneous coordinates

2023-11-21 · historical evidence A · Classical result developed and example checked

Cross-ratio becomes a precise projective invariant once the four-point order and convention are fixed. Harmonicity is then a concrete value of that invariant, not an order-free label.

Definitions and assumptions

For distinct projective points a,b,c,d on P^1, choose nonzero representatives v_a,…,v_d and write [u,v]=det(u,v).

Use CR(a,b;c,d)=[v_a,v_c][v_b,v_d]/([v_b,v_c][v_a,v_d]). In an affine chart this is ((c−a)(d−b))/((c−b)(d−a)).

Projective invariance and harmonic ordering

> Every invertible 2×2 matrix preserves this ordered cross-ratio. Under permutations the possible values lie in {r,1/r,1−r,1/(1−r),r/(r−1),(r−1)/r}.

Derivation or proof

Rescaling any representative multiplies one numerator factor and one denominator factor by the same scalar. Applying M multiplies each bracket by det M; the two determinant factors above and below cancel. Thus CR is both projectively well defined and invariant. The permutation formulas follow by swapping arguments in the affine expression. In particular r=−1 can become 2 or 1/2 after a different ordering, so a harmonic assertion must retain its convention. Homogeneous representatives make the point at infinity routine rather than an exception requiring an invented formula.

Worked example

Take (a,b;c,d)=(−1,1;0,∞) and T(z)=(2z+3)/(z+2).

Using v∞=(1,0) gives CR=−1. The transformation has nonzero determinant one and sends the ordered tuple to (1,5/3;3/2,2). The transformed ratio is [(1/2)(1/3)]/[(-1/6)(1)]=−1, verified with rational arithmetic.

The example simultaneously checks infinity, ordering and projective invariance. A dual statement for four concurrent lines follows only after identifying their pencil with P^1; duality is not a license to replace undefined point coordinates by arbitrary line expressions.

Scope of the result

The four points must be distinct for the displayed nondegenerate ratio. Coinciding points require a limiting or compactified configuration-space interpretation. Over a general field, the listed six values can coincide more than they do over the real numbers.

Further development

Cross-ratio coordinates give a concrete entry into moduli of marked projective lines: normalize three points to 0,1,∞ and the fourth is the remaining parameter. Investigate degenerations by tracking this parameter toward 0,1 or ∞.

M103 A Heisenberg field and its verified local algebra

2023-11-21 · historical evidence A · Classical result developed and example checked

A worked Fock-space model supplies actual operators, a vacuum and a singular operator product. It shows what the abstract vertex-algebra vocabulary is organizing, without claiming that a few words or one OPE alone constitute the full theory.

Definitions and assumptions

Let $F=C[x_1,x_2,\ldots]$, with each polynomial involving finitely many variables. Set $a_{-n}=x_n$ by multiplication and $a_n=n\partial/\partial x_n$ for $n>0$; let a_0 act as zero.

The vacuum is 1. The current is $J(z)=\Sigma_{(n\in Z)}a_n z^{(-n-1)}$, interpreted as a formal field. Normal ordering moves creation modes a_{-n} to the left.

Fock commutator and current contraction

$[a_m,a_n]=m\delta_{(m+n,0)}I$. The singular contraction of $J(z)J(w)$ is $1/(z-w)^2$, in the formal expansion associated with $|z|>|w|$.

Derivation or proof

For positive m,n, derivatives commute and multiplications commute. The only nonzero mixed commutator is $[m\partial_{x_m},x_n]=m\delta_{mn}$. Splitting the field into annihilation and creation parts gives $[J_+(z),J_-(w)]=\Sigma_{(n\geq 1)}n\, z^{(-n-1)}w^{(n-1)}$. Differentiating the formal geometric series $\Sigma_{(n\geq 0)}w^n z^{(-n-1)}=1/(z-w)$ produces the claimed double pole. The normally ordered remainder has no additional singular contraction. This calculation is a complete local commutator derivation in a defined representation.

Worked example

Use the normally ordered stress tensor $T(z)=(1/2):J(z)^2:$ and vacuum states of low degree.

The mode L_{-2} sends 1 to $x_1^2/2$. The only contribution of L_2 on this state is $(1/2)a_1^2$, giving $L_2L_{-2}1=1/2$. With $L_0 1=0$, the Virasoro convention $[L_2,L_{-2}]=4L_0+cI/2$ yields $c=1$. Translation on generators is represented by $T_0=\Sigma_{(n\geq 1)}n\, x_{n+1}\partial_{x_n}$, so $[T_0,a_{-n}]=n\, a_{-(n+1)}$.

The example fixes normalization of the central term and makes the relation between formal fields and modes explicit. The verifier checks the displayed operator relations on a finite set of polynomial states.

Scope of the result

The finite checks do not prove all vertex-algebra axioms. The complete Heisenberg vertex algebra follows from the usual reconstruction construction, with vacuum, translation and locality stated on the full formal space. A coordinate-free chiral-algebra construction additionally needs a curve and coordinate transformation laws.

Further development

Develop conformal blocks for this explicit algebra on a chosen curve, or prove a stated reconstruction theorem. Such a continuation has concrete input data rather than a generic appeal to string theory.

References: R073

M104 Four-site XXX spectrum and the regular one-magnon roots

2023-11-22 · historical evidence A · Classical result developed and example checked

The corrected Bethe equations are tied to one Hamiltonian normalization and checked against a finite matrix. The resulting calculation verifies a sector, while keeping singular-root and completeness questions explicit.

Definitions and assumptions

Take N=4 periodic spin-1/2 sites and H=Σ_(j=0)^3(I−P_(j,j+1)), where P swaps adjacent spins and the last edge wraps around.

In the one-down-spin sector, |j⟩ denotes the down spin at site j. A plane wave has coefficients exp(ipj), with p=2πm/4.

Exact one-magnon diagonalization

H|p⟩=(2−2cos p)|p⟩, so this four-dimensional sector has energies 0,2,4,2. The finite regular Bethe roots are λ=1/2,0,−1/2, giving energies 1/(λ²+1/4)=2,4,2.

Derivation or proof

Only the two bonds incident on the down spin act nontrivially, giving H|j⟩=2|j⟩−|j−1⟩−|j+1⟩. Substitution of the plane wave yields the dispersion. In the periodic XXX convention the regular roots obey ((λ_j+i/2)/(λ_j−i/2))^N=∏_(k≠j)(λ_j−λ_k+i)/(λ_j−λ_k−i). For one root the right side is one. Writing exp(ip)=(λ+i/2)/(λ−i/2) gives λ=(1/2)cot(p/2), and the energy agrees with the plane-wave dispersion. The p=0 state corresponds to the infinite-root limit.

Worked example

Use real vectors in the ordered basis |0⟩,...,|3⟩.

The vectors (1,1,1,1), (1,0,−1,0), (0,1,0,−1) and (1,−1,1,−1) form a basis with eigenvalues 0,2,2,4. Exact integer matrix multiplication verifies all four. For λ=1/2,0,−1/2 the complex Bethe ratio is i,−1,−i respectively; its fourth power is one.

This establishes agreement between a direct operator calculation and the regular Bethe description of the nonuniform one-magnon states. It also demonstrates why omitting the infinite-root state would lose an actual eigenvector.

Scope of the result

The calculation does not prove completeness in other magnetization sectors or for arbitrary N. Roots at ±i/2 and coincident roots need a regularization or a carefully specified admissibility convention. Including the self-index k=j would add a spurious factor and changes the equations.

Further development

For the two-magnon sector, diagonalize the six-dimensional matrix first, then match admissible Bethe solutions and explicitly account for singular solutions and descendants. The matrix spectrum supplies an independent target.

References: R074

M105 A complete two-site Gaudin to scalar-oper calculation

2023-11-22 · historical evidence A · Classical result developed and example checked

A concrete spectral bridge replaces the unsupported Fourier-to-automorphic-form slogan. In the smallest nontrivial sl_2 example, the Bethe vector, Hamiltonian eigenvalues, Miura potential and cancellation of the apparent singularity can all be computed exactly. A rational three-site continuation also gives a complete eight-dimensional joint eigenbasis and two additional scalar-oper certificates.

Definitions and assumptions

On each C^2 take $h=\mathrm{diag}(1,-1)$, $e|\downarrow\rangle=|\uparrow\rangle$, $f|\uparrow\rangle=|\downarrow\rangle$ and $\Omega=(1/2)h\otimes h+e\otimes f+f\otimes e$. Choose marked points $z_1=0, z_2=1$.

Let $H_1=\Omega/(z_1-z_2)=-\Omega$ and $H_2=\Omega$. For weights (1,1), one Bethe root obeys $1/w+1/(w-1)=0$.

Singlet spectral data and apparent-pole cancellation

The root $w=1/2$ yields a singlet with (H_1,H_2) eigenvalues $(3/2,-3/2)$. Define $u(x)=1/(2x)+1/(2(x-1))-1/(x-1/2)$. Then $(\partial-u)(\partial+u)=\partial^2-V$, where $V=u^2-u'=3/[4x^2(x-1)^2]$.

Derivation or proof

The Bethe vector is $w^{-1}|\downarrow\uparrow\rangle+(w-1)^{-1}|\uparrow\downarrow\rangle=2|\downarrow\uparrow\rangle-2|\uparrow\downarrow\rangle$. Direct use of Ω gives eigenvalue $-3/2$, hence the Hamiltonian pair above. Near a root w, write $u=-1/(x-w)+a_0+O(x-w)$. The double pole cancels in u^2-u', and its simple residue is $-2a_0$. Here $a_0=(1/2)(1/w+1/(w-1))=0$ precisely by the Bethe equation. Expanding at the physical points gives $V=3/(4x^2)+3/(4(x-1)^2)+(3/2)/x-(3/2)/(x-1)$. Its simple residues match H_1 and H_2 in the declared normalization.

Worked example

Treat the scalar differential equation $y''-V(x)y=0$ on the punctured projective line.

The apparent point $x=1/2$ is absent from V. One solution is $y_1=(x-1/2)/\mathrm{sqrt}(x(x-1))$. Reduction of order gives $y_2=(x^2-x/2+1/4)/\mathrm{sqrt}(x(x-1))$, with Wronskian one. Both are regular at $x=1/2$. Around

x=0 or x=1 their common square-root factor changes sign, so monodromy is scalar and projectively trivial. The vacuum sector, with no root, instead has simple residues (−1/2,1/2), matching the triplet eigenvalues. For sites (0,1,8/3), the one-root equation reduces to $3w^2-(22/3)w+8/3=0$, with roots 2 and 4/9. Their eigenvalue triples are (−3/16,6/5,−81/80) and (25/16,−8/5,3/80). The highest vectors have one-down-spin coefficients (1/2,1,−3/2) and (9/4,−9/5,−9/20). Applying total lowering once to each gives two doublets; applying it zero through three times to the vacuum gives a quartet with eigenvalues (−11/16,1/5,39/80). All eight rational vectors are independent and satisfy all three full 8×8 Hamiltonians. For each regular root, with $g=\prod(x-z_i)$, $u=g'/(2g)-1/(x-w)$ gives $V=(3/4)\Sigma_i(x-z_i)^{-2}+\Sigma_i E_i/(x-z_i)$. Exact residual polynomials vanish. Since $g'(w)=0$, reduction of order has no logarithm: $y_1=(x-w)/\sqrt{g}$, $y_2=[-g(w)+(g''(w)/2)(x-w)^2+(x-w)^3/2]/\sqrt{g}$. Their Wronskian is one, and both ODE residuals vanish identically.

The calculation supplies an actual rank-one instance of the Gaudin-oper relationship: eigenvalues are encoded in coefficients of a scalar differential operator. The simple finite example is classical; its explicit replay is an exposition and verification result. The three-site continuation settles this complete finite joint-spectrum example, including its scalar equations, rather than leaving it as a proposed computation.

Scope of the result

This is an sl_2 example, not a proof of the full Gaudin-algebra spectrum theorem, a global automorphic correspondence or arbitrary Bethe completeness. Oper terminology and Langlands duality must retain the relevant group, singularity and monodromy conditions.

Further development

S010 develops the arbitrary-site one-magnon theorem, S013 completes the three-site exceptional-point analysis, and S014 classifies every distinct-complex-site one-magnon Jordan block with closed matrix and characteristic formulas. Remaining extensions concern multi-magnon sectors, higher representations or physical-site collisions; they require additional hypotheses and independent checks. This is separate from full Gaudin completeness.

References: R010

M106 Joint reliability is a dependence problem

2023-11-06 · historical evidence A · Classical result developed and example checked

The arithmetic correction is only the first step. An all-success probability cannot be identified from marginal accuracies alone. Exact dependence constructions show the full gap between what the individual probabilities say and what independence adds.

Definitions and assumptions

Let $A_1,\ldots,A_n$ be success events with $P(A_i)=p_i$. The target is $P(\cap A_i)$, a joint probability.

Mutual independence means $P(\cap_{i\in S}A_i)=\prod_{i\in S}p_i$ for every subset S. Pairwise independence checks only subsets of size two.

Fréchet bounds and an independence counterexample

$\max(0,\Sigma p_i-(n-1))\le P(\cap A_i)\le \min_i p_i$. If all 104 marginals are 0.95, these bounds are 0 and 0.95, whereas independence gives $0.95^{104}\approx 0.00482230805327$.

Derivation or proof

The upper bound follows because the intersection is a subset of every A_i. For the lower bound apply the union bound to the failure events: $P(\cap A_i)=1-P(\cup A_i^c)\geq 1-\Sigma(1-p_i)$. Probabilities are also nonnegative. The product formula follows only after imposing mutual independence. To see why pairwise independence is insufficient, let X,Y be independent fair bits and set Z=X xor Y. Any pair is independent, but P(X=Y=Z=1)=0 rather than 1/8.

Worked example

Construct joint laws for 104 events, each with success probability 0.95.

For the maximal case take every $A_i=\{U\leq 0.95\}$ for one uniform U; then all-success probability is 0.95. For a zero case let failures be arcs of length 0.05 on the unit circle, starting at i/104. Since adjacent starts are less than 0.05 apart, the arcs cover the circle, so at least one failure always occurs, although each event still succeeds with probability 0.95. Independent Bernoulli variables give the intermediate product.

The same individual accuracy is compatible with no chance of complete correctness, a 95 percent chance, or the independent-model value. Replacing a dependency model by repeated multiplication can therefore be a major modeling error even after arithmetic is repaired.

Scope of the result

The numerical examples are abstract probability constructions, not estimated properties of a genealogy or any real audit. Real dependence requires observations or structural assumptions. Pairwise correlations alone generally do not determine the required high-order intersection.

Further development

For a specified dependency graph, derive an interval using available joint moments or solve a finite linear program over atomic outcomes. Report the identified interval if the joint probability remains underdetermined.

M107 Nonlinearity does not imply chaos or identifiability

2023-11-24 · historical evidence A · Classical result developed and example checked

A rigorous counterexample is more informative than another catalogue of nonlinear applications. A simple nonlinear ODE has a fully solvable, globally convergent flow and an observation that permanently hides the sign of its initial state.

Definitions and assumptions

Consider $\dot{x}=-x^3$ on the real line for $t\geq 0$. The state is x, and first choose the observable $y=x^2$.

An observation model is identifiable on a set if no two admissible initial states generate exactly the same observed time history. This is distinct from state stability.

Global convergence with an exact observation collision

For every initial x_0, $x(t)=x_0/\mathrm{sqrt}(1+2x_0^2t)$. All trajectories converge to zero, while x_0 and $-x_0$ give identical $y(t)=x_0^2/(1+2x_0^2t)$.

Derivation or proof

For $x_0 \neq 0$, differentiate x^{-2} to obtain $d(x^{-2})/dt = 2$ and solve; $x_0 = 0$ gives the equilibrium. The denominator stays positive, giving a global forward solution. With $V = x^2/2$, $\dot{V} = -x^4 \leq 0$, and the explicit solution proves convergence. Squaring removes the initial sign for all times. Equivalently y satisfies $\dot{y} = -2y^2$ with the same initial y_0 for both states. Thus the nonlinear system is elementary and convergent, while the chosen observation remains structurally ambiguous.

Worked example

Compare $x_0 = 1$ and $x_0 = -1$, recording $y = x^2$ at any finite or infinite collection of times.

At t=0,1,2 the common observations are 1,1/3,1/5. No increase in temporal sampling resolves the sign because the entire continuous histories coincide. Adding the observable z=x distinguishes the two immediately. The exact rational y values and the observation evolution equation are checked by the verifier.

Complexity of the evolution law, richness of sampling and distinguishability of states are separate questions. This example refutes a generic inference from nonlinear equations to chaos and a generic inference from many measurements to complete state reconstruction.

Scope of the result

This proof concerns one scalar system and one observation. It is not a general criterion for absence of chaos in higher dimensions. Nor does adding a sign-sensitive observation establish practical robustness when measurement noise is large.

Further development

Classify observation collisions under a symmetry group: if both dynamics and output are sign-equivariant or sign-invariant in the appropriate way, determine whether the output can at best recover a symmetry orbit. Then quantify the additional measurement needed to break it.

M108 The Klein-Gordon dispersion relation and its massless limit

2023-12-02 · historical evidence A · Classical result developed and example checked

The mathematical comparison starts with two explicitly normalized differential operators. Their plane-wave dispersions, propagation equations and interpretation can then be compared without confusing a classical field equation with the procedure of quantization.

Definitions and assumptions

Use coordinates (t,x), speed c and $\Box = c^{-2}\partial_t^2 - \Delta$. The free Klein-Gordon equation is $(\Box + (mc/\hbar)^2)\varphi = 0$.

A plane wave is $\exp(i(k\cdot x - \omega t))$. Its wave vector k has inverse-length units, while ω has inverse-time units.

Massive and massless plane-wave dispersions

Substitution gives $\omega^2 = c^2|k|^2 + m^2c^4/\hbar^2$. Setting m=0 gives the classical massless wave dispersion $\omega^2 = c^2|k|^2$.

Derivation or proof

The second time derivative multiplies the wave by $-\omega^2$ and the Laplacian by $-|k|^2$. The PDE therefore reduces to $-\omega^2/c^2+|k|^2+(mc/\hbar)^2=0$. Multiplying by $\hbar^2$ and writing $E=\hbar\omega$ and $p=\hbar k$ gives $E^2=c^2|p|^2+m^2c^4$. For $m>0$, differentiating the positive branch gives group velocity $v_g=c^2k/\omega$ with magnitude below c; at high frequency it approaches c. These are consequences of the chosen PDE and Fourier ansatz, not additional quantum postulates.

Worked example

In units $c=\hbar=1$ take $m=2$ and spatial mode $k=(3,0,0)$.

The frequencies are $\omega=\pm\mathrm{sqrt}(13)$, and each wave has zero PDE residual because $-13+9+4=0$. The positive-branch group speed is $3/\mathrm{sqrt}(13)$. On a periodic box of side L, allowed wave vectors are $2\pi n/L$, producing a countable family of uncoupled harmonic oscillators at the classical level.

Mass changes the relation between frequency and wave number. The periodic Fourier decomposition also identifies the finite-mode truncations that can be computed exactly; it does not by itself choose a Fock space or turn amplitudes into operators.

Scope of the result

Negative-frequency solutions do not by themselves establish a negative-probability theory. Quantization and the choice of conserved inner product require their own construction. Interactions, curved backgrounds and boundary conditions change the analysis.

Further development

For a specified potential or curved metric, determine whether a positive self-adjoint spatial operator supplies a stable mode expansion. Retain the operator domain and boundary conditions rather than transferring the flat-space dispersion unchanged.

References: R075

M109 A controlled formal route from Dirac to Pauli

2023-12-02 · historical evidence A · Classical result developed and example checked

The nonrelativistic limit must preserve spin and the magnetic coupling. Removing the rest-energy phase and eliminating the small spinor component gives the leading Pauli equation, together with an explicit statement of the scale assumptions.

Definitions and assumptions

Let $\pi=p-qA$ with $p=-i\hbar\nabla$ and $B=\nabla\times A$. Use $H_D=c\alpha\cdot\pi+\beta mc^2+q\Phi$ and a four-spinor $\psi=e^{(-imc^2t/\hbar)}(\phi,\chi)^{\mathsf{T}}$.

The nonrelativistic regime requires typical $|\pi|/(mc)\ll 1$ and energies and time variation small compared with mc^2, together with fields smooth enough for the expansion.

Leading spin-sensitive Hamiltonian

At leading order $\chi\approx(\sigma\cdot\pi)\phi/(2mc)$, and $i\hbar\dot{\phi}=[\pi^2/(2m)-(q\hbar/(2m))\sigma\cdot B+q\Phi]\phi$.

Derivation or proof

After extracting the rest-energy phase, the two components satisfy $i\hbar\dot{\phi}=q\Phi\phi+c\sigma\cdot\pi\chi$ and $i\hbar\dot{\chi}=(q\Phi-2mc^2)\chi+c\sigma\cdot\pi\phi$. The large $2mc^2$ denominator permits the stated leading elimination of χ when the remaining terms are small. Pauli-matrix multiplication gives $(\sigma\cdot\pi)^2=\pi^2+(i/2)\Sigma\varepsilon_{ijk}\sigma_k[\pi_i,\pi_j]$. Since $[\pi_i,\pi_j]=iq\hbar\varepsilon_{ijk}B_k$, the second term is $-q\hbar\sigma\cdot B$. Inserting this identity into the first equation produces the Hamiltonian and fixes the sign convention.

Worked example

First take $A=\Phi=0$ and a plane wave of momentum magnitude p.

The exact positive relativistic energy is $\sqrt{m^2c^4+c^2p^2}=mc^2+p^2/(2m)-p^4/(8m^3c^2)+O(p^6/(m^5c^4))$. Removing mc^2 leaves the Schrödinger kinetic term at leading order. For a magnetic field, the additional Pauli spin term survives at that same effective order; a scalar wave equation would miss it.

The result specifies both the limiting equation and the leading neglected kinetic correction. Its small parameter is $p/(mc)$, not simply the statement that an experiment looks classical.

Scope of the result

This is a formal asymptotic derivation for a prescribed background field. A uniform error theorem over long times needs operator estimates and additional regularity. Strong fields and pair-production regimes invalidate the elementary elimination.

Further development

Choose a bounded momentum sector and smooth bounded fields, then derive a finite-time norm estimate comparing projected Dirac dynamics with the effective Pauli evolution. That is the appropriate next step toward a controlled approximation theorem.

References: R076

M110 Maxwell equations as field equations and identities

2023-12-02 · historical evidence A · Classical result developed and example checked

Differential forms separate what follows from a potential from what must be imposed as a sourced equation. Gauge redundancy becomes a precise kernel of the map from potentials to field strength, with topology supplying an important qualification.

Definitions and assumptions

On an oriented spacetime with a fixed metric, A is a one-form and $F=dA$ is a two-form. The Hodge star depends on that metric and orientation.

Use normalization $dF=0$ and $d\star F=J$, where J is a source three-form. Constants can be restored according to electromagnetic unit conventions.

Gauge invariance and current closure

$A\mapsto A+d\chi$ leaves F unchanged. Whenever $d\star F=J$ holds, $dJ=0$. Locally, every closed F is a potential field on a contractible coordinate patch.

Derivation or proof

The identity $d^2=0$ gives $d(A+d\chi)=dA$. Applying d to the sourced equation gives $dJ=d^2\star F=0$, so source conservation is a compatibility condition. The local converse $dF=0\Rightarrow F=dA$ is the Poincaré lemma under the contractibility assumption. It is not a global theorem on arbitrary manifolds: nontrivial degree-two cohomology can obstruct a single globally defined potential. The coordinate form of $dF=0$ is the homogeneous Maxwell system; the Hodge-star equation contains the metric-dependent dynamical content.

Worked example

On flat coordinates (t,x,y,z), choose $A=a\cos(kx-\omega t)dy$ and use units $c=1$.

Then $F=-ak\sin(kx-\omega t)dx\wedge dy+a\omega\sin(kx-\omega t)dt\wedge dy$. Direct exterior differentiation gives $dF=0$ because the two mixed terms cancel. The vacuum sourced equation requires $\omega^2=k^2$. Adding $d\chi$ for $\chi=\lambda tx$ changes components of A but contributes no change to F.

The example distinguishes the automatic homogeneous identity, the dispersion imposed by the vacuum field equation and a gauge change of the potential. It also shows why covariant notation alone is not a quantization procedure.

Scope of the result

Global potentials, boundary conditions and gauge fixing require separate analysis. A flat connection can have nontrivial global holonomy while being locally pure gauge. A globally exact potential $A=d\chi$ with single-valued χ has no such loop holonomy; these cases must not be conflated.

Further development

Develop the same problem on a torus or a punctured domain and calculate closed forms that are not exact. This makes the topology of gauge data explicit before introducing a quantum theory.

References: R077

M111 Ward-Takahashi constrains a vertex without determining it

2023-12-02 · historical evidence A · Classical result developed and example checked

The identity is made concrete by verifying it at tree level, constructing a longitudinal vertex for a prescribed inverse propagator, and displaying the remaining transverse freedom. This explains both the power and the limitation of the constraint.

Definitions and assumptions

Factor the charge from the full electromagnetic vertex Γ^μ. With consistent momentum assignments, the identity is $q_\mu\Gamma^\mu(p+q,p)=S^{-1}(p+q)-S^{-1}(p)$.

A transverse addition T^μ satisfies $q_\mu T^\mu=0$. Such an addition cannot be detected by this contracted identity alone.

An explicit longitudinal construction

For a differentiable matrix-valued inverse propagator $K(p)=S^{-1}(p)$, $\Gamma_L^\mu=\int_0^1(\partial K/\partial p_\mu)(p+tq)dt$ satisfies the difference identity on any regular path. It is not the unique satisfying vertex.

Derivation or proof

Contract with q and use the chain rule: $q_\mu \partial K(p+tq)/\partial p_\mu = dK(p+tq)/dt$. The fundamental theorem of calculus gives $K(p+q)-K(p)$. Adding any transverse T preserves the contraction. At tree level $K(p)=\text{slash}(p)-m$ and $\Gamma^\mu=\gamma^\mu$, reducing the identity to $\text{slash}(q)=\text{slash}(p+q)-\text{slash}(p)$. These are direct algebraic checks of the kinematic relation; they do not assert that an arbitrarily chosen K and Γ arise from a complete interacting quantum field theory.

Worked example

For an illustrative polynomial matrix function, take $K(p)=\text{slash}(p)-m-\alpha p^2 I$.

The construction gives $\Gamma_L^\mu=\gamma^\mu-\alpha(2p^\mu+q^\mu)I$. Contracting yields $\text{slash}(q)-\alpha(2p\cdot q+q^2)I$, exactly $K(p+q)-K(p)$. One may add $\kappa(q^2 p^\mu-(p\cdot q)q^\mu)I$, whose contraction is zero. The verifier checks this example in a two-coordinate Euclidean Clifford representation with rational inputs.

The identity constrains differences of inverse propagators and the longitudinal vertex. It leaves dynamical transverse information undetermined. The polynomial example tests the logic without making a false renormalization claim.

Scope of the result

In QED the full identity is a statement about regularized and renormalized correlation functions. A symmetry-breaking regulator, inconsistent counterterms or an anomalous symmetry can require additional terms or restoration conditions. The algebraic construction above does not settle those issues.

Further development

Specify an actual perturbative action and regulator, calculate the one-loop self-energy and vertex in the same convention, and check the renormalized identity including counterterms. Retain transverse form factors as independent output.

References: R078

M112 A complete finite-dimensional perturbation calculation

2023-12-02 · historical evidence A · Classical result developed and example checked

A zero-dimensional quartic model permits an exact derivation of propagator contractions, vacuum normalization, connected coefficients and a rigorous asymptotic remainder. It supplies a fully checkable model of perturbative logic without claiming to construct a continuum field theory.

Definitions and assumptions

Let X be a real standard Gaussian and define $Z(g)=E[\exp(-gX^4/24)]$ for $g\geq 0$. The normalized two-point function is $G_2(g)=E[X^2\exp(-gX^4/24)]/Z(g)$.

Gaussian moments are $E[X^{2r}]=(2r-1)!!$ and odd moments vanish. Wick pairings count those double factorials.

Quartic coefficients and a controlled remainder

$Z(g)=1-g/8+35g^2/384+O(g^3)$, $\log Z(g)=-g/8+g^2/12+O(g^3)$, and $G_2(g)=1-g/2+2g^2/3+O(g^3)$ as $g\downarrow 0$.

Derivation or proof

Expand the interaction exponential to a fixed order and integrate each Gaussian moment. The nth coefficient of Z is a_n=(−1)^n(4n−1)!!/(24^n n!). Multiplying and dividing the resulting formal series gives the displayed connected and normalized coefficients. For g≥0, Taylor’s integral remainder for exp(−y) has magnitude at most y^(N+1)/(N+1)!; taking $y=gX^4/24$ and integrating bounds the partition-function remainder by the next absolute coefficient times g^(N+1). This proves a finite-order asymptotic statement without exchanging an infinite series and the integral.

Worked example

Calculate through second order using integer Gaussian moments.

$E[X^4]=3$, $E[X^8]=105$ and $E[X^{10}]=945$. Thus the numerator of G_2 is $1-5g/8+105g^2/128+O(g^3)$; division by Z gives $1-g/2+2g^2/3$. Meanwhile |a_(n+1)/a_n|=(4n+3)(4n+1)/(24(n+1)), which grows without bound.

The perturbative series has zero radius of convergence although the original integral exists for every g≥0. Correct finite-order coefficients and an explicit asymptotic bound are therefore stronger and more honest deliverables than a claim that the infinite diagram sum converges.

Scope of the result

This toy integral has no spacetime ultraviolet divergences, locality or scattering theory. It does not replace renormalization of a continuum model. Its value is the exact separation of algebraic combinatorics, normalization and asymptotic control.

Further development

Add several Gaussian variables with a stated covariance and derive connected cumulants. For a continuum problem, introduce a cutoff and identify which bounds survive its removal rather than assuming the finite-dimensional argument is uniform.

References: R079

M113 An exact RG toy flow and an invariant local exponent

2023-12-02 · historical evidence A · Classical result developed and example checked

The beta function must be specified before discussing running couplings. Simple autonomous flows can be solved exactly, and the derivative at a fixed point gives a local quantity that survives smooth invertible reparametrization of the coupling.

Definitions and assumptions

Write $t=\log(\mu/\mu_0)$ and $dg/dt=\beta(g)$. A fixed point g^* satisfies $\beta(g^*)=0$.

A scheme coordinate change is a smooth locally invertible h=h(g). Its beta function is β_h(h(g))=h′(g)β(g). This is a coordinate statement for an exact flow, not a guarantee about

truncated calculations. The nonzero-initial-condition formula below assumes $g_0 \neq 0$; $g_0 = 0$ is the constant equilibrium.

Flow solutions and fixed-point linearization

For $\beta = bg^2$, $g(t) = g_0/(1 - bg_0 t)$. For $\beta = -\varepsilon g + bg^2$ with $\varepsilon, b > 0$ and $g^* = \varepsilon/b$, $g(t) = g^*/[1 + (g^*/g_0 - 1)e^{(\varepsilon t)}]$ wherever its denominator is nonzero. At a fixed point, $d\beta_h/dh = \beta'(g^*)$.

Derivation or proof

The first solution follows by differentiating $1/g$. For the second, substitution or separation of variables verifies the displayed logistic-type formula. Linearization gives $\delta\dot{g} = \beta'(g^*)\delta g$. Differentiating $\beta_h = h'\beta$ with respect to h yields $(h''\beta + h'\beta')/h'$. At a fixed point $\beta = 0$, so the derivative equals β'. Thus the local exponent is invariant under the declared regular change of variable, although the numerical coordinate used to label the fixed point can change.

Worked example

Take $\varepsilon = b = 1$ and $g_0 = 1/2$.

Then $g(t) = 1/(1 + e^{t})$. It approaches the Gaussian fixed point zero as μ increases and the nonzero fixed point one as μ decreases. The derivatives are $\beta'(0) = -1$ and $\beta'(1) = 1$, giving the stated ultraviolet and infrared stability directions. For $\beta = g^2$ with $g_0 = 1/2$, the toy solution has a pole at $t = 2$.

These statements are exact for the selected differential equations. A pole in a one-loop approximation does not prove that an underlying physical theory has an actual singularity there; it marks where the approximation can no longer justify extrapolation.

Scope of the result

A field-theoretic beta function depends on the specified action, regulator, renormalization prescription and perturbative order. Multiple couplings require a Jacobian matrix, and marginal directions require nonlinear analysis. An arbitrary scalar cost function is not automatically an RG flow.

Further development

For a specified two-coupling beta vector field, solve the fixed-point equations, classify its Jacobian and compare the invariant eigenvalues under an explicit analytic coordinate change. Report the truncation order with every numerical prediction.

References: R078

M114 Moving level sets and the singularity of threshold coordinates

2023-12-16 · historical evidence A · Classical result developed and example checked

An implicit boundary can be differentiated only where its defining level is regular. The same calculation quantifies sensitivity and gives an exact example where a boundary is born and its graph velocity diverges.

Definitions and assumptions

Let $f(x,t)$ be continuously differentiable and consider the level set $f(x,t)=c$. A regular spatial level has $\nabla_x f \neq 0$.

For a one-dimensional graph $r(t)$, the condition is $f_r(r(t),t)\neq 0$. Choose the unit normal $n=\nabla f/|\nabla f|$ when discussing normal velocity.

Implicit velocity and perturbation sensitivity

Along a regular graph $r'=-f_t/f_r$. In several dimensions a moving level set has normal velocity $v_n=-f_t/|\nabla f|$. A small additive level perturbation δf produces first-order normal displacement $-\delta f/|\nabla f|$.

Derivation or proof

Differentiate $f(r(t),t)=c$ to obtain $f_r r'+f_t=0$. For a path $X(t)$ on a spatial level, $\nabla f\cdot X'+f_t=0$; taking the normal component gives the velocity. For the perturbation, Taylor-expand $f(X+\delta r\, n,t)+\delta f=c$ and use $\nabla f\cdot n=|\nabla f|$. The denominator therefore measures conditioning as well as existence of the local graph. Tangential motion is a parametrization choice and is not fixed by the level equation.

Worked example

Take $f(x,y,t)=x^2+y^2-t$ and $c=0$.

For $t>0$ the boundary is a circle of radius $\sqrt{t}$, and its outward normal speed is $1/(2\sqrt{t})$. At $t=0$ the spatial gradient vanishes at the origin. For $t<0$ there is no real level set, at zero there is one point, and for $t>0$ there is a circle. The one-dimensional version $r^2-t=0$ has two branches $\pm\sqrt{t}$, neither a differentiable graph through the critical event.

The diverging speed is not a contradiction in the implicit-function theorem; it occurs exactly where its nonzero-gradient hypothesis fails. The same inverse-gradient factor explains why threshold localization becomes unstable near a flat or critical level.

Scope of the result

These are local regularity and first-order sensitivity statements. They do not guarantee a global graph, a unique topology, or robustness under large perturbations. Nonsmooth defining functions need generalized notions of normal and solution.

Further development

For a specified smooth family, locate critical levels by solving $f=c$ and $\nabla f=0$ simultaneously, classify the local Hessian, and track which topology changes are compatible with those critical events.

Complete entry index

M001 Separate the meanings of Q

M002 p-adic completion with certified digits

M003 Affine dynamics on Z_p

M004 A valid probability model and identifiable parameters

M005 Specify a hard problem before using arithmetic complexity

M006 Zeno through a convergent series and a time law

M007 Octonions with a declared multiplication convention

M008 The sharp octonion regrouping bound

M009 Hyperbolic identities that actually verify a wave

M010 Gudermannian coordinates with the measure retained

M011 KdV and KP with matched conventions

M012 A complete two-soliton construction in a fixed KdV convention

M013 Radial conservation and stereographic geometry

M014 q-differentiation of the Gudermannian

M015 Quantitative Jacobian bounds for deep compositions

M016 Minimal universal recurrences on an observable quotient

M017 The reachable and observable parts of a realization

M018 Exact finite closure and a polynomial obstruction

M019 Hardy-BMO duality from atoms

M020 A complete heat-signature collision

M021 Exact modularity phases on a four-cycle

M022 Ensemble assumptions determine the spectral baseline

M023 A reversible q-dependent chain with an exact gap

M024 Continuous q-Hermite polynomials through a Jacobi operator

M025 A graph game with explicit equilibria

M026 Relative entropy along a finite Markov chain

M027 Neat groups and a smooth modular-curve example

M028 Minimal and toroidal compactifications in an explicit chart

M029 General type and log-general type on modular curves

M030 A projective embedding from a degree bound

M031 Adjunction fixes the quintic dimension

M032 Formal inversion has a public coefficient algorithm

M033 The A_2 Weyl denominator expanded exactly

M034 An affine A_1 identity with its normalization proved

M035 Young tableaux counted by recursion and hooks

M036 A nonconstant Kazhdan-Lusztig calculation in S_4

M037 An independently replayable BCH certificate

M038 A finite Peter-Weyl calculation with an exact truncation error

M039 Rephasing invariance and a sharp three-flavor CP bound

M040 A complete small persistence calculation

M041 A two-level Berry holonomy with a gauge check

M042 Crossing symmetry and an exactly opened spectral gap

M043 A Moore spectrum as a precise chromatic example

M044 A conservative positive diffusion stencil with a sharp local condition

M045 Reflection equivariance as an operator identity

M046 Affine coordinate changes and time evolution are different operations

M047 Mutual information detects dependence rather than a causal channel

M048 One measurement can miss coherence that another detects

M049 An observable timescale that saturates the energy-time bound

M050 Modularity from lattice reindexing and a diagnostic zero

M051 A classical abelian Chern-Simons calculation with its global boundary

M052 Explicit genus-one modular data with the anomaly retained

M053 Two-oscillator locking with stable and unstable branches

M054 Invariant gates and an exact identifiability obstruction in conductance ODEs

M055 Decision thresholds and genuine strategic equilibria

M056 Infinite utilities through controlled finite limits

M057 A Cauchy model has no ordinary mean or variance

M058 Feedback instability is a mechanism to specify

M059 Frobenius energy across disjoint blocks

M060 Population weights versus amplitude weights

M061 Covariance inequalities with finite moments

M062 A quadratic program with singular cases included

M063 Open-system dynamics must preserve physical states

M064 Population growth and the actual logarithmic time

M065 A unified equation needs shared variables and invariances

M066 Bessel versus Parseval

M067 Hölder and Young with equality cases

M068 Matrix norms and the quantity being bounded

M069 Lyapunov conditions in the right direction

M070 Units of the Einstein equation

M071 Why the coefficient is 8π

M072 Sixteen entries but ten independent components

M073 Covariant conservation is a structural condition

M074 Grothendieck universes and their logical cost

M075 Stirling partitions do not supply torsional energy

M076 Vector motion and rotation conventions

M077 A proposed analogy needs a dimensionless correspondence

M078 Residual energy and exact Gaussian calibration

M079 The chain rule as a statement about local linear maps

M080 Graph heat flow and the stability of a discrete gradient step

M081 Exact exponent arithmetic before approximation

M082 Competing response laws and identifiable parameters

M083 Tensor order and matrix rank are different invariants

M084 Directional cancellation as a limited shearlet illustration

M085 A fully normalized spin one-half Wigner representation

M086 Exact polynomial Moyal products

M087 A spectral definition of fractional quantum evolution

M088 Lagrange identity and its equality geometry

M089 Winding number and Gauss-Bonnet with boundaries explicit

M090 An explicit marked-torus deformation

M091 Certified first logistic-map period doublings

M092 Parameter loci, orbit closures and dissipative flows

M093 Convex conjugates with their actual domains

M094 The CHSH bound and a finite-sample test

M095 An exact free resolution and a separate geometric blowup

M096 A complete two-dimensional toric subdivision

M097 Hodge numbers constrained by geometry

M098 A rational parametrization and a nonexistence certificate

M099 A typed glossary that prevents invalid compositions

M100 Bernstein-Sato certificates and a minimality proof

M101 Conic polarity as a typed point-to-line map

M102 Cross-ratio with a fixed ordering and homogeneous coordinates

M103 A Heisenberg field and its verified local algebra

M104 Four-site XXX spectrum and the regular one-magnon roots

M105 A complete two-site Gaudin to scalar-oper calculation

M106 Joint reliability is a dependence problem

M107 Nonlinearity does not imply chaos or identifiability

M108 The Klein-Gordon dispersion relation and its massless limit

M109 A controlled formal route from Dirac to Pauli

M110 Maxwell equations as field equations and identities

M111 Ward-Takahashi constrains a vertex without determining it

M112 A complete finite-dimensional perturbation calculation

M113 An exact RG toy flow and an invariant local exponent

M114 Moving level sets and the singularity of threshold coordinates

Verification scope

Chapter a: 17 scoped records.

M002 · Hensel simple-root lift · passed

M003 · Finite affine fixed-point classification · passed

M004 · Probability range counterexample · passed

M004 · Logistic three-design example · passed

M005 · Toy linear inversion attack · passed

M006 · Exact geometric partial sums · passed

M007 · Octonion associator and all alternation coefficients · passed

M007 · Moufang identity checked by complete coefficient polarization · passed

M008 · Sharp unit associator bound attained · passed

M009 · KdV residual polynomial identity · passed

M010 · Gudermannian identities and inverse · passed

M012 · Two-soliton Hirota identity · passed

M014 · q-Gudermannian secant bounds · passed

M015 · Twenty saturated scalar layers · passed

M016 · Observable universal recurrence · passed

M017 · Reachable-visible quotient example · passed

M018 · Polynomial Koopman degree-growth obstruction · passed

Chapter a supplement: 4 scoped records.

S007 · Minimal invariant triangular feature spaces · passed

S007 · Resonance Jordan chains and minimal polynomials · passed

S007 · Closed iterate formula including zero resonance · passed

S007 · Invertible coupled degree-growth counterexample · passed

Chapter b: 19 scoped records.

M019 · Atomic Hardy-BMO example · passed

M020 · Exact nonisomorphic heat-signature collision · passed

M021 · C4 modularity phase diagram · passed

M022 · Ensemble moments checked by exhaustive graph expectation · passed

M023 · Three-state reversible q-chain · passed

M024 · Continuous q-Hermite finite orthogonality certificate · passed

M025 · C4 binary coordination equilibria and exact potential · passed

M026 · Shannon entropy can decrease while relative entropy contracts · passed

M027 · Principal modular-curve small-level data · passed

M029 · General type differs from log-general type · passed

M030 · Very ample threshold and curve sections · passed

M028 · Cusp uniformizer and fan smoothness · passed

M031 · Smooth quintic characteristic-class calculation · passed

M032 · Formal inverse coefficient certificate · passed

M033 · A2 Weyl denominator · passed

M034 · Affine A1 Jacobi triple-product finite certificate · passed

M035 · Young-tableau count by two independent routes · passed

M036 · A nonconstant type-A Kazhdan-Lusztig calculation · passed

M037 · Binary narrow-sense BCH complete finite certificate · passed

Chapter b supplement: 1 scoped records.

S011 · Observation order and sharp aligned heat-entry margin · passed

Chapter c: 21 scoped records.

M038 · Finite Fourier coefficients and exact tail energy · pass

M038 · SU(2) Haar integrals · pass

M039 · Unitary Fourier matrix attains rephasing-invariant bound · pass

M040 · Boundary ranks reproduce the entire filtration · pass

M041 · Berry holonomy and sampled gauge cancellation · pass

M042 · Exact two-level square and gap · pass

M043 · Moore-spectrum cofiber argument · proof-only

M044 · Exact conservation contraction and positivity counterexample · pass

M045 · Reflection commutator on a basis and asymmetric witness · pass

M046 · Affine composition order · pass

M047 · Direct joint-table information and cascade loss · pass

M048 · A second effect separates otherwise identical probabilities · pass

M049 · Observable timescale saturates Robertson-derived bound · pass

M050 · Eisenstein coefficients and forced-zero numerical residual · pass

M051 · Torus action and a periodic small gauge change · pass

M052 · Torus anomaly and complete two-label fusion table · pass

M053 · Kuramoto branches and nonlocking witness · pass

M054 · Exact conductance nonidentifiability witness · pass

M054 · Gate closed-form invariant interval samples · pass

M055 · Finite decision threshold and mixed best-response equality · pass

M056 · Nonuniform prior-and-utility limit · pass

Chapter c supplement: 9 scoped records.

S009 · Rational incidence and cut-flow certificate · pass

S009 · Exact sine-cosine feasibility · pass

S009 · Exact rational node phasors · pass

S009 · Positive tree Hessian certificate · pass

S009 · Complete eight-branch Morse classification · pass

S009 · Capacity sensitivity and critical-slowing bounds · pass

S009 · Exact relative and center-only observability ranks · pass

S009 · Exact-form nonlinear center blindness replay · pass

S009 · Finite exhaustive cut-flow replay · pass

Chapter d: 21 scoped records.

M057 · Cauchy averaging characteristic function · pass

M058 · Jordan-block transient formula · pass

M059 · Block-energy accounting · pass

M060 · Probability versus amplitude weighting · pass

M061 · Covariance Gram matrix · pass

M062 · PSD quadratic programs and nonnegative KKT certificate · pass

M063 · Amplitude damping Kraus channel · pass

M064 · Exponential target-time correction · pass

M065 · Gradient-flow coordinate transport · pass

M066 · Bessel residual · pass

M067 · Young inequality and sharp equality · pass

M068 · Distinct matrix norms · pass

M069 · Exact Lyapunov certificate · pass

M070 · Einstein-equation dimensional exponents · pass

M071 · Newtonian normalization factor · pass

M072 · Component and canonical counts · pass

M073 · FLRW continuity exponents · pass

M074 · Finite-rank closure illustration only · pass

M075 · Partition counts and discrete continuum energy · pass

M076 · Exact frame conversion and speed · pass

M077 · Periodic two-band dispersion and gap · pass

Chapter d supplement: 12 scoped records.

S008 · All coordinate panels exhaustively enumerated · pass

S008 · Unique exact optimum · pass

S008 · Every cheaper panel has a failed target · pass

S008 · Optimal energies and robust margins · pass

S008 · No single channel is cost-dominated · pass

S008 · Every declared target constraint is necessary · pass

S008 · Every channel appears in an inclusion-minimal feasible panel · pass

S008 · Declared greedy heuristic replay · pass

S008 · Strict finite greedy gap · pass

S008 · Robust lower bounds are attained · pass

S008 · Withheld-direction exact blindness · pass

S008 · Projection identity · pass

Chapter e: 22 scoped records.

M078 · Whitened discrepancy and unit invariance · passed

M079 · Composite derivative · passed

M080 · Graph heat time-step threshold · passed

M081 · Certified mantissa bracket · passed

M082 · Noiseless power-family identification · passed

M083 · Signed dyadic decomposition · passed

M084 · Directional cancellation illustration · passed

M085 · Normalized qubit kernel reconstruction · passed

M086 · Exact polynomial Moyal algebra · passed

M087 · Fractional periodic spectral evolution · passed

M088 · Lagrange identity · passed

M089 · Winding and sphere calibration · passed

M090 · Marked torus stretch · passed

M091 · Certified logistic thresholds · passed

M092 · State-versus-parameter examples · passed

M093 · Convex conjugate and boundary optimizer · passed

M094 · CHSH deterministic exhaustion · passed

M095 · Syzygy and strict-transform checks · passed

M096 · Toric lattice certificate · passed

M097 · Hodge consistency examples · passed

M098 · Conic parametrization and obstruction · passed

M099 · Typed composition and Hermitian positivity · passed

Chapter e supplement: 5 scoped records.

S012 · Nontrivial output-quotient counterexample · passed

S012 · Ridge separation of range and nullspace · passed

S012 · Infeasibility by a dual separating direction · passed

S012 · Rank and active-constraint parameter strata · passed

S012 · Degenerate optimizer sets · passed

Chapter e independent: 2 scoped records.

S014 · Independent quartic complete Jordan certificate · passed

S014 · Independent eight-site characteristic polynomial by determinants · passed

Chapter f: 16 scoped records.

M100 · Bernstein-Sato coefficient identities · passed

M101 · Conic polarity and tangent-locus test · passed

M102 · Ordered cross-ratio under a projectivity · passed

M103 · Heisenberg mode algebra on polynomial states · passed

M104 · Four-site XXX spectrum and Bethe roots · passed

M105 · Gaudin matrices and exact oper residual polynomials · passed

M105 · Complete rational three-site Gaudin spectrum and opers · passed

M106 · Independence arithmetic and dependence bounds · passed

M107 · Nonlinear observation collision · passed

M108 · Klein-Gordon dispersion residual · passed

M109 · Pauli algebra and nonrelativistic energy expansion · passed

M110 · Exterior derivative and plane-wave checks · passed

M111 · Ward difference identity and transverse null direction · passed

M112 · Gaussian quartic perturbation coefficients · passed

M113 · Toy beta flow and fixed-point slopes · passed

M114 · Regular-level velocity and critical-level failure · passed

Chapter f supplement: 5 scoped records.

S010 · Complete rational 2-site one-magnon sector · passed

S010 · Complete rational 3-site one-magnon sector · passed

S010 · Complete rational 4-site one-magnon sector · passed

S010 · Exact exceptional three-site Jordan point · passed

S010 · Independent replay of the full eight-dimensional rational example · passed

Chapter f spectral: 4 scoped records.

S013 · Exceptional Gram and constant-projector identities · passed

S013 · All three exact nonzero square-zero Jordan parts · passed

S013 · Projector blowup with bounded sum · passed

S013 · Tracked spectral-cover label exchange · passed

Chapter f jordan: 4 scoped records.

S014 · Multiplicity-three critical point and complete length-three chain · passed

S014 · Closed characteristic formula and fractional-linear Hamiltonian identities for four sites · passed

S014 · Mixed multiplicity example with exact characteristic factorization · passed

S014 · Closed characteristic formula and fractional-linear identities in the mixed case · passed

Chapter syntheses: 12 scoped records.

S001 · Exact second-order recurrence · passed

S001 · Minimality witness · passed

S001 · Subsampling alias · passed

S002 · Finite-bin Fisher information after Gudermannian quantization · passed

S002 · Gudermannian chart and Jacobian identities · passed

S003 · Observation class counts · passed

S003 · Exact Lagrange indicator basis · passed

S004 · Noncommuting symmetric generators satisfy forward and inverse example bounds · passed

S005 · Exact stochasticity and entropy example on a four-cycle · passed

S005 · Positivity endpoint can retain a periodic mode · passed

S006 · Retained and invisible Gaussian shifts · passed

S006 · Worst-case robust noncentrality is attained · passed

Mathematical references

R001 Boyd and Vandenberghe Introduction to Applied Linear Algebra

https://stanford.edu/~boyd/vmls/

R002 NIST DLMF Gudermannian

https://dlmf.nist.gov/4.23#viii

R003 Berkeley Score Function and Fisher Information

https://stat210a.berkeley.edu/fall-2025/reader/score-fisher.html

R004 Higham Functions of Matrices Matrix Logarithm

https://epubs.siam.org/doi/10.1137/1.9780898717778.ch11

R005 Stanford CS265 Spectral Analysis of Markov Chains

https://web.stanford.edu/class/archive/cs/cs265/cs265.1232/Lectures/Lecture13/class13.pdf

R006 Brunton and coauthors Koopman invariant subspaces and finite linear representations of nonlinear dynamical systems for control

https://arxiv.org/abs/1510.03007

R007 Finite Dimensional Koopman Form of Polynomial Nonlinear Systems

https://arxiv.org/abs/2301.06557

R008 Dörfler Chertkov and Bullo Synchronization in complex oscillator networks and smart grids PNAS 2013 classical tree case

https://pmc.ncbi.nlm.nih.gov/articles/PMC3568350/

R009 Dörfler and Bullo Exploring Synchronization in Complex Oscillator Networks

https://arxiv.org/abs/1209.1335

R010 Frenkel Gaudin model and opers

https://arxiv.org/abs/math/0407524

R011 Brouwer Shrikhande graph reference and strongly regular parameters

https://aeb.win.tue.nl/graphs/Shrikhande.html

R012 Boyd and Vandenberghe Convex Optimization, conjugate functions and equality-constrained quadratic programming

https://web.stanford.edu/~boyd/cvxbook/

R013 NIST DLMF q calculus

https://dlmf.nist.gov/17.2

R014 Keith Conrad p-adic course materials

https://kconrad.math.uconn.edu/math3094f16/

R015 Keith Conrad Hensel’s Lemma

https://kconrad.math.uconn.edu/blurbs/gradnumthy/hensel.pdf

R016 John Baez The Octonions

https://arxiv.org/abs/math/0105155

R017 Baez Cayley-Dickson construction

https://math.ucr.edu/home/baez/octonions/node5.html

R018 Biondini and Chakravarty Soliton solutions of the KP II equation

https://arxiv.org/abs/nlin/0511068

R019 On tau-functions for the KdV hierarchy

https://arxiv.org/abs/1812.08488

R020 NIST DLMF q derivatives

https://dlmf.nist.gov/17.2#iv

R021 Stephen Boyd EE263 lecture materials

https://web.stanford.edu/class/archive/ee/ee263/ee263.1082/lectures.html

R022 University of Bonn Hardy and BMO notes

https://www.math.uni-bonn.de/ag/ana/WiSe1415/V4B5_hardy.pdf

R023 Newman Finding community structure using eigenvectors

https://arxiv.org/abs/physics/0605087

R024 Terence Tao Topics in Random Matrix Theory

https://terrytao.wordpress.com/books/topics-in-random-matrix-theory/

R025 Levin and Peres Markov Chains and Mixing Times

https://pages.uoregon.edu/dlevin/MARKOV/

R026 NIST DLMF continuous q-Hermite polynomials

https://dlmf.nist.gov/18.28#vi

R027 Nash Equilibrium Points in N-Person Games

https://doi.org/10.1073/pnas.36.1.48

R028 J S Milne Introduction to Shimura Varieties

https://www.jmilne.org/math/articles/2005aX.pdf

R029 J S Milne Toroidal Compactification of Shimura Varieties

https://www.jmilne.org/math/articles/1990aT.pdf

R030 Stacks project algebraic curves

https://stacks.math.columbia.edu/download/curves.pdf

R031 Stacks project very ample line bundles on curves

https://stacks.math.columbia.edu/tag/0H2V

R032 Ravi Vakil Foundations of Algebraic Geometry adjunction

https://math.stanford.edu/~vakil/216blog/FOAGjul2724public.pdf

R033 Eisenbud and Harris 3264 and All That

https://dash.harvard.edu/bitstreams/7312037e-7c66-6bd4-e053-0100007fdf3b/download

R034 MIT Lie algebra course notes

https://ocw.mit.edu/courses/18-745-lie-groups-and-lie-algebras-i-fall-2020/lists/lecture-notes/

R035 NIST DLMF Jacobi triple product

https://dlmf.nist.gov/17.8

R036 SageMath tableaux reference

https://doc.sagemath.org/html/en/reference/combinat/sage/combinat/tableau.html

R037 SageMath Kazhdan-Lusztig polynomials

https://doc.sagemath.org/html/en/reference/combinat/sage/combinat/kazhdan_lusztig.html

R038 SageMath BCH code reference

https://doc.sagemath.org/html/en/reference/coding/sage/coding/bch_code.html

R039 Eckhard Meinrenken Lie groups and Lie algebras compact-group representation background

https://www.math.toronto.edu/mein/teaching/LectureNotes/lie.pdf

R040 Particle Data Group neutrino mixing review

https://pdg.lbl.gov/2024/reviews/rpp2024-rev-neutrino-mixing.pdf

R041 Allen Hatcher Algebraic Topology homology chapters

https://pi.math.cornell.edu/~hatcher/AT/ATchapters.html

R042 Michael Berry Quantal phase factors accompanying adiabatic changes

https://doi.org/10.1098/rspa.1984.0023

R043 Jacob Lurie Chromatic Homotopy Theory course

https://people.math.harvard.edu/~lurie/252x.html

R044 Claude Shannon A Mathematical Theory of Communication

https://www.princeton.edu/~wbialek/rome/refs/shannon_48.pdf

R045 Albeverio and Motovilov Quantum speed limits for time evolution of a system subspace

https://arxiv.org/abs/2011.02778

R046 J S Milne Modular Functions and Modular Forms

https://www.jmilne.org/math/CourseNotes/MF.pdf

R047 Edward Witten Quantum field theory and the Jones polynomial

https://doi.org/10.1007/BF01217730

R048 Andersen and Ueno Modular functors are determined by their genus zero data

https://arxiv.org/abs/math/0611087

R049 Daffertshofer and van Wijk On the influence of amplitude on connectivity between phases

https://pmc.ncbi.nlm.nih.gov/articles/PMC3139941/

R050 Hodgkin and Huxley original quantitative conductance model

https://pmc.ncbi.nlm.nih.gov/articles/PMC1392413/

R051 John Nash Equilibrium points in n-person games

https://pmc.ncbi.nlm.nih.gov/articles/PMC1063129/

R052 NIST Cauchy distribution reference

https://itl.nist.gov/div898/handbook/eda/section3/eda3663.htm

R053 Verdú The Cauchy Distribution in Information Theory

https://pmc.ncbi.nlm.nih.gov/articles/PMC9955388/

R054 Boyd and coauthors Linear Matrix Inequalities in System and Control Theory

https://web.stanford.edu/~boyd/lmibook/index.html

R055 Microscopic derivation of one-qubit amplitude and phase damping Kraus operators

https://arxiv.org/abs/1606.01145

R056 Solution of the Lindblad equation in the Kraus representation

https://doi.org/10.1103/PhysRevA.74.062113

R057 MIT OpenCourseWare Functional Analysis Hilbert spaces

https://ocw.mit.edu/courses/18-102-introduction-to-functional-analysis-spring-2021/

R058 Sean Carroll General Relativity lecture notes

https://www.preposterousuniverse.com/wp-content/uploads/2015/08/grtinypdf.pdf

R059 Sean Carroll General Relativity lecture notes

https://preposterousuniverse.com/wp-content/uploads/grnotes-one.pdf

R060 SGA 4 Exposé I universes and strongly inaccessible cardinals

https://grothendiecksga.com/read/sga4/en/I-A-5.html

R061 Stacks Project constructing categories of schemes

https://stacks.math.columbia.edu/tag/000H

R062 Asboth Oroszlany and Palyi A Short Course on Topological Insulators

https://arxiv.org/abs/1509.02295

R063 Kutyniok, Lemvig and Lim, Shearlets and Optimally Sparse Approximations

https://arxiv.org/abs/1106.1325

R064 Tilma and coauthors, Wigner Functions for Arbitrary Quantum Systems

https://arxiv.org/abs/1601.07772

R065 Rundle and Everitt, Overview of phase-space quantum mechanics

https://arxiv.org/abs/2102.11095

R066 Lizzi and Vitale, Matrix Bases for Star Products

https://sigma-journal.com/2014/086/

R067 Nick Laskin, Fractional Schrodinger equation

https://arxiv.org/abs/quant-ph/0206098

R068 Howard Masur, Teichmuller geometry of moduli space

https://www.math.uchicago.edu/~masur/handbook1.pdf

R069 Yuri Kuznetsov, Elements of Applied Bifurcation Theory, flip theorem

https://www.ma.imperial.ac.uk/~dturaev/kuznetsov.pdf

R070 Singular official tutorial, free resolutions and Schreyer methods

https://www.singular.uni-kl.de/ftp/pub/Math/Singular/doc/tutor.pdf

R071 Álvarez Montaner and coauthors Bernstein-Sato functional equations

https://web.mat.upc.edu/josep.alvarez/pdf/Vfiltration.pdf

R072 Saito Introduction to a theory of b-functions

https://arxiv.org/abs/math/0610783

R073 Frenkel and Ben-Zvi Vertex Algebras and Algebraic Curves

https://math.berkeley.edu/~frenkel/BOOK/

R074 Faddeev How Algebraic Bethe Ansatz works for integrable model

https://arxiv.org/abs/hep-th/9605187

R075 Tong Free Fields lecture chapter

https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S2.html

R076 Tong Spinor Fields lecture chapter

https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html

R077 Tong Gauge Theory electromagnetism notes

https://www.damtp.cam.ac.uk/user/tong/gaugetheory/1em.pdf

R078 Beisert Quantum Field Theory II notes

https://people.phys.ethz.ch/~nbeisert/lectures/QFT2-17FS-Notes.pdf

R079 Tong Interacting Fields lecture chapter

https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S3.html